

Probability Distributions

by

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Topic No. 1

Random Experiment, Outcomes, Sample Space & Events

02:32

Random Experiment

Any experiment or process in which the outcome is unpredictable that is eligible to be called a random experiment.

03:04

Outcomes

The various possible results of a random experiment.

Example 1: Head, Tail

Example 2: Single, Married, Separated, Divorced, Widowed

00:49

Random Experiment

Any experiment or process in which the outcome is unpredictable that is eligible to be called a random experiment.

03:34

Sample Space

It is the set of all possible outcomes of a random experiment.

01:08

Example 1:

Tossing of a Coin => a Head or a Tail.

However, until it lands on the floor, we do not know which of these two will turn up.

04:01

Example 1: Head , Tail

Example 2: Toss a single die
 $1, 2, 3, 4, 5, 6$

Sample space will be the set of all of these taken together written as

$$S = \{1, 2, 3, 4, 5, 6\}$$

01:45

Example 2:

Survey of the Employees of an Organization:

Question on Marital Status:

Possible outcomes:

single, married, separated, divorced, widowed

However, until we obtain a reply from the respondent, we do not know what is his/her marital status.

04:42

In case of two fair dice, the sample space consists of the 36 ordered pairs:

i.e. $\{(1, 1), (1, 2), (1, 3), \dots, (6, 5), (6, 6)\}$.

05:55

Event

Any subset of a sample space is called an Event.

00:18

Concept of a Random Variable

It is that variable which is associated with a random experiment.

Example:

Let the random experiment be the toss of a fair coin and the possible outcomes are Head and Tail. And the sample space associated with the experiment denoted by C be $C = \{H, T\}$, where H and T represent heads and tails, respectively.

06:20

Example 1

The set of ordered pairs which yield a sum which is an even number

denoted as $A = \{(1,1), (1,3), (1,5), (2,2), (2,4), (2,6), \dots, (6,6)\}$.

01:13

Let X be a function such that $X(T) = 0$ and $X(H) = 1$. Thus X is a real-valued function defined on sample space C .

That X is a function defined on the sample space C , $X(T) = 0$ and $X(H) = 1$ which takes us from the sample space C to a space of real numbers $D = \{0, 1\}$.

07:24

Example 2

The set of ordered pairs in which both the numbers are same

denoted as $B = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\}$

02:58

C is H and T and applying the function the image we obtain i.e the domain and range concept, So, if the domain here is H and T then the range will be 0 and 1.

07:44

$A = \{(1,1), (1,3), (1,5), (2,2), (2,4), (2,6), \dots, (6,6)\}$.

$B = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\}$

Event B is a subset of event A .

03:37

Definition

Consider a random experiment with a sample space C . A function X , which assigns to each element $c \in C$ one and only one number $X(c) = x$, is called a **random variable**.

The **space** or **range** of X is the set of real numbers

$D = \{x : x = X(c), c \in C\}$.

00:09

Topic No. 02

Definition of Random Variable

04:53

Given a random variable X , its range D becomes the space of interest. Besides inducing the space D , X also induces a probability which we call the distribution of X .

We are interested in knowing the probability of getting a 0 and 1, and as we already know that if it's a fair coin then the probability of getting a 0 is half and also for the probability of getting a 1 is half as well.

05:37

But, generally in this case, we can say that the set D is a finite or countable set or an interval of real numbers.

When a set is finite then, we are referring to discrete random variables. We call random variables of the first type **discrete** random variables, while those of the second type **continuous** random variables.

06:18

Discrete random variable has two situation, one is exactly the same I mentioned earlier where a set D is a finite set that has the elements $\{0, 1\}$ or the other, if we toss a fair die we have the elements $\{1, 2, 3, 4, 5, 6\}$.

The second situation, which also gives you a discrete random variable that is when it is the **Countably infinite set**.

07:00

Example:

D is the set consisting of the numbers 1, 2, 3, and so on, which mean that they are countable but not finite. So, in this situation, we say that this is a Discrete random variable or if they are finite the elements may be called $d_1, d_2, \dots, d_m$ then also it is a discrete random variable.

00:00

Topic No. 3

Example of Random Variable

00:12

Example:

Let a card be selected from an ordinary deck of playing cards. The outcome is one of these 52 cards. Suppose that P assigns a probability of $1/52$ to each outcome c .

Let $X(c)=4$ if c is an ace, let $X(c)=3$ if c is a king, let $X(c)=2$ if c is a queen, let $X(c)=1$ if c is a jack and let $X(c)=0$ otherwise.

02:25

04:28

Now, we can develop a table with two columns.

The first column consisting of the values of the random variable X i.e. 0, 1, 2, 3, 4 and the second are the probabilities.

The classical definition of probability i.e. favorable over the total.

The sum of probability should be equal to 1.

x	$p_X(x)$
0	$\frac{36}{52}$
1	$\frac{4}{52}$
2	$\frac{4}{52}$
3	$\frac{4}{52}$
4	$\frac{4}{52}$
	$\sum_{x \in D} p_X(x) = \frac{52}{52} = 1$

00:09

Topic No. 04

Discrete Random Variable

00:12

Definition

(Discrete Random Variable)

We say a random variable is a discrete random variable if its space is either finite or countable.

00:24

Example:

Consider a sequence of independent flips of a coin, each resulting in a head (H) or a tail (T). Moreover, on each flip, we assume that H and T are equally likely; that is, $P(H)=P(T)=1/2$.

x	$p(x)$
0	1/2
1	1/2
sum	1

This is that case, when the space is Finite.

01:43

Example of a Countably infinite set:

Let the random variable X equal the number of tosses needed to obtain the first head. The space of X is $D= \{1,2,3,4,\dots\}$.

Consider that $X=1$ when the sequence begins with a H.

02:58

Likewise, $X=2$ when the sequence begins with TH.
In the same way when $X=3$ the sequence is TTH. See we are getting a head in the third toss.

04:54

Suppose that we are interested in finding the probabilities of the various possible values of X.

As shown in the table:

Although it is an infinite sequence of probabilities the sum has to be 1.

x	P(x)
1	$\left(\frac{1}{2}\right)^1$
2	$\left(\frac{1}{2}\right)^2$
3	$\left(\frac{1}{2}\right)^3$
4	$\left(\frac{1}{2}\right)^4$
⋮	⋮
⋮	⋮
⋮	⋮
∞	
Σ	1

05:51

$$P(X=x)=\left(\frac{1}{2}\right)^{x-1}\left(\frac{1}{2}\right)=\left(\frac{1}{2}\right)^x, \quad x=1,2,3,\dots$$

Here, X is the total no. of tosses that was required in order to get the first Head.

07:01

Now, let $X=1$ we see,

$$P(X=x)=\left(\frac{1}{2}\right)^{x-1}\left(\frac{1}{2}\right)=\left(\frac{1}{2}\right)^x, \quad x \in \{1 \text{ or } 2 \text{ or } 3 \text{ or } \dots\}$$

$$P(X=1)=\left(\frac{1}{2}\right)^{1-1}\left(\frac{1}{2}\right)=\left(\frac{1}{2}\right)^1$$

$$P(X=2)=\left(\frac{1}{2}\right)^{2-1}\left(\frac{1}{2}\right)=\left(\frac{1}{2}\right)^2$$

$$P(X=3)=\left(\frac{1}{2}\right)^{3-1}\left(\frac{1}{2}\right)=\left(\frac{1}{2}\right)^3$$

Hence, the sequence of the probability will be

08:54

$$\left\{\left(\frac{1}{2}\right)^1+\left(\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^3+\dots\right\}$$

Obviously, that is an infinite geometric series

$$a=\frac{1}{2}, \quad r=\frac{\left(\frac{1}{2}\right)^2}{\frac{1}{2}}=\left(\frac{1}{2}\right)$$

as we know that

$$\text{Geometric series}=\frac{a}{1-r}$$
$$P[X \in \{1,2,3,\dots\}]=\frac{1/2}{1-(1/2)}=1$$

00:09

Topic No. 5

Concept of Probability Mass Function

00:19

What is Probability Mass Function?

Consider a random experiment with a sample space, finite or countably infinite. A function X, which assigns to each element $c \in C$ one and only one number $X(c) = x$, is called a random variable. If we express the probabilities against each random values of X algebraically, then, we can say that this is the probability mass function of X.

01:00

Example:

Toss a fair die, the possible values are 1,2,3,4,5,6.

Algebraically:

$$P(X=x)=\frac{1}{6}, \quad x=1,2,3,4,5,6.$$

01:41

Definition

The probability mass function (pmf) of X is given by

$$p_x(x)=P[X=x], \quad \text{for } x \in D. \quad (1)$$

02:01

The PMFs satisfy two properties:

- 1) $0 \leq p_X(x) \leq 1, \quad x \in D$
- 2) $\sum_{x \in D} p_X(x) = 1 \tag{2}$

07:08

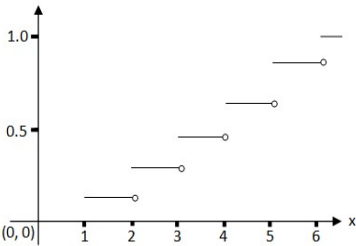


Figure 1: Distribution Function

The height of each ‘Jump’ yields the probability of that particular value of X.

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CDF of a Discrete Random Variable

Topic No. 6
Example of Probability Mass Function

03:28

00:15

Example:

Suppose we roll a fair die with the numbers 1 through 6 on it. Let X be the upface of the roll. Then the space of X is {1,2,...,6} and its pmf is $p_X(i) = 1/6, \text{ for } i = 1, 2, \dots, 6.$

Example:
Consider an urn which contains slips of paper each with one of the numbers 1,2,...,100 on it. Suppose there are i slips with the number i on it for $i = 1, 2, \dots, 100$. For example, there are 25 slips of paper with the number 25. Assume that the slips are identical except for the numbers. Suppose one slip is drawn at random. Let X be the number of slip.
a) Suppose that X has the pmf $p(x) = x/5050, x = 1, 2, 3, \dots, 100$, zero elsewhere
b) Compute $P(X \leq 50)$.

04:43

01:32

$$F(6) = P(X \leq 6) = 1/6$$

or

$$F(4) = P(X \leq 4) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{4}{6}$$

Solution:
a) Suppose that X has the pmf $p(x) = x/5050, x = 1, 2, 3, \dots, 100$, zero elsewhere.

i	
1	1 slip containing the no. 1
2	2 slip containing the no. 2
3	3 slip containing the no. 3
.	
.	
.	
100	100 slip containing the no. 100

Total number of slips = $1+2+3+\dots+100$

06:30

02:22

x	$p(x)$	$F(x)$
1	1/6	1/6
2	1/6	$=1/6+1/6=2/6$
3	1/6	$=2/6+1/6=3/6$
4	1/6	$=3/6+1/6=4/6$
5	1/6	$=4/6+1/6=5/6$
6	1/6	$5/6+1/6=6/6=1$

‘Classical definition’ = m/n

The sum of the first n natural numbers $= \frac{n(n+1)}{2}$
Here, total no. of slips $= \frac{100(101)}{2} = 5050$
As the slips are identical i.e. equally likely to be drawn
Therefore, the probability $= \frac{m}{5050}$

04:30

X	Frequency (f)	$\frac{f}{\sum f} = p(x) = \frac{m}{n}$
1	1	1/5050
2	2	2/5050
3	3	3/5050
4	4	4/5050
-	-	-
-	-	-
-	-	-
100	100	100/5050
	5050	5050/5050=1

Now, to express mathematically,

$$P(X = x) = \frac{x}{5050}, \quad x=1,2,3,...,100$$

05:26

b) Compute $P(X \leq 50)$.
Solution:

$$P(X \leq 50) = \frac{1+2+3+...+50}{5050}$$

or

$$P(X \leq 50) = \frac{\frac{n(n+1)}{2}}{5050} = \frac{\frac{50(51)}{2}}{5050} = \frac{51}{202}$$

$$\therefore P(X \leq 50) = 0.2525$$

01:57

- In reality it is a some value between 63.5 and 64.5, if we use a more refined instrument and after measuring we say that it is actually 64.2 inches. Even this is not the actual truth, it is the height somewhere between 64.15 and 64.25 inches.
- It all depend on the measuring instrument that how much precision can that instrument measures.
- Theoretically there can be an infinite number of decimal places after the decimals. This is the basic concept that variable can assume any value.

03:39

Cumulative Distribution Function $F(x)$

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

00:09

Topic No. 7

Continuous Random Variable

04:40

The function $f_X(t)$ is called a **probability density function (pdf)** of X. If $f_X(x)$ is also continuous, then the Fundamental Theorem of Calculus implies that

$$\frac{d}{dx} F_X(x) = f_X(x)$$

00:53

Concept of a Continuous Random Variable

In the continuous case, we cannot define probability on one particular point, instead, it is the probability defined on an interval.

05:00

The PDFs satisfy the two properties:

$$1) \int_{-\infty}^{\infty} f_X(x) dt = 1 \quad \text{and}$$

$$2) f_X(x) \geq 0$$

01:20

Example:

If we are measuring a height of a student. It depends on the refinement of that measuring instrument.

Suppose if we say that this person is 5 feet 4 inches tall, or in other words we are saying he is 64 inches tall. This is the measurement at a crude level.

06:09

In the Continuous case, the area under the curve of that function gives you the probability.

If you are interested in computing the probabilities, then, they can be obtained by integration, i.e.,

$$P(a < X \leq b) = \int_a^b f_X(t) dt = F_X(b) - F_X(a)$$

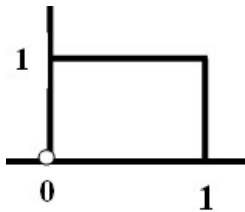
07:18

Example:

Suppose that, we randomly select a number between 0 and 1. Recall that the cdf of X is $F_X(x) = x$, for $0 < x < 1$. Hence the pdf of X is given by

$$f_X(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

08:14



This is called a Uniform or rectangular distribution where, $0 < x < 1$.

00:09

Topic No. 8

**Example of
Probability
Density Function**

00:15

Example:

In continuous case, consider the following simple experiment: choose a real number at random from the interval $(0,1)$.

Let X be the number chosen. In this case the space of X is $D=(0,1)$.

00:42

The number is chosen at random it is reasonable to assign

$$P_X[(a,b)] = b - a, \\ \text{for } 0 < a < b < 1.$$

0:56

It follows that the pdf of X is

$$f_X(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

For example, the probability that X is less than an eighth or greater than seven by eighths is

$$P\left[\left\{X < \frac{1}{8}\right\} \cup \left\{X > \frac{7}{8}\right\}\right] = \int_0^{\frac{1}{8}} dx + \int_{\frac{7}{8}}^1 dx \\ = x \Big|_0^{\frac{1}{8}} + x \Big|_{\frac{7}{8}}^1$$

03:00

$$= \left(\frac{1}{8} - 0\right) + \left(1 - \frac{7}{8}\right) \\ = \frac{1}{8} + \frac{1}{8} \\ = \frac{2}{8} \\ \therefore P\left[\left\{X < \frac{1}{8}\right\} \cup \left\{X > \frac{7}{8}\right\}\right] = \frac{1}{4}$$

00:15

Topic No. 9

Probability Generating Function

00:37

Probability Generating Function

- In probability theory, the **probability generating function** of a discrete random variable is a power series representation (the generating function) of the probability mass function of the random variable.

01:12

Definition

Univariate case

If X is a discrete random variable taking values in the non negative integers $\{0,1, \dots\}$, then the *probability generating function* of X is defined as

$$G(z) = E(z^X) = \sum_{x=0}^{\infty} p(x) z^x,$$

where p is the probability mass function of X .

02:38

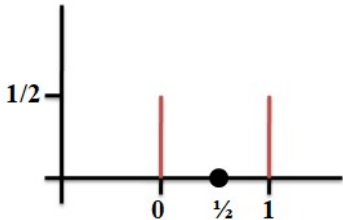
Example:

The probability generating function of a Bernoulli random variable with parameter p . So the probability generating function of X , where X represents the number of heads that we obtain when tossing a fair coin once is given by

$$G(z) = \frac{1}{2} + \frac{z}{2}$$

04:16

x	p(x)	z^x	$z^x p(x)$
0	1/2	1	1/2
1	1/2	z	$z/2$
Sum			$\frac{1}{2} + \frac{z}{2}$



06:27 Example:

For Poisson Distribution = $G(z) = e^{\lambda(z-1)}$

x	$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$	z^x	$z^x p(x)$
0	$\frac{e^{-\lambda} \lambda^0}{0!} = e^{-\lambda}$	$z^0 = 1$	$e^{-\lambda}$
1	$\frac{e^{-\lambda} \lambda^1}{1!}$	$z^1 = z$	$e^{-\lambda} \lambda z$
2	$\frac{e^{-\lambda} \lambda^2}{2!}$	$z^2 = z^2$	$\frac{e^{-\lambda} (\lambda z)^2}{2!}$
3	$\frac{e^{-\lambda} \lambda^3}{3!}$	$z^3 = z^3$	$\frac{e^{-\lambda} (\lambda z)^3}{3!}$
.	.	.	.
.	.	.	.
.	.	.	.
∞	∞	∞	∞

09:53

$$\begin{aligned} \sum p(x) z^x &= e^{-\lambda} + \frac{e^{-\lambda} (\lambda z)^1}{1!} + \frac{e^{-\lambda} (\lambda z)^2}{2!} + \dots \\ \sum p(x) z^x &= e^{-\lambda} \left[1 + (\lambda z)^1 + \frac{(\lambda z)^2}{2!} + \frac{(\lambda z)^3}{3!} + \dots \right] \\ G(z) &= e^{-\lambda} \left[e^{\lambda z} \right] \\ &= e^{-\lambda + \lambda z} = e^{\lambda(z-1)} \quad \therefore e^{\theta} = 1 + \theta + \frac{\theta^2}{2!} + \frac{\theta^3}{3!} + \dots \end{aligned}$$

00:10

Topic No. 10

How to utilize Probability
Generating Function
and
Mean and Variance through
PGF

00:43

The **probability mass function** of X is recovered by taking derivatives of G ,

$$p(k) = P(X = k) = \frac{G^{(k)}(0)}{k!} \tag{1}$$

01:15

Example:

The probability generating function of a Bernoulli random variable with parameter p . So the probability generating function of X , where X represents the number of heads that we obtain when tossing a fair coin once is given by

$$G(z) = 1/2 + z/2$$

01:42

With the help of eq. (1), we take the derivative

$$\begin{aligned} \lim_{x \rightarrow \infty} G(z) &= \frac{1}{2} + \frac{z}{2} \\ P(0) = P(X=0) &= \frac{G^{(0)}(0)}{0!} = \frac{\left(\frac{1}{2} + \frac{z}{2} \right)}{0!} \bigg|_{z=0} = \frac{1}{2} \\ P(1) = P(X=1) &= \frac{G^{(1)}(0)}{1!} = \frac{\left(\frac{1}{2} + \frac{z}{2} \right)}{1!} \bigg|_{z=0} = \frac{1}{2} \end{aligned}$$

04:14

MEAN
and VARIANCE
through the
Probability generating
function

04:30

The expectation of X is given by

$$E(X) = G'(1^-) \tag{1}$$

where,
 $G'(1^-) = \lim_{z \rightarrow 1} G'(z)$ from below.

05:30

Example:

Poisson Distribution (Mean)

$$G(z) = e^{\lambda(z-1)}$$

$$\therefore G'(z) = \lambda e^{\lambda(z-1)}$$

$$G'(1^-) = \lim_{z \rightarrow 1} G'(z)$$

$$= \lambda e^{\lambda(1-1)} = \lambda e^{\lambda(0)} = \lambda e^0$$

$$\therefore G'(1^-) = \lambda$$

01:28

(ii) Bernoulli r.v. - if

$p_1 = p$, $p_0 = 1 - p = q$, $p_k = 0$, $k \neq 0$ or 1,
then

$$G_X(z) = E(z^X) = q + pz$$

07:00

So the variance of X is given by

$$\text{Var}(X)$$

$$= G''(1^-) + G'(1^-) - [G'(1^-)]^2 \quad (2)$$

02:12

(iii) Geometric r.v. - if

$p_k = pq^{k-1}$, $k = 1, 2, \dots$; $q = 1 - p$, then

$$G_X(z) = \frac{pz}{1 - qz} \text{ if } |z| < q^{-1}$$

07:20

Example:

Poisson Distribution (Variance)

$$G''(z) = \lambda \cdot \lambda e^{\lambda(z-1)} = \lambda^2 e^{\lambda(z-1)}$$

$$\therefore G''(1^-) = \lambda^2 e^{\lambda(1-1)} = \lambda^2 e^{\lambda(0)} = \lambda^2$$

$$G'(1^-) = \lambda \quad \leftarrow \text{From eq. (2)}$$

$$\therefore V(X) = \lambda^2 + \lambda - [\lambda]^2$$

$$\therefore V(X) = \lambda$$

02:38

(iv) Binomial r.v. - if

$X \sim \text{Bin}(n, p)$, then

$$G_X(z) = (q + pz)^n, \quad (q = 1 - p)$$

00:15

03:45

Topic No. 11

Algebraic Expressions of Some Well-known PGFs

(v) Poisson r.v. - if

$X \sim \text{Poisson}(\lambda)$, then

$$G_X(z) = \sum_{k=0}^{\infty} \frac{1}{k!} \lambda^k e^{-\lambda} z^k = e^{\lambda(z-1)}$$

00:50

03:58

For some of the more common
distributions, the PGFs are as
follows:

(i) Constant r.v. -

if $p_c = 1$, $p_k = 0$, $k \neq c$, then

$$G_X(z) = E(z^X) = z^c$$

(vi) Negative binomial r.v. - if

$X \sim \text{NegBin}(n, p)$, then

$$G_X(z) = \sum_{k=0}^{\infty} \binom{k-1}{n-1} p^n q^{k-n} z^k$$

$$= \left(\frac{pz}{1 - qz} \right)^n \text{ if } |z| < q^{-1} \text{ and } p + q = 1$$

04:41 Discrete Uniform Distribution

Example:
Random selection of one of the ten digits 0,1,2,...,9

X	$p(x)$	z^x	$z^x p(x)$
0	1/10	z^0	1/10
1	1/10	$z^1 = z$	$z / 10$
2	1/10	z^2	$z^2 / 10$
3	1/10	z^3	$z^3 / 10$
⋮	⋮	⋮	⋮
9	1/10	z^9	$z^9 / 10$
	10/10=1		

06:50

Therefore,
$$G(z) = \frac{1}{10} [1 + z + z^2 + z^3 + \dots + z^9]$$

This is a finite geometric series in which the first term a is 1 and the common ratio $r = z$ and $n = 10$
So,
$$Sum = \frac{a(1 - r^n)}{1 - r} = \frac{1(1 - z^{10})}{1 - z}$$

$$G(z) = \frac{1}{10} \left(\frac{1 - z^{10}}{1 - z} \right)$$

00:09

Topic No. 12

A Linear Combination
of
PDFs is a PDF

In mathematics , a linear combination is an expression constructed from a say of terms by multiplying each term by a constant and adding those results
(e.g. a linear combination of x and y would be any expression of the form ax+by, where a and b are constants).

01:25

Let,
1: Consider k continuous-type distributions with the following characteristics:
pdf $f_i(x)$,
mean μ_i , $i=1,2,\dots,k$.
If $c_i \geq 0, i=1,2,\dots,k$,
and
 $c_1 + c_2 + \dots + c_k = 0$
 $w(x) = c_1 f_1(x) + c_2 f_2(x) + \dots + c_k f_k(x) \geq 0$

02:35

To prove
Let,
$$\int_{-\infty}^{\infty} w(x) dx = 1$$

Proof :
$$\begin{aligned} L.H.S &= \int_{-\infty}^{\infty} [c_1 f_1(x) + c_2 f_2(x) + \dots + c_k f_k(x)] dx \\ &= \int_{-\infty}^{\infty} c_1 f_1(x) dx + \int_{-\infty}^{\infty} c_2 f_2(x) dx + \dots + \int_{-\infty}^{\infty} c_k f_k(x) dx \\ \int_{-\infty}^{\infty} w(x) dx &= c_1 \int_{-\infty}^{\infty} f_1(x) dx + c_2 \int_{-\infty}^{\infty} f_2(x) dx + \dots + c_k \int_{-\infty}^{\infty} f_k(x) dx \quad (1) \end{aligned}$$

05:10

But $f_i(x)$ is a pdf
$$\int_{-\infty}^{\infty} f_i(x) dx = 1$$

eq.(1) re-written
$$\begin{aligned} L.H.S &= c_1 (1) + c_2 (1) + \dots + c_k (1) \\ &= c_1 + c_2 + \dots + c_k \end{aligned}$$

But by our choice
$$c_1 + c_2 + \dots + c_k = 1$$

$$\therefore L.H.S = 1 = R.H.S$$

06:30

Show that the mean of the distribution having
pdf $c_1 f_1(x) + c_2 f_2(x) + \dots + c_k f_k(x)$ are mean $\mu = \sum_{i=1}^k c_i \mu_i$
$$\begin{aligned} E(\text{new density}) &= \int_{-\infty}^{\infty} w(x) dx = 1 \\ E(X) &= \int_{-\infty}^{\infty} x [c_1 f_1(x) + c_2 f_2(x) + \dots + c_k f_k(x)] dx \\ &= \int_{-\infty}^{\infty} [c_1 x f_1(x) + c_2 x f_2(x) + \dots + c_k x f_k(x)] dx \\ &= c_1 \int_{-\infty}^{\infty} x f_1(x) dx + c_2 \int_{-\infty}^{\infty} x f_2(x) dx + \dots + c_k \int_{-\infty}^{\infty} x f_k(x) dx \\ E(X) &= c_1 \mu_1 + c_2 \mu_2 + \dots + c_k \mu_k \end{aligned}$$

Hence, we get
$$\mu = \sum_{i=1}^k c_i \mu_i$$

In other words, the mean of a linear combination of PDFs is that same Linear combination of the means of those PDFs

00:08

Topic No. 13

Concept of
Cumulative Distribution Function
(discrete and continuous)

00:26

Definition: (Cumulative Distribution Function)

Let X be a random variable. Then its cumulative distribution function (cdf) is defined by $F_X(x)$, given in eq. (1) which you can see on the screen

$$F_X(x) = P_X((-\infty, x]) = P(\{c \in C : X(c) \leq x\}). \quad (1)$$

00:57

As above, we shorten

$$P\left(\left\{c \in C : X(c) \leq x\right\}\right) \text{ to } P(X \leq x).$$

Also,

$F_X(x)$ is often called simply the **distribution function (df)**.

10:08

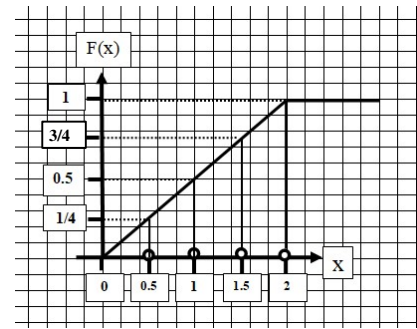


Figure 2: Distribution Function

01:36

- The graph for the discrete case it is like a staircase, which is also called a **Step function**.
- On the other hand, in the continuous case it is a continuous **curve or a line**.

00:08

Topic No. 14

Example of the CDF of a Discrete Random Variable

01:58

For example:

Let $p_X(x)$ be the pmf of a random variable X . Find the cdf $F(x)$ of X and sketch its graph with that of $p_X(x)$ if:

a) $p_X(x) = 1/3, x = -1, 0, 1$, zero elsewhere.

x	p(x)	F(x)
-1	$\frac{1}{3}$	$\frac{1}{3}$
0	$\frac{1}{3}$	$\frac{1}{3} + \frac{1}{3} = \frac{2}{3}$
1	$\frac{1}{3}$	$\frac{2}{3} + \frac{1}{3} = \frac{3}{3} = 1$

04:40

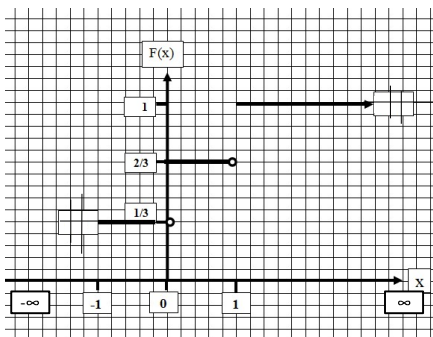


Figure 1: Distribution Function

09:06

Example 2:

Let X denotes a real number chosen at random between 0 and 2. we now obtain the cdf of X . First, if $x < 0$, then $P(X \leq x) = 0$.

Next, $x \geq 2$, then $P(X \leq x) = 1$. Finally, if $0 < x < 2$ it follows from equation 2 that $P(X \leq x) = P(0 < X \leq x) = x - 0 = x$. Hence the cdf of X is

$$F_X(x) = \begin{cases} 0 & \text{if } x < 0 \\ x/2 & \text{if } 0 \leq x < 2 \\ 1 & \text{if } x \geq 2 \end{cases} \quad (3)$$

$$P_X[(a, b)] = b - a, \quad \text{for } 0 < a < b < 1. \quad (2)$$

00:26

Example:

Consider an urn which contains slips of paper each with one of the numbers $1, 2, \dots, 100$ on it. suppose there are i slips with the number i on it for $i = 1, 2, \dots, 100$. or example, there are 25 slips of paper with the number 25.

Assume that the slips are identical except for the numbers. Suppose one slip is drawn at random.

Let X be the number on the slip

00:26

Show that the cdf of X is $F(x) = [x]([x]+1)/10100$, for $1 \leq x \leq 100$, where $[x]$ is the greatest integer in x .

Let X be the number on the slip

(a)

02:10

x	$p(x)$
1	1/5050
2	2/5050
3	3/5050
-	-
-	-
-	-
100	100/5050
Sum	5050/5050=1

04:23

x	p(x)	F(x)
1	$\frac{1}{5050}$	$\frac{1}{5050}$
2	$\frac{2}{5050}$	$\frac{(1+2)}{5050}$
3	$\frac{3}{5050}$	$\frac{(1+2+3)}{5050}$
4	$\frac{4}{5050}$	$\frac{(1+2+3+4)}{5050}$
.	.	.
.	.	.
.	.	.
100	$\frac{100}{5050}$	$\frac{(1+2+\dots+100)}{5050}$

Show that the cdf of X is $F(x) = [x]([x]+1)/10100$, for $1 \leq x \leq 100$, where $[x]$ is the greatest integer in x.

Therefore,

$$F(x) = \frac{[x]([x]+1)}{2 \times 5050}$$
$$F(x) = \frac{[x]([x]+1)}{10100}$$

00:24

PDF of the Exponential Distribution

with mean=0.5, 1, 1.5

- $f(x) = \left(\frac{1}{0.5}\right)e^{-\left(\frac{x}{0.5}\right)}, \quad 0 < x < \infty$ → orange
- $f(x) = e^{-(x)}, \quad 0 < x < \infty$ → purple
- $f(x) = \left(\frac{1}{1.5}\right)e^{-\left(\frac{x}{1.5}\right)}, \quad 0 < x < \infty$ → skyblue

00:07

Topic No. 15

Obtaining the PMF from the CDF

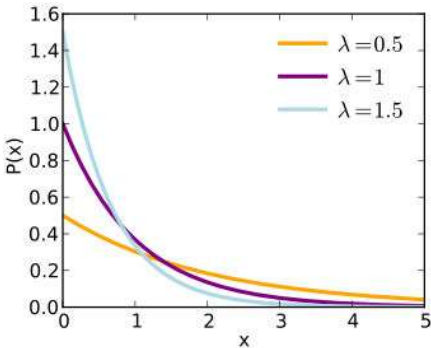
00:36

For each of the following CDFs $F(x)$, find the PDF of $f(x)$:

(a) $F(x) = \sum_{j=1}^x \left(\frac{1}{2}\right)^j$

x	$F(x) = \sum_{j=1}^x \left(\frac{1}{2}\right)^j$	P(x)
1	$\left(\frac{1}{2}\right)^1$	$\left(\frac{1}{2}\right)^1$
2	$\left(\frac{1}{2}\right)^1 + \left(\frac{1}{2}\right)^2$	$\left(\frac{1}{2}\right)^2$
3	$\left(\frac{1}{2}\right)^1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3$	$\left(\frac{1}{2}\right)^3$
4	$\left(\frac{1}{2}\right)^1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4$	$\left(\frac{1}{2}\right)^4$
.	.	.
.	.	.
.	.	.
∞	$\left(\frac{1}{2}\right)^1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4 + \dots$	1

00:00

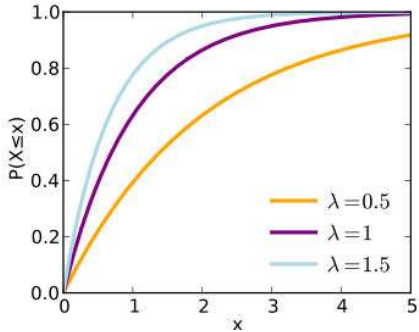


00:00

CDF of the Exponential Distribution with mean=0.5, 1, 1.5

$$F(x) = 1 - e^{-x}, \quad 0 < x < \infty$$

- $F(x) = 1 - e^{-\frac{x}{0.5}}, \quad 0 < x < \infty$
- $F(x) = 1 - e^{-x}, \quad 0 < x < \infty$
- $F(x) = 1 - e^{-\frac{x}{1.5}}, \quad 0 < x < \infty$



00:06

Topic No. 16

Example of the CDF of a Continuous Random Variable

00:25

For a continuous random variable X , the CDF is as follows:

- $F(x) = 0$ for all x values which are less than zero
- $F(x) = x$ for all those x values that lie between 0 and 1.
- $F(x) = 1$ for all those x values that are greater than 1.

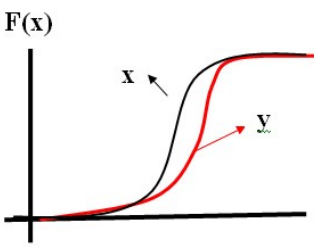
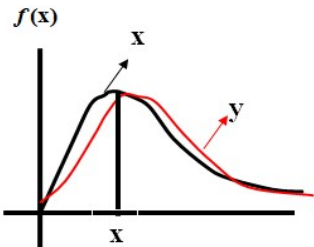
00:08

Topic No. 17

Concept of two Random Variables being Equal in Distribution

00:38

Let us first consider that situation when two random variables X and Y are not equal in distribution



$$F_X(x) \neq F_Y(x)$$

02:21

We say that X and Y are equal in distribution $X \stackrel{D}{=} Y$ if and only if

$$F_X(x) = F_Y(x), \text{ for all } x \in R.$$

Mathematically we express it as

$$X \stackrel{D}{=} Y$$

It is important to note while X and Y may be equal in distribution, they may be quite different.

04:07

Example

Recall that X denotes a real number chosen at random between 0 and 1. we now obtain the cdf of X . First, if $x < 0$, the $P(X \leq x) = 0$.

Next, if $x \geq 1$, then $P(X \leq x) = 1$. Finally, $0 < x < 1$, it follows from expression (5.3) that $P(X \leq x) = P(0 < X \leq x) = x - 0 = x$.

Hence the cdf of X is

$$F_X(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x \geq 1 \end{cases} \quad (5.6)$$

04:50

For instance, in the above example define the random variable Y and transform it as $Y = 1 - X$.

Then $Y \neq X$.

For example, let, $X = 0.2$ then, $Y = 1 - 0.2 = 0.8$.

So, when X is 0.2 then Y is 0.8 which is not the same.

Therefore it is interesting to note that the space of Y is the interval $(0,1)$, the same as X .

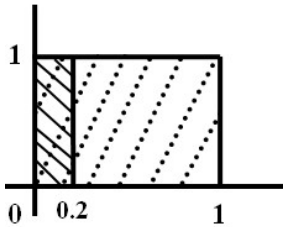
06:28

X	Y=1-X
0	1
0.2	0.8
0.3	0.7
0.5	0.5
0.6	0.4
0.8	0.2
1	0
Y≠X	

$$F(0.8) = 1 - P(X < 1 - 0.8) = 1 - P(X < 1 - 0.2) = 1 - 0.2 = 0.8$$
$$F(0.6) = 1 - P(X < 1 - 0.6) = 1 - P(X < 1 - 0.4) = 1 - 0.4 = 0.6$$

07:41

But we know that for the Uniform distribution defined on $(0,1)$, $P(X < 0.2) = 0.2$



Therefore, $P(X < 1 - y) = 1 - y$

08:11

Further, the cdf of Y is 0 for $y < 0$; 1 for $y \geq 1$; and for $0 \leq y < 1$, it is

$$F_Y(y) = P(Y \leq y) = P(1 - X \leq y) \\ = P(X \geq 1 - y) = 1 - P(X < 1 - y) \\ = 1 - (1 - y) = y.$$

Hence, $F_Y(y) = y$, $0 \leq y \leq 1$ which is of exactly the same form as the CDF of X .

Therefore, it is clarified that two random variables may not be the same yet, sometimes they may be equal in distribution.

$$\text{i.e., } Y \stackrel{D}{=} X, \text{ but } Y \neq X.$$

00:31

Topic No. 18

First property of Cumulative Distribution Function and its Proof

00:39

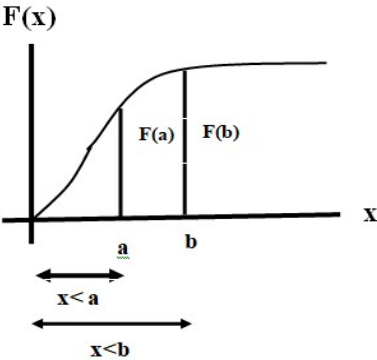
Theorem 1:

Let X be a random variable with cumulative distribution function $F(x)$.

Then

(a) For all a and b , if $a < b$, the $F(a) \leq F(b)$ (F is nondecreasing).

01:40



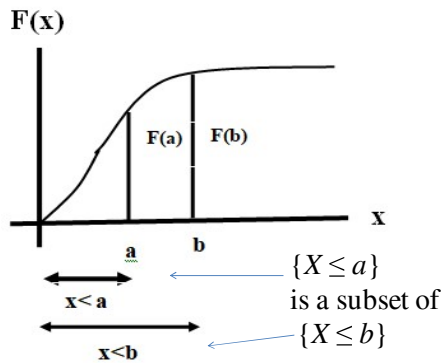
03:27

Proof:

Part (a):

Because of the fact that $a < b$, we can see that the interval being $\{X \leq a\} \subset \{X \leq b\}$

03:51



01:57

Mathematically,

$$\lim_{x \rightarrow -\infty} F(x) = 0.$$

05:00

Theorem:

If C_1 and C_2 are events such that $C_1 \subset C_2$, then, $P(C_1) \leq P(C_2)$.

So, applying this theorem we say that since the interval $X \leq a$ is contained in the interval $X \leq b$, therefore the probability of $X \leq a$ is less than or equal to the probability $X \leq b$, i.e. $F(a) \leq F(b)$.

00:05

Topic No. 20

Third property of Cumulative Distribution Function and its Proof

00:05

Topic No. 19

Second property of Cumulative Distribution Function and its Proof

00:16

Theorem 1:

Let X be a random variable with cumulative distribution function $F(x)$.

Then,

(c) $\lim_{x \rightarrow \infty} F(x) = 1$ (the upper limit of F is 1).

00:16

Theorem 1:

Let X be a random variable with cumulative distribution function $F(x)$.

Then,

(b) $\lim_{x \rightarrow -\infty} F(x) = 0$ (the lower limit of F is 0).

00:40

Proof:

We know that

$$F(x) = P(X \leq x)$$

so obviously

$$F(\infty) = P(X \leq \infty) \text{ is a sure event.}$$

Therefore, $F(\infty) = 1$

$$\lim_{x \rightarrow \infty} F(x) = 1.$$

00:54

We know that

$$F(x) = P(X \leq x)$$

so obviously

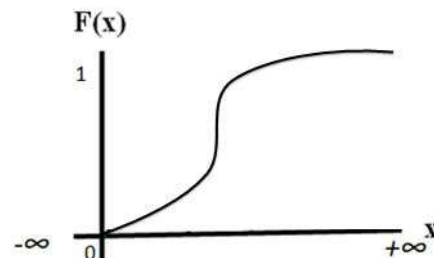
$$F(-\infty) = P(X \leq -\infty)$$

which is an impossible event.Therefore, $F(-\infty) = 0$

02:04

Graphically, the $F(x)$ of any random variable will always start from level zero and rise upto level 1.

Subsequent to that point 1 it goes upto to infinity.



00:08

Topic No. 21

Fourth property of Cumulative Distribution Function and its Proof

00:24

Theorem 1:

Let X be a random variable with cumulative distribution function $F(x)$.

Then

(d) $\lim_{x \downarrow x_0} F(x) = F(x_0)$ (F is right continuous).

01:08

Explanation:

In case of a discrete distribution, we take an example of a Discrete Uniform distribution.

- Toss a fair die (1,2,3,4,5,6).
- The probability of getting a 1 or 2 or 3 or 4 or 5 or 6 are equally likely to occur.

02:52

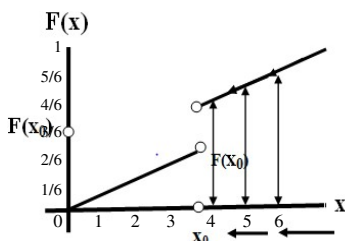
x	$P(x)$	$F(x)$
1	1/6	1/6
2	1/6	1/6+1/6=2/6
3	1/6	2/6+1/6=3/6
4	1/6	3/6+1/6=4/6
5	1/6	4/6+1/6=5/6
6	1/6	5/6+1/6=1

06:24

Consider a CDF that is continuous everywhere other than at the point x_0

(ref. CDF's that have a few points of discontinuity)

As x tends to x_0 from the R.H.S, the ordinate $F(x)$ tends to the ordinate $F(x_0)$.



00:00

t (d):

Let $\{x_n\}$ be any sequence of real numbers such that $x_n \downarrow x_0$. Let $C_n = \{X \leq x_n\}$. Then the sequence of sets $\{C_n\}$ is decreasing and $\bigcap_{n=1}^{\infty} C_n = \{X \leq x_0\}$

Hence, by Theorem 3.6,

To be displayed on the screen.

$$\lim_{n \rightarrow \infty} F(x_n) = P\left(\bigcap_{n=1}^{\infty} C_n\right) = F(x_0),$$

which is the desired result.

00:20

Topic No. 22

Evaluating Probabilities using CDF

00:20

Theorem 1:

Let X be a random variable with the CDF F_x .

Then,
for $a < b$,

$$P[a < X \leq b] = F_x(b) - F_x(a).$$

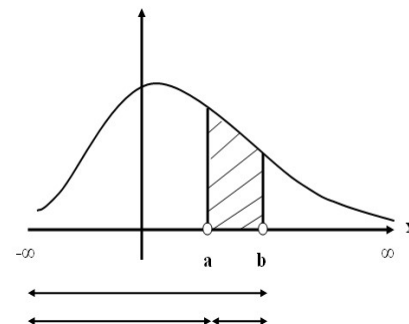
01:00

Proof:

Since the interval

$$\{-\infty < X \leq b\} = \{-\infty < X \leq a\} \cup \{a < X \leq b\}.$$

Since these two interval are mutually exclusive (not overlapping).



01:49

So, therefore,

we can apply the addition theorem of probability and can say that the probability of the union is equal to the sum of the probabilities.

$$P\{-\infty < X \leq b\} = P\{-\infty < X \leq a\} + P\{a < X \leq b\}.$$

$$\text{or } P\{a < X \leq b\} = P\{-\infty < X \leq b\} - P\{-\infty < X \leq a\}$$

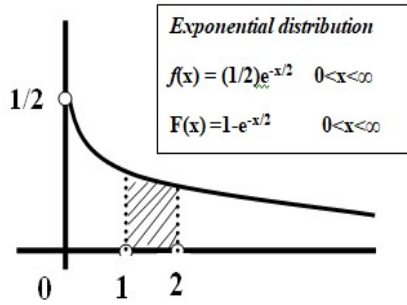
03:35

Example:

Now, find $P(1 < X < 2)$, $\theta=2$

Then required probability

$$\begin{aligned} &= F(2) - F(1) = 1 - 1 - e^{-1} + e^{-1/2} \\ &= e^{-1/2} - e^{-1} = (1/\sqrt{e}) - (1/e) \\ &= (1/1.64872) - (1/2.71828) \\ &= 0.6065 - 0.3679 = 0.2386 \\ &= 23.86\% \end{aligned}$$



00:08

Topic No. 24

In the continuous case,
**probability does not Exist
at any particular point**

00:35

Topic No. 23

Derivative Of the CDF is the PDF

01:00

The function $f_X(x)$ is called a probability density function (pdf) of X . If $f_X(x)$ is also continuous, the Fundamental Theorem of Calculus implies that

$$\frac{d}{dx} F_X(x) = f_X(x) \quad (1)$$

01:44

Example:

Consider the CDF of the exponential distribution with mean = 1

$$\text{i.e. } F(x) = 1 - e^{-x}$$

Then, the PDF is given by

$$\frac{d}{dx} F_X(x) = \frac{d}{dx} (1 - e^{-x}) = 0 - (-1)e^{-x} = e^{-x}, \quad x \geq 0$$

04:26

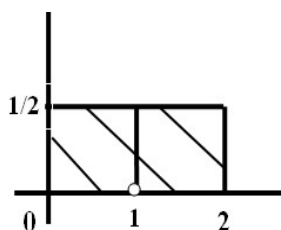
2: Another Example

$$\text{If } F_X(x) = x/2, \quad 0 < x < 2$$

Then pdf is

$$\frac{d}{dx} F_X(x) = \frac{d}{dx} \left(\frac{x}{2} \right) = \frac{1}{2} \frac{d}{dx} (x) = \frac{1}{2}, \quad 0 < x < 2$$

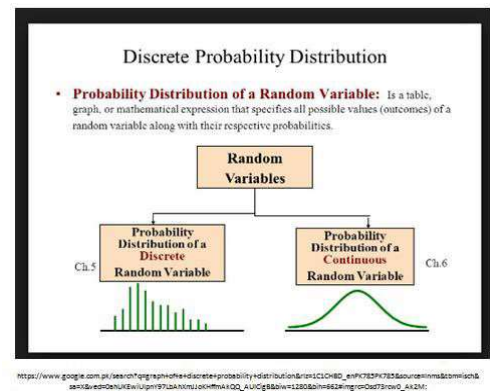
i.e. Uniform distribution.



00:33

In the discrete case probability is given by the
length of the ordinate erected on top of any
particular point.

00:59



01:45

- On the other hand, in the continuous case, probability is not given by the ordinate erected on top of any particular point (on the contrary, it is given by the area between any two ordinates).
- If $f(x)$ is the pdf of x , then the probability x belongs to A , where A is some interval, is given by the integral of $f(x)$ over that interval, that is:

$$P(X \in A) = \int_A f(x) dx$$

02:55

- If $f(x)$ is the pdf of x , then the probability x belongs to A , where A is some interval, is given by the integral of $f(x)$ over that interval, that is:

$$P(X \in A) = \int_A f(x) dx$$

03:28

WHY do we say that probability does not exist at any particular point ?

i.e.

$$P(X = x_0) = 0$$

We say this because it is not possible to have a measurement exactly equal to X_0 whatever we are saying is an approximation to the truth.

06:00

What is $P(X=1/2)$?

Solution:

It is a straightforward integration to see that the probability is 0:

In fact, in general, if X is continuous, the probability that X takes on any specific value x is 0.

That is, when X is continuous, $P(X = x) = 0$ for all x in the support.

$$\int_{1/2}^{1/2} 3x^2 dx = \left[x^3 \right]_{x=1/2}^{x=1/2} = \frac{1}{8} - \frac{1}{8} = 0$$

07:46

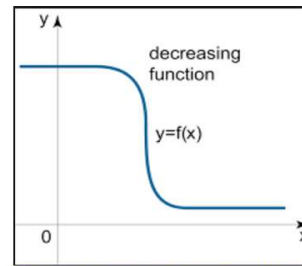
An implication of the fact that $P(X=x) = 0$ for all x when X is continuous is that you can be careless about the endpoints of intervals when finding probabilities of continuous random variables.

That is:

$$P(a \leq X \leq b) = P(a < X \leq b) = P(a \leq X < b) = P(a < X < b)$$

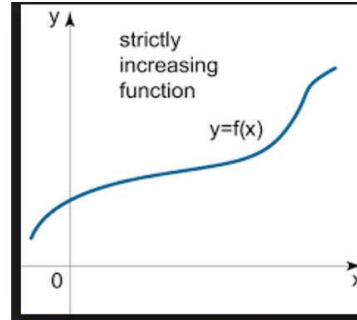
for any constant a and b .

Monotonically Decreasing Function



https://www.google.com.pk/search?q=monotonically+decreasing+function&rlz=1C1CHBD_enPK785PK785&source=lnms&btm=sch&sa=X&ved=0ahUKEw6idSa_LLbAhUkKsAKHYBwANQQAUICjgB&biw=1380&bih=662&imgref=5420ACuHg65M

Strictly Increasing Function

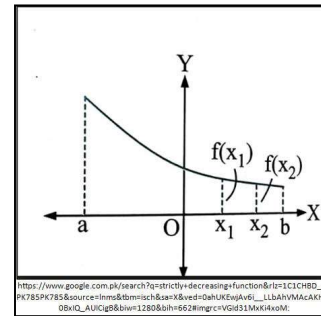


https://www.google.com.pk/search?q=strictly+increasing+function&rlz=1C1CHBD_enPK785PK785&source=lnms&btm=sch&sa=X&ved=0ahUKEw6idSa_LLbAhUkKsAKHYBwANQQAUICjgB&biw=1380&bih=662&imgref=5420ACuHg65M

A function $f(x)$ is said to be strictly increasing on an interval I if $f(b) > f(a)$ for all $b > a$, where $a, b \in I$.

On the other hand, if $f(b) \geq f(a)$ for all $b > a$, the function is said to be (nonstrictly) increasing.

Strictly decreasing Function



https://www.google.com.pk/search?q=strictly+decreasing+function&rlz=1C1CHBD_enPK785PK785&source=lnms&btm=sch&sa=X&ved=0ahUKEw6idSa_LLbAhUkKsAKHYBwANQQAUICjgB&biw=1380&bih=662&imgref=5420ACuHg65M

Topic No. 25

The concept of Monotonicity

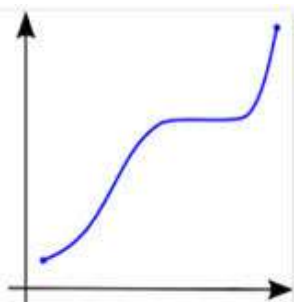
Focus on the word

“Mono-tone”

Monotonically Increasing Function

In calculus, a function defined on a subset of the real numbers with real values is called **monotonic** if and only if it is either entirely non-increasing, or entirely non-decreasing.

That is, as per Fig., a **function** that increases **monotonically** does not exclusively have to **increase**, it simply must not decrease.



The concept of the CDF, will always be monotonically increasing function.

Topic No. 26

Total probability is 1

Total probability is 1

The total probability associated with a random variable X of the discrete type with pmf $p_X(x)$

$$\sum_{x \in D} p_X(x) = 1$$

or

the continuous type with pdf $f_X(x)$ is 1, i.e.

$$\int_D f_X(x) dx = 1,$$

where D is the space of X .

The definition for the pdf of a continuous random variable differs from the definition for the pmf of a discrete random variable by simply changing the summations that appeared in the discrete case to integrals in the continuous case.

It is very well-known that pmfs satisfy the two properties

$$(i) p_X(x) \geq 0 \quad \text{and} \quad (ii) \sum_{x \in D} p_X(x) = 1$$

Therefore, the minimum possible value of any probability is 0 and the maximum possible value of any probability is equal to 1.

It is very well-known that pdfs satisfy the two properties

$$(i) f_X(x) \geq 0 \quad \text{and} \quad (ii) \int_{-\infty}^{\infty} f_X(t) dt = 1$$

(ii) The second property, of course, follows from $F_X(\infty) = 1$

- In the continuous case the $f(x)$ represents the ordinate against the value x , and the ordinate can be greater than 1 (although area under the curve can never be greater than 1).

Example:

Let X be a continuous random variable whose probability density function is:

$$f(x) = 3x^2$$

for $0 < x < 1$. First, note that $f(x) \neq P(X=x)$.

For example, $f(0.9) = 3(0.9)^2 = 2.43$, which is clearly not a probability!

In the continuous case, $f(x)$ is instead the height of the curve at $X=x$, so that the total area under the curve is 1.

In the continuous case, it is areas under the curve that define the probabilities.

Now, let's first start by verifying that $f(x)$ is a valid probability density function.

- The function $f(x) = 3x^2$ is not a Probability density function that is why we are getting $f(x)$ greater than 1. So, let me show you that it is a PDF.

$$\int_0^1 3x^2 dx = 3 \left. \frac{x^3}{3} \right|_0^1 = x^3 \Big|_0^1 = 1^3 - 0^3 = 1$$

Topic No. 27

Example

showing the determination of an unknown constant c

Such that $f(x)$ or $p(x)$ is the PMF

Example:

Find the constant c so that $p(x)$ satisfies the condition of being a pmf of one random variable X .

$$p(x) = \begin{cases} c \left(\frac{2}{3} \right)^x, & x = 1, 2, 3, \dots \\ 0 & \text{elsewhere} \end{cases}$$

- If $p(x)$ to be a probability mass function, then:

$$\begin{aligned} 1 &= \sum_{x=1}^{\infty} p(x) = \sum_{x=1}^{\infty} c \left(\frac{2}{3} \right)^x = c \left(\frac{2}{3} \right)^1 + c \left(\frac{2}{3} \right)^2 + c \left(\frac{2}{3} \right)^3 + \dots \\ &= c \left(\left(\frac{2}{3} \right)^1 + \left(\frac{2}{3} \right)^2 + \left(\frac{2}{3} \right)^3 + \dots \right) \end{aligned}$$

What we have inside the bracket is an Infinite Geometric series,

Therefore, applying the formula $S = \frac{a}{1-r}$, we obtain

$$1 = c \left(\frac{\left(\frac{2}{3} \right)}{1 - \left(\frac{2}{3} \right)} \right) = c \left(\frac{\left(\frac{2}{3} \right)}{\frac{1}{3}} \right) = c(2) \quad \text{and hence} \quad c = \frac{1}{2}.$$

As an illustration, suppose we want to find a probability of $x=4$.

Then putting $c = \frac{1}{2}$, we have $p(x) = \frac{1}{2} \left(\frac{2}{3} \right)^x$.

now putting $x=4$,

we obtain,

$$p(4) = \frac{1}{2} \left(\frac{2}{3} \right)^4 = \frac{1}{2} \left(\frac{16}{81} \right) = \frac{8}{81} = 0.099$$

Topic No. 28

Example

showing the determination of an unknown constant c
Such that $f(x)$ is a the PDF

Example

Suppose X has the pdf

$$f_X(x) = \begin{cases} cx^3 & 0 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

for a constant c

Solution:

If $f(x)$ is to be a pdf, then,

$$1 = \int_0^2 cx^3 dx = c \left[\frac{x^4}{4} \right]_0^2 = c \left[\frac{2^4}{4} - \frac{0}{4} \right] = c \left[\frac{16}{4} \right] = c[4] = 4c,$$

and hence, $c=1/4$.

So, now the pdf will be as follows

$$f_X(x) = \begin{cases} \frac{x^3}{4} & 0 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

For illustration of the computation of a probability involving X , we have

$$P\left(\frac{1}{4} < X < 1\right) = \int_{1/4}^1 \frac{x^3}{4} dx = \frac{255}{4096} = 0.0623.$$

Topic No. 29

Concept of Transformation of a Discrete variable

(when the transformation is one-to-one)

We have a discrete random variable X and we know its distribution.

We are interested, though, in a random variable Y which is some transformation of X , say, $Y = g(X)$.

In particular, we want to determine the distribution of Y .

Assume X is discrete with space D_x . We consider two cases.

Then the space of Y is

$$D_y = \{g(x) : x \in D_x\}.$$

Let us assume that g is one-to-one.

Then, clearly, the pmf of Y is obtained as

$$P[Y = y] = P[g(X) = y] = P[X = g^{-1}(y)].$$

Example

Let X have the pmf

$$P(x) = (1/2)^{|x|}, \quad x = -1, -2, -3, \dots$$

Find the pmf of $Y = X^4$

Solution:

The transformation,

$$y = g(x) = x^4$$

$$\text{maps } D_x = \{x : x = -1, -2, -3, \dots\} \text{ onto}$$

$$D_y = \{y : y = 1, 16, 81, \dots\}$$

We have

$$y = g(x) = x^4 \Rightarrow x = g^{-1}(y) = \sqrt[4]{y}$$

So, utilizing the equation

$$P[Y = y] = P[g(X) = y] = P[X = g^{-1}(y)].$$

we get the single-valued inverse function as follows:

$$\therefore P[Y = y] = P(X = \sqrt[4]{y}) = \left(\frac{1}{2}\right)^{|\sqrt[4]{y}|}, \quad y = 1, 16, 81, \dots$$

So, this is the PMF of $Y = X^4$.

Topic No. 30

Concept of Transformation of a Discrete Variable (when the transformation is not one-to-one)

Consider a sequence of independent flips of a **fair** coin, each resulting in a head H or a tail T .

Let the random variable X denote the no. of flips needed to obtain the first Head.

Then, X is a **geometric** random variable.

If $X=x$, where $x=1,2,3,4,\dots$, there must be a string of $x-1$ tails followed by a head:

that is, $TT\dots TH$, where there are $x-1$ tails in $TT\dots T$.

Thus, from independence, we have a geometric sequence of probabilities, namely,

$$P(X=x) = (1/2)^{x-1}(1/2) = (1/2)^x, \quad x = 1, 2, 3, \dots, \quad (1)$$

Suppose we are playing a game in which if the first head appears on an odd number of flips, we lose one dollar whereas, if the first head appears on an even number of flips, we win one dollar.

Let Y denote our net gain.

Then the space of Y is $\{-1, 1\}$.

An interesting event is that the first head appears on an odd number of flips;

i.e., $X \in \{1, 3, 5, \dots\}$.

The probability of this event is

$$P[X \in \{1 \text{ or } 3 \text{ or } 5 \text{ or } \dots\}] = \sum_{x=1}^{\infty} \left(\frac{1}{2}\right)^{2x-1}$$

Next,

$$= (1/2)^1 + (1/2)^3 + (1/2)^5 + \dots$$

And infinite geometric series, $\frac{a}{1-r} = a = (1/2), r = (1/2)^2 = 1/4$

$$\text{Required probability} = \sum_{x=1}^{\infty} \left(\frac{1}{2}\right)^{2x-1} = \frac{\frac{1}{2}}{1 - \left(\frac{1}{4}\right)} = \frac{\frac{1}{2}}{\left(\frac{3}{4}\right)} = \frac{2}{3}.$$

We have already determined that the probability that X is odd is $2/3$.

Hence, the distribution of Y is given by

$$p_Y(-1) = 2/3 \text{ and } p_Y(1) = 1/3.$$

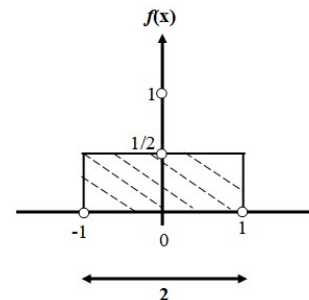
So, the probability that player A will win 1\$ is $1/3$ and the probability that he will lose 1 \$ is $2/3$.

Example:

Let $f_X(x) = 1/2, -1 < x < 1$,

zero elsewhere,

be the pdf of a random variable X .



From the graph it is obvious that we are dealing with continuous uniform distribution defined on the interval $(-1, 1)$.

Now, let us suppose that we are interested in determining the PDF of X^2 .

Let the random variable Y be $Y = X^2$. Now to find the pdf of Y .

The point to be understood is that, if $y \geq 0$, then

$$X^2 \leq y \Rightarrow -\sqrt{y} \leq X \leq \sqrt{y}$$

e.g. $0 \leq X^2 \leq 0.5 \Rightarrow -0.25 \leq X \leq 0.25$
(because square can never be negative)

From the table it is obvious that the range of the variable Y is from 0 to 1.

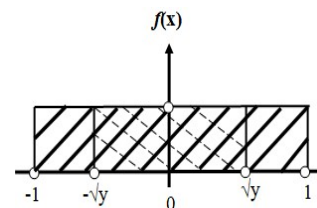
$X^2=Y$	X
0	0
0.04	± 0.2
0.25	± 0.5
0.49	± 0.7
1.00	± 1.0

Now finding the pdf of Y .

The cdf of Y , $F_Y(y) = P(Y \leq y)$, is given by the cdf of Y ,

$F_Y(y) = P(Y \leq y)$, is given by

$$\begin{aligned} P(X^2 < y) &= P(-\sqrt{y} < X < \sqrt{y}) \\ &= \int_{-\sqrt{y}}^{\sqrt{y}} f(x) dx = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} dx \end{aligned}$$



For the range $0 \leq y \leq 1$, we have

$$F(y) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} dx = \frac{1}{2} x \Big|_{-\sqrt{y}}^{\sqrt{y}} = \frac{1}{2} [\sqrt{y} - (-\sqrt{y})] = \frac{2\sqrt{y}}{2} = \sqrt{y}$$

$$F_Y(y) = \begin{cases} 0 & y < 0 \\ \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} dx = \sqrt{y} & 0 \leq y < 1 \\ 1 & 1 \leq y \end{cases}$$

Topic No. 31

Concept of Transformation of a Continuous Variable using the CDF technique)

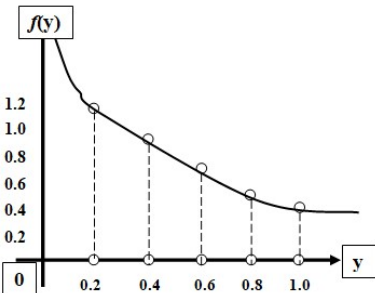
Hence, the pdf of Y is given by

$$f_Y(y)=\begin{cases} \frac{1}{2\sqrt{y}} & 0 < y < 1 \\ 0 & elsewhere. \end{cases}$$

$$f(y)=\frac{d}{dy}F(y)=\begin{cases} \frac{d}{dy}(0)=0, & y < 0 \\ \frac{d}{dy}\sqrt{y}=\frac{d}{dy}\sqrt{y}=\frac{1}{2}y^{\frac{1}{2}-1}=\frac{1}{2}y^{-\frac{1}{2}}=\frac{1}{2\sqrt{y}}, & 0 \leq y \leq 1 \\ \frac{d}{dy}(1)=0, & y \geq 1 \end{cases}$$

$$or\ f(y)=\begin{cases} \frac{1}{2\sqrt{y}}, & 0 \leq y \leq 1 \\ 0 & elsewhere. \end{cases}$$

y	1/2√y
0	∞
0.2	1.12
0.4	0.79
0.6	0.65
0.8	0.56
1	0.50



Hence we can see that the shape of the PDF of the transformed variable is very different from the shape of the PDF of the original variable.

Topic No. 32

Transformation of
a Continuous Variable
using the **Jacobian of
transformation**

Theorem 1:

Let X be a continuous random variable with pdf $f_X(x)$ and **support** S_X .

Let $Y=g(X)$, where $g(x)$ is **one-to-one differentiable function**, on the support of X, S_X .

Denote the inverse of g by $x=g^{-1}(y)$ and let $dx/dy=d[g^{-1}(y)]/dy$.

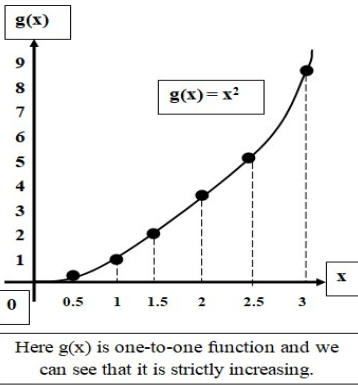
Then the PDF of Y is given by

$$f_Y(y)=f_X(g^{-1}(y)) \left| dx / dy \right|, \text{ for } y \in S_Y, \tag{1}$$

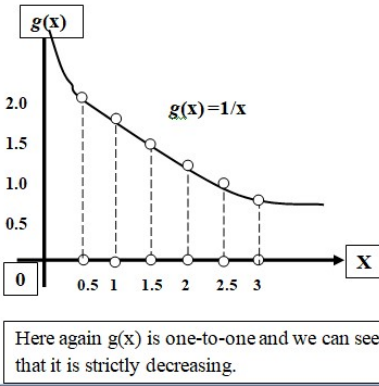
where the support of Y is the set $S_Y=\{y=g(x):x \in S_X\}$.

Proof:

If $g(x)$ is one-to-one and continuous, it is either **strictly** increasing or **strictly** decreasing.



This can be regarded as a case of **Direct relationship** between X and g(x).



This can be regarded as a case of **Inverse relationship** between X and g(x).

Case 1:

Let us assume that it is strictly **increasing**.

Then the CDF of Y is given by

$$\begin{aligned} F_Y(y) &= P[Y < y] = P[g(x) < y] \\ &= P[X < g^{-1}(y)] = F_X(g^{-1}(y)). \end{aligned} \tag{2}$$

Hence, the pdf of Y is

$$f_Y(y) = \frac{d}{dy}(F_Y(y)) = \frac{d}{dy}(F_X(g^{-1}(y))) = f_X(g^{-1}(y)) \left(\frac{dx}{dy} \right). \tag{3}$$

where dx/dy is the derivative of the function $x = g^{-1}(y)$.

In this case, **because g is increasing, $dx/dy > 0$** .

Hence, we can write $dx/dy =|dx/dy|$.

Case 2:

Suppose $g(x)$ is strictly decreasing.

Then

$$F_Y(y) = P[Y < y] = P[g(x) < y] = P[X < g^{-1}(y)] = F_X(g^{-1}(y)). \quad (2)$$

becomes $F_Y(y) = 1 - F_X(g^{-1}(y))$.

Hence, the pdf of Y is

$$f_Y(y) = 0 - f_X(g^{-1}(y)) \left(\frac{dx}{dy} \right) = f_X(g^{-1}(y)) \left(-\frac{dx}{dy} \right).$$

But since g is decreasing, $dx/dy < 0$ and,

hence, $-dx/dy = |dx/dy|$.

Thus eq. 1 is true in both cases.

- Henceforth, we refer to $dx/dy = (d/dy)g^{-1}(y)$ as Jacobian (denoted by J) of the transformation.
- In most mathematical areas, $J = dx/dy$ is referred to as the Jacobian of the inverse transformation $x = g^{-1}(y)$.

The transformation $y = g(x) = -2\log x$ is one-to-one between these sets.

The inverse of the transformation is $x = g^{-1}(y) = e^{-y/2}$

The Jacobian of the Transformation is

$$J = \frac{de^{-y/2}}{dy} = -\frac{1}{2}e^{-y/2}$$

Accordingly, the pdf of $Y = -2 \log X$ is

$$f_Y(y) = \begin{cases} f_X(e^{-y/2}) |J| = 1 \cdot \frac{1}{2} e^{-y/2} = \frac{1}{2} e^{-y/2} & 0 < y < \infty \\ 0 & \text{elsewhere} \end{cases}$$

This is the Exponential Distribution with Mean = 2

The above is an example of

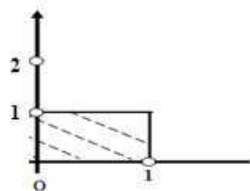
- **Transformation of Continuous Uniform Random Variable to an Exponentially distributed random variable**

Topic No. 33

Example of Transformation of a Continuous Variable (using the Jacobian of transformation)

Example : Let X have the pdf

$$f(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$



Consider the random variable $Y = -2\log X$.

The support sets of X and Y are given by $(0,1)$ and $(0,\infty)$, respectively.

Topic No. 34

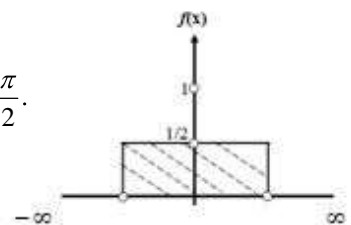
Another example of Transformation of a Continuous variable (using the Jacobian of transformation)

Example:

Let X have the uniform pdf

$$f_X(x) = \frac{1}{\pi}, \quad \text{for } -\frac{\pi}{2} < x < \frac{\pi}{2}.$$

Find the pdf of $Y = \tan X$.



Let $y=\tan x$

When, $x \rightarrow -\pi/2, y = -\infty, x \rightarrow \pi/2, y = \infty$

$y = \tan x \Rightarrow x = \tan^{-1}y \Rightarrow dx/dy = 1/(1+y^2)$

$|J| = |dx/dy| = |1/(1+y^2)| = 1/(1+y^2)$

Accordingly, the pdf of $Y=\tan X$ is

$$f_Y(y) = \begin{cases} f_X(\tan^{-1}(y))|J| = \frac{1}{\pi(1+y^2)} & -\infty < y < \infty \\ 0 & elsewhere \end{cases}$$

This is the pdf of a Cauchy distribution which is one of the well-known distributions.

In other words, the above is an example of Transformation of Continuous Uniform Random variable to a Cauchy random variable

Topic No. 35

Mode of Discrete Random Variable

Definition

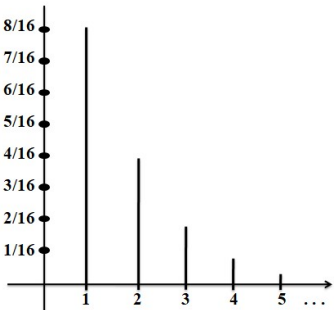
A mode of the distribution of a random variable X is a value of x that maximizes the pdf or pmf.

If there is only one such x , it is called the **mode** of the distribution.

Find the mode of the following distribution.

a) $p(x) = (1/2)^x, x = 1, 2, 3, \dots, \text{zero elsewhere.}$

X	P(x)
1	$(1/2)^1$
2	$(1/2)^2$
3	$(1/2)^3$
4	$(1/2)^4$
-	-
-	-



Therefore, simply by inspection we say that the mode = 1.

Topic No. 36

Mode of a Continuous Random Variable

Definition

A mode of the distribution of a random variable X is a value of x that maximizes the pdf.

If there is only one such x , it is called the mode of the distribution.

Procedure:

We will be following the procedure that is adopted when determining the maxima and minima of a function.

1. First derivative to be equated to zero and the equation thus obtained to be solved for x .
2. The value of the second derivative to be evaluated at that value of x which was obtained in step 1.
3. In case the second derivative turns out to be less than zero, that particular value of ' x ' will be regarded as the mode of the distribution.

Find the mode of the following distribution.

$$c) f(x) = \begin{cases} \frac{1}{2}x^2e^{-x}, & 0 < x < \infty, \\ \text{zero elsewhere.} \end{cases}$$

Solution:

Taking the first derivative w.r.t. 'x'

$$f'(x) = \frac{d}{dx} \left[\frac{1}{2}x^2e^{-x} \right]$$

$$= \frac{1}{2} \frac{d}{dx} [x^2e^{-x}]$$

$$f'(x) = \frac{1}{2} [x^2e^{-x}(-1) + e^{-x}2x]$$

$$= -\frac{1}{2}x^2e^{-x} + \frac{1}{2}2xe^{-x}$$

$$f'(x) = xe^{-x} - \frac{1}{2}x^2e^{-x}$$

Now, equating it to zero, we get

$$\Rightarrow xe^{-x} - \frac{1}{2}x^2e^{-x} = 0$$

$$\Rightarrow xe^{-x} = \frac{1}{2}x^2e^{-x} \Rightarrow 2 = \frac{1}{xe^{-x}} x^2e^{-x}$$

$$\therefore x = 2$$

$$f'(x) = xe^{-x} - \frac{1}{2}x^2e^{-x}$$

Now, taking the second derivative

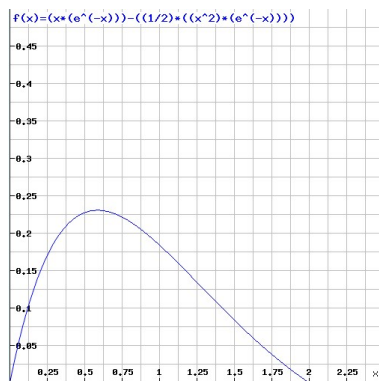
$$f''(x) = (xe^{-x}(-1) + e^{-x}(1)) - \frac{1}{2}(x^2e^{-x}(-1) + e^{-x}2x) = -xe^{-x} + e^{-x} + \frac{1}{2}x^2e^{-x} - xe^{-x}$$

$$f''(x) = e^{-x} - 2xe^{-x} + \frac{1}{2}x^2e^{-x}$$

putting the value of x in the second derivative, we get

$$f''(2) = e^{-2} - 2(2)e^{-2} + \frac{1}{2}(2)^2e^{-2} = e^{-2} - 4e^{-2} + 2e^{-2} = -e^{-2} = -0.135 \text{ which is less than zero}$$

Therefore, the **MODE** of the distribution is equal to **2**.



Topic No. 37

**Median
Of a
Discrete Random Variable**

Definition

The median of a distribution of a random variable X of the discrete or continuous type is a value of x such that

$$P(X < x) \leq \frac{1}{2} \text{ and } P(X \leq x) \geq \frac{1}{2}$$

If there is only one such x, it is called the **median** of the distribution.

Example: Find the median of the following distribution:

$$a) p(x) = \left(\frac{4!}{x!(4-x)!} \right) \left(\frac{1}{4} \right)^x \left(\frac{3}{4} \right)^{4-x}, \quad x = 0, 1, 2, 3, 4 \quad \text{zero elsewhere}$$

x	P(x)
0	$\frac{4!}{0!(4-0)!} \left(\frac{1}{4} \right)^0 \left(\frac{3}{4} \right)^{4-0} = 1 \times 1 \times \left(\frac{3}{4} \right)^4 = \frac{81}{256}$
1	$\frac{4!}{1!(4-1)!} \left(\frac{1}{4} \right)^1 \left(\frac{3}{4} \right)^{4-1} = \frac{4!}{3!} \times \frac{1}{4} \times \left(\frac{3}{4} \right)^3 = \frac{108}{256}$
2	$\frac{4!}{2!(4-2)!} \left(\frac{1}{4} \right)^2 \left(\frac{3}{4} \right)^{4-2} = \frac{4!}{2!2!} \times \frac{1}{16} \times \left(\frac{3}{4} \right)^2 = \frac{54}{256}$
3	$\frac{4!}{3!(4-3)!} \left(\frac{1}{4} \right)^3 \left(\frac{3}{4} \right)^{4-3} = \frac{4!}{3!1!} \times \frac{1}{64} \times \left(\frac{3}{4} \right)^1 = \frac{12}{256}$
4	$\frac{4!}{4!(4-4)!} \left(\frac{1}{4} \right)^4 \left(\frac{3}{4} \right)^{4-4} = \frac{4!}{4!0!} \times \frac{1}{256} \times \left(\frac{3}{4} \right)^0 = \frac{1}{256}$

The median of a distribution of a random variable X of the discrete or continuous type is a value of x such that



$$P(X < x) \leq \frac{1}{2} \text{ and } P(X \leq x) \geq \frac{1}{2}$$

Trial and Error

$$P(X < 2) = \frac{81}{256} + \frac{108}{256} = \frac{189}{256} = 0.74 > 0.5$$

$\therefore 2$ cannot be the median

Try

$$P(X < 1) = \frac{81}{256} = 0.32 < 0.5$$

So,

$$P(X \leq 1) = \frac{81}{256} + \frac{108}{256} = \frac{189}{256} = 0.74 > 0.5$$

Both requirements fulfilled $\Rightarrow \tilde{x} = 1$

Topic No. 38

**Median of a
Continuous Random Variable**

Definition

The median of a distribution of a random variable X of the discrete or continuous type is a value of x such that $P(X < x) \leq \frac{1}{2}$ and $P(X \leq x) \geq \frac{1}{2}$.

If there is only one such x , it is called the **median** of the distribution.

Procedure:

In the continuous case, the median is obtained by integrating the pdf from $-\infty$ to M and equating it to $1/2$;

by solving this equation for M , we obtain the median.

Find the median of the following distribution:

c) $f(x) = 2/\pi(1+x^2)$, $0 < x < \infty$

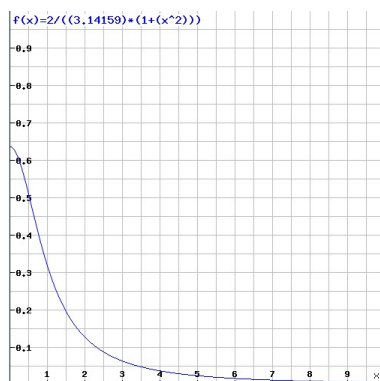
Solution:

$$f(x) = \int_0^M \frac{2}{\pi(1+x^2)} dx = \frac{1}{2} \Rightarrow \frac{2}{\pi} \int_0^M \frac{1}{(1+x^2)} dx = \frac{1}{2} \Rightarrow \frac{2}{\pi} \tan^{-1}(x) \Big|_0^M = \frac{1}{2}$$

$$\Rightarrow \frac{2}{\pi} [\tan^{-1}(M) - \tan^{-1}(0)] = \frac{1}{2}$$

$$\Rightarrow [\tan^{-1}(M) - 0] = \frac{\pi}{4} \Rightarrow \tan^{-1}(M) = \frac{\pi}{4}$$

$$\Rightarrow M = \tan\left(\frac{\pi}{4}\right) = 1 \leftarrow \text{Median}$$



Topic No. 39

Concept of (100p)th percentile (quantile of order p) of a Continuous Random Variable

Definition

Let $0 < p < 1$. A quantile of order p of the distribution of a random variable X is a value **ZAAI p** ξ_p such that $P(X < \xi_p) \leq p$ and $P(X \leq \xi_p) \geq p$.

Note:

A quantile of order p can also be regarded as the $(100 \times p)^{\text{th}}$ percentile.

Explanation:

By a quantile of order p , we mean that point on the x -axis to the left of which the area under the curve of the probability density function is equal to p or in other words is equal to $100 \times p$ %.

Now, we know that, the point to the left of which the area is, say, 20% is known as the 20th percentile, the point to the left of which the area is, say, 35% is known as the 35th percentile and so on.

Therefore, that point to the left of which the area under the curve is equal to $100 \times p$ % will be known as the $(100 \times p)^{\text{th}}$ percentile.

Explanation of the way in which these two equations are fulfilled in the case of a continuous variable.

$$P(X < \xi_p) \leq p \quad (1)$$

$$P(X \leq \xi_p) \geq p \quad (2)$$

(1) & (2) need to be fulfilled simultaneously. But the LHS's of both (1) and (2) are the same $P(X < c) = P(X \leq c)$

Hence it is obviously that both (1) & (2) can hold simultaneously of and only of the two inequalities are replaced by the equal sign

$$P(X < \xi_p) = P(X \leq \xi_p) = p$$

$$\text{or} \quad \int_{-\infty}^{\xi_p} f(x) dx = p$$

Find the 20th percentile of the distribution that has pdf $f(x)=4x^3$, $0 < x < 1$, zero elsewhere.

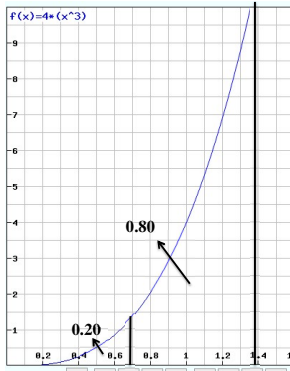
Hint: With a continuous-type random variable X , $P(X < \xi_p) = P(X \leq \xi_p)$ and hence that common value must equal p .

$\xi_{0.20}$:

$$\int_0^{\xi_{0.20}} 4x^3 dx = 0.20$$

$$\Rightarrow 4 \frac{x^4}{4} \Big|_0^{\xi_{0.20}} = 0.20 \Rightarrow \xi_{0.20}^4 - 0^4 = 0.20 \Rightarrow \xi_{0.20}^4 = 0.20 \Rightarrow \xi_{0.20} = (0.20)^{\frac{1}{4}}$$

$$\Rightarrow \xi_{0.20} = 0.6687 \approx 0.67$$



Topic No. 40

Real-Life Example

of the computation of the (100p)th percentile
(quantile of order p)
of a Continuous Random Variable

Let X be the number of gallons of ice cream that is requested at a certain store on a hot summer day.

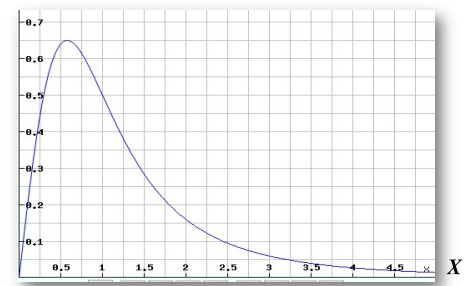
Assume that X follows the log-logistic distribution with $\alpha=1$ and $\beta=2$.

In other words, the pdf of X is given by

$$f(x) = \frac{2x}{(1+x^2)^2}, \quad 0 < x < \infty$$

How many gallons of ice cream should the store have on hand each of these days, so that the probability of exhausting its supply on a particular day is 0.10?

Solution: $f(x)$



where X represents the number of gallons requested i.e., the demand

The store will be exhausting its supply only if the demand exceeds the amount in hand.

We want the probability of this circumstance to be only 0.10.

Hence, it is obvious that we are talking about 90th percentile of the distribution of the random variable X .

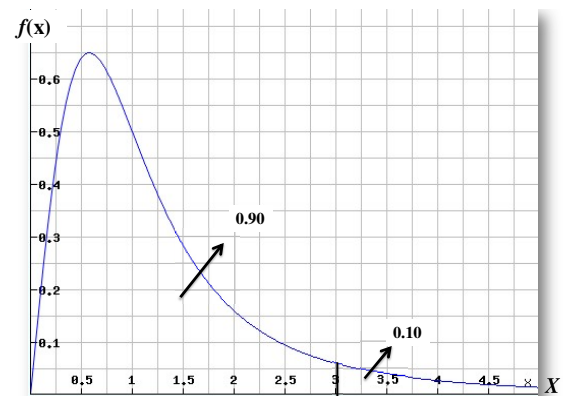
The pdf of X is given by $f(x) = \frac{2x}{(1+x^2)^2}$, $0 < x < \infty$, when $\alpha=1$ and $\beta=2$

$$f(x) = \int_0^{\xi_{0.90}} \frac{2x}{(1+x^2)^2} dx = 0.90 \Rightarrow -\frac{1}{1+x^2} \Big|_0^{\xi_{0.90}} = 0.90$$

$$\Rightarrow \left(\frac{1}{1+\xi_{0.90}^2} - \frac{1}{1+0} \right) = -0.90 \Rightarrow \frac{1}{1+\xi_{0.90}^2} - 1 = -0.90$$

$$\Rightarrow \frac{1}{1+\xi_{0.90}^2} = 1 - 0.90 \Rightarrow \frac{1}{0.10} = 1 + \xi_{0.90}^2 \Rightarrow \frac{1}{0.10} - 1 = \xi_{0.90}^2 \Rightarrow 9 = \xi_{0.90}^2 \Rightarrow \sqrt{\xi_{0.90}^2} = \sqrt{9}$$

$$\therefore \xi_{0.90} = 3$$



Topic No. 41

Inverse CDF or Quantile function

$$Q(p; \lambda) = -\frac{\ln(1-p)}{\lambda}, \text{ for } 0 < p < 1.$$

- The **quantile function** is also called the **inverse cumulative distribution function**.

The quartiles are therefore,

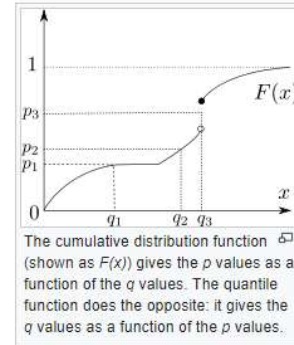
first quartile ($p = 1/4$) = $\ln(4/3) / \lambda$,

median ($p = 2/4$) = $\ln(2) / \lambda$

and **third quartile** ($p = 3/4$) = $\ln(4) / \lambda$

For a given probability distribution, the **quantile function** specifies the quantile of order p for all values of p lying between 0 and 1.

In other words, the **quantile function** specifies the **value** on the x -axis for which the probability of the random variable X being less than or equal to this value is p (for all values of p lying between 0 and 1).



Example

Suppose we have the Exponential distribution

$$f(x) = \lambda e^{-\lambda x} \quad 0 < x < \infty, \lambda > 0$$

Expected value or (mean) $1/\lambda$

Topic No. 42

Example of a Random Variable being Stochastically Larger than another random variable

The **cumulative distribution function** of Exponential Distribution

$$F(x) = \begin{cases} 1 - e^{-\lambda x} & x > 0 \\ 0 & x < 0 \end{cases}$$

The quantile function for Exponential(λ) is derived by finding the value of x for which

$$\Rightarrow p = 1 - e^{-\lambda x} \Rightarrow e^{-\lambda x} = 1 - p$$

Taking log on both sides

$$\Rightarrow \ln e^{-\lambda x} = \ln(1-p) = -\lambda x \ln e = \ln(1-p)$$

$$\Rightarrow -\lambda x = \ln(1-p) \quad \because \ln e = 1$$

$$\therefore x = -\frac{\ln(1-p)}{\lambda}$$

Definition

A random variable X is said to be **stochastically larger** than another random variable Y if

$$P(X > z) \geq P(Y > z),$$

for all real values of z , with **strict inequality** holding for at least one z value.

This requires that the CDFs enjoy the following property:

$$F_X(z) \leq F_Y(z),$$

for all real z , with strict inequality holding for at least one z value.

This requires that the CDFs enjoy the following property:

$$F_Y(z) \leq F_X(z),$$

$$\text{Now, } F_X(z) = P(X \leq z)$$

$$\text{whereas, } F_Y(z) = P(Y \leq z) = P(X + \Delta \leq z) = P(X \leq z - \Delta)$$

$$P(X \leq z - \Delta) < P(X \leq z) \quad (1)$$

eq. (1) can be re-write as

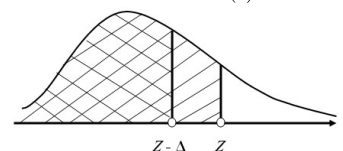
$$P(X + \Delta \leq z) < P(X \leq z)$$

or

$$P(Y \leq z) < P(X \leq z)$$

or

$$F_Y(z) \leq F_X(z).$$



Example:

Let X be a continuous random variable with support $(-\infty, \infty)$. Consider the random variable $Y = X + \Delta$, where $\Delta > 0$. Suppose that we want to show that Y is stochastically larger than X .

Solution:

First and foremost, let us re-state the above definition interchanging the roles of X and Y .

This requires that the CDFs enjoy the following property:

$$F_Y(z) \leq F_X(z),$$

Now, $F_X(z) = P(X \leq z)$

whereas, $F_Y(z) = P(Y \leq z) = P(X + \Delta \leq z) = P(X \leq z - \Delta)$

$$P(X \leq z - \Delta) < P(X \leq z) \quad (1)$$

eq. (1) can be re-write as

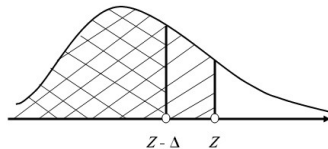
$$P(X + \Delta \leq z) < P(X \leq z)$$

or

$$P(Y \leq z) < P(X \leq z)$$

or

$$F_Y(z) \leq F_X(z).$$



Therefore the random variable $Y = X + \Delta$, where $\Delta > 0$, has showed that Y is stochastically larger than X .

Topic No. 43

Concept of Mathematical Expectation

for Discrete and Continuous Random Variables

Definition 8.1. (Expectation). Let X be a random variable. If X is a continuous random variable with pdf $f(x)$ and

$$\int_{-\infty}^{\infty} |x| f(x) dx < \infty,$$

then the expectation of X is

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx.$$

If X is a discrete random variable with pmf $p(x)$ and

$$\sum_x |x| p(x) dx < \infty$$

Then the expectation of X is

$$E(X) = \sum_x x p(x).$$

Sometimes the expectation $E(X)$ is called the mathematical expectation of X , the expected value of X , or the mean of X . When the mean designation is used, we often denote the $E(X)$ by μ ; i.e., $\mu = E(X)$.

Example

Suppose that we have a discrete random variable X given by -1, 0, and 1. and the probabilities of $P(-1) = 1/4$, $P(0) = 1/2$, $P(1) = 1/4$.

So,

$$P(-1) + P(0) + P(1) = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = \frac{1+2+1}{4} = \frac{4}{4} = 1.$$

Now, to find the Mean

So,

$$Mean = -\frac{1}{4} + 0 + \frac{1}{4} = 0.$$

x	$p(x)$	$xp(x)$
-1	1/4	-1/4
0	1/2	0
1	1/4	1/4

Now to fulfill the condition to get a finite number

So,

$$\sum_x |x| p(x) dx < \infty$$

$$= \frac{1}{4} + 0 + \frac{1}{4} = \frac{2}{4} = \frac{1}{2} = 0.5.$$

x	$p(x)$	$xp(x)$	$ x $	$ x p(x)$
-1	1/4	-1/4	1	1/4
0	1/2	0	0	0
1	1/4	1/4	1	1/4

Topic No. 44

Concept of Mathematical Expectation

of a Function of a Random Variable X
(for discrete and continuous random variables)

Theorem :

Let X be a random variable and let $Y = g(x)$ for some function g .

a) Suppose X is continuous with pdf $f_X(x)$.

If $\int_{-\infty}^{\infty} |g(x)| f_X(x) dx < \infty$, then the expectation of Y

exists and it is given by

$$E(Y) = \int_{-\infty}^{\infty} g(x) f_X(x) dx. \quad (1)$$

b) Suppose X is discrete with pmf $p_X(x)$. Suppose the support of

X is denoted by S_X . If $\sum_{x \in S_X} |g(x)| p_X(x) < \infty$, then the

expectation of Y exists and it is given by

$$E(Y) = \sum_{x \in S_X} g(x) p_X(x). \quad (2)$$

Topic No. 45

Concept of Linear combination of the Expected Values of two different functions of a Random Variable 'X' (for discrete and continuous random variables)

Theorem 1: Let $g_1(X)$ and $g_2(X)$ be functions of a random variable X . Suppose the expectations of $g_1(X)$ and $g_2(X)$ exists. Then for any constants k_1 and k_2 , the expectation of $k_1 g_1(X) + k_2 g_2(X)$ exists and it is given by

$$E[k_1 g_1(X) + k_2 g_2(X)] = k_1 E[g_1(X)] + k_2 E[g_2(X)]$$

Suppose that we have a discrete random variable X that have the values -1, 0, 1 and the probabilities of three values are $P(-1) = 1/4$, $P(0) = 1/2$ and $P(1) = 1/4$.

Now let us consider two different functions of the r.v. 'X'

Let $g_1(X)$ be given by $X+2$ and $g_2(X)$ be given by $X+5$

And $k_1 = 2$ and $k_2 = 5$,

So, let us construct a table:

x	$p(x)$	$g_1(X)$ $=X+2$	$k_1 g_1(X)$	$g_2(X)$ $=X+5$	$k_2 g_2(X)$	$k_1 g_1(X)$ $+ k_2 g_2(X)$	$p(x)(k_1 g_1(X) + k_2 g_2(X))$
-1	1/4	1	2	4	20	22	-22/4
0	1/2	2	4	5	25	29	29/2
1	1/4	3	6	6	30	36	36/4
							72/4=18

In the same way, you can find the R.H.S of the formula and will find

$$E[k_1 g_1(X) + k_2 g_2(X)] = k_1 E[g_1(X)] + k_2 E[g_2(X)]$$

That both sides are equal as it was required.

00:49

Topic No. 47

Explanation of the fact that in general the expected value of a product is not equal to the product of the expected values

01:09

Suppose that we have two random variables X and Y .

Then, in general

$$E(XY) \neq E(X)E(Y).$$

Reason:

X times Y is a non-linear function

and the expectation operator doesn't go inside non-linear functions, it only goes inside linear functions such as $X + Y$.

03:24 Example:

Let us divide, at random, a horizontal line segment of length 5 into two parts.

If X is the length of the left-hand part, then $5 - X$ will be the length of the right-hand part.

04:35

Since we are dividing the line segment into two parts at random, therefore it is reasonable to assume that the length X is uniformly distributed i.e.

$$f(x) = \begin{cases} \frac{1}{5}, & 0 < x < 5 \\ 0 & \text{elsewhere} \end{cases}$$

05:57

Now, the expected length of the left-hand part is given by

$$\begin{aligned} E(X) &= \int_0^5 xf(x)dx = \int_0^5 x \left(\frac{1}{5}\right) dx \\ &= \int_0^5 \left(\frac{x}{5}\right) dx = \left[\frac{x^2}{10}\right]_0^5 = \frac{25}{10} - \frac{0}{10} = \frac{25}{10} = \frac{5}{2} \end{aligned}$$

07:40

Similarly, the expected length of the right-hand part is given by

$$\begin{aligned} E(5 - X) &= \int_0^5 (5 - x)f(x)dx = \int_0^5 (5 - x) \left(\frac{1}{5}\right) dx \\ &= \int_0^5 \left(\frac{5 - x}{5}\right) dx = \int_0^5 \left(1 - \frac{x}{5}\right) dx \\ &= \left[x - \frac{x^2}{10}\right]_0^5 = 5 - \frac{25}{10} = 5 - \frac{5}{2} = \frac{5}{2} \end{aligned}$$

09:52 On the other hand, the expected length of the product is calculated as follows:

$$\begin{aligned} E[X(5 - X)] &= E(5X - X^2) = \int_0^5 (5x - x^2)f(x)dx \\ &= \int_0^5 (5x - x^2) \left(\frac{1}{5}\right) dx = \int_0^5 \left(\frac{5x - x^2}{5}\right) dx = \int_0^5 \left(x - \frac{x^2}{5}\right) dx \\ &= \left[\frac{x^2}{2} - \frac{x^3}{15}\right]_0^5 = \frac{25}{2} - \frac{125}{15} - 0 + 0 = \frac{375 - 250}{30} = \frac{125}{30} = \frac{25}{6} \end{aligned}$$

11:12

$$\text{Now, } E(X)E(5 - X) = \frac{5}{2} \times \frac{5}{2} = \frac{25}{4}$$

whereas

$$E[X(5 - X)] = \frac{25}{6}$$

so that

$$E[X(5 - X)] \neq E(X)E(5 - X).$$

Hence, it is easy to see that, in general, the expected value of a product is not equal to the product of the expected values.

12:46

However, we have the following important result:

If X and Y are independent random variables, then

$$E(XY) = E(X)E(Y).$$

Topic No. 48

Example of computing mathematical Expectation of a Function of a Discrete Random Variable 'X'

Example: Let the pmf $p(x)$ be positive at $x = -1, 0, 1$ and zero elsewhere.

a) If $p(0) = 1/4$, find $E(X^2)$.

x	x^2
-1	1
0	0
1	1

Solution:

First we find the pmf for X^2 . The random variable X^2 takes on the values 0 and 1.

Now,

$$P(X^2 = 0) = P(X = 0) = \frac{1}{4},$$

and

$$\begin{aligned}
 P(X^2 = 1) &= P[(X = -1) \cup P(X = 1)] \\
 &= p(-1) + p(1) \\
 &= 1 - p(0) \\
 &= \frac{3}{4}
 \end{aligned}$$

It then follows that

$$E(X^2) = 0.P(X^2 = 0) + 1.P(X^2 = 1) = \frac{3}{4}.$$

04:39

Therefore, the sum of the probabilities are

$$\begin{aligned}
 p(-1) + p(0) + p(1) &= \frac{1}{4} + \frac{1}{4} + \frac{1}{2} \\
 &= \frac{1+1+2}{4} = \frac{4}{4} = 1.
 \end{aligned}$$

00:32

Topic No. 48

Another Example of computing mathematical Expectation of a Function of a Discrete Random Variable 'X'

00:42

Example: Let the pmf $p(x)$ be positive at $x = -1, 0, 1$ and zero elsewhere.

b) If $p(0) = 1/4$ and if $E(X) = 1/4$, determine $p(-1)$ and $p(1)$.

01:27

Solution:

From

$$\begin{aligned}
 E(X) &= (-1) \cdot p(-1) + 0 \cdot p(0) + 1 \cdot p(1) \\
 &= (-1) \cdot p(-1) + 1 \cdot p(1) = \frac{1}{4},
 \end{aligned}$$

and

$$p(-1) + p(0) + p(1) = 1$$

03:39

we obtain the equations

$$\begin{cases} p(-1) - p(1) = -1/4, & (1) \\ p(-1) + p(1) = 3/4 & (2) \end{cases}$$

Solving the equations simultaneously yields

$$p(-1) = \frac{1}{4},$$

and

$$p(1) = \frac{1}{2}.$$

02:05

Topic No. 49

Yet another example of computing Mathematical Expectation of a Function of a Continuous Random Variable X

02:20 **Example:**

Let X be a non-negative random variable of the continuous type with pdf $f(x), x > 0$.

Show that

$$E(X) = \int_0^{\infty} (1 - F(x)) dx$$

where $F(x)$ is the cdf of X .

03:56

Solution:

$$1 - F(x) = 1 - P(X \leq x) = P(X > x) = \int_x^{\infty} f_X(y) dy$$

So

$$\int_0^{\infty} (1 - F(x)) dx = \int_0^{\infty} \int_x^{\infty} f_X(y) dy dx$$

07:55

Now,

$$1 - F(x) = 1 - P(X \leq x) = P(X > x) = \int_x^{\infty} f_X(y) dy$$

$$\text{So } \int_0^{\infty} (1 - F(x)) dx = \int_0^{\infty} \int_x^{\infty} f_X(y) dy dx$$

Note that in the above, we are integrating over the domain $0 < x < \infty$ and $x < y < \infty$.

08:54

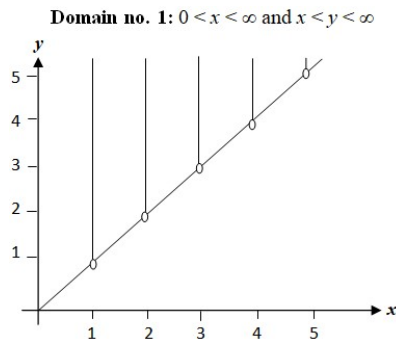
00:09

Topic No. 50

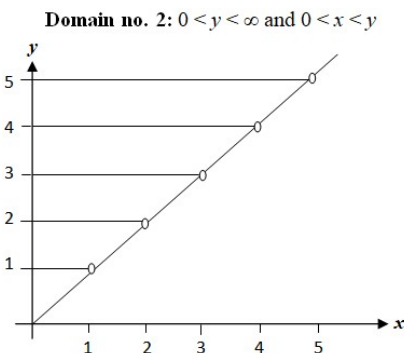
Now, integrating over the domain $0 < x < \infty$ and $x < y < \infty$
is the same as integrating over the domain $0 < y < \infty$ and $0 < x < y$.

Proof of the fact
that $E(X^2)$
is greater than or
equal to $[E(X)]^2$

09:42 Explanation:



11:19



12:30

So, integrating over the domain $0 < x < \infty$ and $x < y < \infty$
is the same as integrating over the domain $0 < y < \infty$ and $0 < x < y$.

$$\begin{aligned}\text{So } \int_0^{\infty} (1 - F(x)) dx &= \int_0^{\infty} \int_x^{\infty} f_X(y) dy dx \\ &= \int_0^{\infty} \int_0^y f_X(y) dx dy = \int_0^{\infty} \left(\int_0^y f_X(y) dx \right) dy\end{aligned}$$

14:22

Now,

$$\int_0^y f_X(y) dx = f_X(y) \int_0^y 1 dx = f_X(y) [x]_0^y = f_X(y)(y - 0) = y f_X(y)$$

Therefore

$$\int_0^{\infty} (1 - F(x)) dx = \int_0^{\infty} (y f_X(y)) dy = \int_0^{\infty} y f_X(y) dy = E(X).$$

00:14

Theorem

If the variance of the random variable X exists, then

$$E(X^2) \geq [E(X)]^2$$

00:39

Now

$$\begin{aligned}\text{Var}(X) &= E[X - E(X)]^2 \\ &= E[X^2 - 2XE(X) + [E(X)]^2] \\ &= E[X^2] - 2E(X)E(X) + [E(X)]^2 \\ &= E[X^2] - 2[E(X)]^2 + [E(X)]^2 \\ &= E[X^2] - [E(X)]^2\end{aligned}$$

03:40

Hence, it is established that

$$E(X^2) - [E(X)]^2 = E[X - E(X)]^2$$

But $[X - E(X)]^2 \geq 0$
as the square of any quantity has to be nonnegative.

Therefore, $E(X - E(X))^2 \geq 0$

$$E(X^2) \geq [E(X)]^2 \Rightarrow E(X^2) - [E(X)]^2 \geq 0$$

Hence proved.

00:09

Topic No. 51

Mean, Variance and
Standard deviation
of a Random Variable

00:25

Mean

First, let X be a random variable of the discrete type with pmf $p(x)$.

Then

$$E(X) = \sum_x xp(x).$$

01:01

If the support of X is $\{a_1, a_2, a_3, \dots\}$, it follows that

$$E(X) = a_1p(a_1) + a_2p(a_2) + a_3p(a_3) + \dots$$

This sum of products is seen to be a “weighted average” of the values of $a_1, a_2, a_3, \dots$ the “weight” associated with each a_i being $p(a_i)$.

02:50

Let $a_1, a_2, a_3, \dots, a_n$

$$\begin{aligned} &= \frac{a_1 + a_2 + a_3 + \dots + a_n}{n} = \frac{a_1}{n} + \frac{a_2}{n} + \frac{a_3}{n} + \dots + \frac{a_n}{n} \\ &= \left(a_1 \frac{1}{n} + a_2 \frac{1}{n} + a_3 \frac{1}{n} + \dots + a_n \frac{1}{n} \right) \end{aligned}$$

This suggests that we call $E(X)$ the arithmetic mean of the values of X , or, more simply,

the mean value of X (or the mean value of the distribution).

05:15

Variance

Let X be a discrete random variable with support $\{a_1, a_2, a_3, \dots\}$, and with pmf $p(x)$, then

$$E[(X - \mu)^2] = \sum_x (x - \mu)^2 p(x) = (a_1 - \mu)^2 p(a_1) + (a_2 - \mu)^2 p(a_2) + \dots$$

06:38

This sum of products may be interpreted as a “weighted average” of the squares of the deviations of the numbers $a_1, a_2, a_3, \dots$ from the mean value μ ,

where the “weight” associated with each $(a_i - \mu)^2$ is $p(a_i)$.

07:16

Standard deviation

The positive square root of the variance is called Standard deviation.

$$\sigma = \sqrt{E[(X - \mu)^2]}$$

07:36

Short-cut formula

$$\sigma^2 = E[(X - \mu)^2] = E(X^2 - 2\mu X + \mu^2);$$

and since E is a linear operator,

$$\begin{aligned} \sigma^2 &= E(X^2) - 2\mu E(X) + \mu^2 \\ &= E(X^2) - 2\mu^2 + \mu^2 \\ &= E(X^2) - \mu^2, \end{aligned}$$

This frequently affords an easier way of computing the variance of X .

00:45

Topic No. 52

The Concept of Degenerate Random Variable and one of its basic properties

00:51

Definition

A degenerate distribution is a probability distribution with support only on **a single point**.

02:57

Examples

- A coin both sides of which are showing a head
- A die all six sides of which are showing the same number.

This distribution satisfies the definition of “**random variable**” even though it does not appear **random** in the everyday sense of the word; hence it is considered **degenerate**.

06:30

Mean and Variance of a Degenerate Random Variable

If the space contains only one point k for which

Then
$$p(k) > 0,$$

$$p(k) = 1, \mu = k, \text{ and } \sigma^2 = 0.$$

07:08

Explanation:

Let us consider another example, suppose that a teacher administers a test out of 10 in a class and $k = 20$.

If every single student get 7 out of 10 then, the mean mark will be 7.

In this case, the mean of the random variable X is also equal to k and the variance of the random variable X is equal to zero.

00:19

Topic No. 53

Proof of the fact that the Mean of a symmetric distribution lies at the point of symmetry

00:51

Example

Let a random variable X of the continuous type have a pdf $f(x)$ whose graph is symmetric with respect to $x = c$.

If the mean value of X exists, show that

$$E(X) = c.$$

01:58

Hint: Show that $E(X - c)$ equals zero

by writing $E(X - c)$ as the sum of two integrals:

Where one from $-\infty$ to c and the other from c to ∞ .

In the first, let $y = c - x$; and, in the second, $z = x - c$.

Finally, use the symmetry condition $f(c - y) = f(c + y)$ in the first.

04:45

Solution:

Given that $f(c-x) = f(c+x)$, we will show that $E(X - c) = E(X) - c = 0$.

$$\begin{aligned} E[X - c] &= \int_{-\infty}^{\infty} (x - c) f(x) dx \\ &= \int_{-\infty}^c (x - c) f(x) dx + \int_c^{\infty} (x - c) f(x) dx. \end{aligned}$$

06:01

In the first integral,

$$\text{let } y = c - x \Rightarrow dy = -dx$$

$$\text{and } x \rightarrow -\infty \Rightarrow y \rightarrow \infty, x \rightarrow c \Rightarrow y \rightarrow 0$$

and in the second integral,

$$\text{let } z = x - c \Rightarrow dz = dx$$

$$\text{and } x \rightarrow c \Rightarrow z \rightarrow 0, x \rightarrow \infty \Rightarrow z \rightarrow \infty.$$

08:30

We have

$$E(X - c) = \int_{-\infty}^c (x - c) f(x) dx + \int_c^{\infty} (x - c) f(x) dx.$$

Applying the above transformations, we obtain

$$\begin{aligned} E(X - c) &= -\int_{\infty}^0 (-y) f(c - y) dy + \int_0^{\infty} z f(c + z) dz \\ &= -\int_0^{\infty} y f(c - y) dy + \int_0^{\infty} z f(c + z) dz \end{aligned}$$

10:20

Now, let us focus on the first integral:

Since $f(\cdot)$ is symmetric about c ,

$$\text{therefore } f(c - y) = f(c + y)$$

$$\begin{aligned} \Rightarrow -\int_0^{\infty} y f(c - y) dy &= -\int_0^{\infty} y f(c + y) dy \\ &= -\int_0^{\infty} z f(c + z) dz \end{aligned}$$

using the concept of dummy variable.

Hence

$$E(X - c) = -\int_0^{\infty} z f(c + z) dz + \int_0^{\infty} z f(c + z) dz = 0.$$

13:16

Hence, we conclude that if the density function for a random variable X is symmetric about c , then $\mu = E[X] = c$.

00:14

Topic No. 54

**Proof of the fact that the
Mean of a Standardized variable
is zero and the variance is 1.**

01:23

Proof:

Let the random variable X have mean μ and standard deviation σ ,

$-h < t < h$. Show that

$$E\left(\frac{X-\mu}{\sigma}\right)=0, \quad E\left[\left(\frac{X-\mu}{\sigma}\right)^2\right]=1.$$

02:08

Using the linear properties of expected value and the definition of $\mu = E[X]$, we calculate

$$E\left[\frac{X-\mu}{\sigma}\right] = \frac{1}{\sigma} E[X-\mu] = \frac{1}{\sigma} (E[X]-\mu) = \frac{\mu-\mu}{\sigma} = 0,$$

which verifies the first equation.

04:06

Using the linear properties of expected value again and the definition of $\sigma^2 = E[(X-\mu)^2]$, we calculate

$$E\left[\left(\frac{X-\mu}{\sigma}\right)^2\right] = E\left[\frac{(X-\mu)^2}{\sigma^2}\right] = \frac{E[(X-\mu)^2]}{\sigma^2} = \frac{\sigma^2}{\sigma^2} = 1,$$

which verifies the second equation.

05:46

Note:

$$E\left(\frac{X-\mu}{\sigma}\right)=0, \quad E\left[\left(\frac{X-\mu}{\sigma}\right)^2\right]=1$$

implies that

$$\text{Var}\left(\frac{X-\mu}{\sigma}\right)=1.$$

06:05

Proof:

$\text{Var}(w) = E[(w)^2] - [E(w)]^2$, we have

$$\begin{aligned} \text{Var}\left(\frac{X-\mu}{\sigma}\right) &= E\left[\left(\frac{X-\mu}{\sigma}\right)^2\right] - \left[E\left(\frac{X-\mu}{\sigma}\right)\right]^2 \\ &= 1 - 0^2 = 1 - 0 = 1. \end{aligned}$$

Hence, the mean and standard deviation of a standardized variable are 0 and 1 respectively.

00:11

Topic No. 55

**The Concept of
Moments**

00:34

m^{th} Moment about Arbitrary origin

Consider the expression $E(X-a)^m$ where m is a positive integer.

$$E(X-a)^m = \int_{-\infty}^{\infty} (x-a)^m f(x) dx$$

or

$$E(X-a)^m = \sum_x (x-a)^m p(x)$$

depending on whether X is a continuous or a discrete random variable and it is known as the m^{th} moment about an Arbitrary origin.

02:44

**Notation
for m^{th} Moment about
Arbitrary origin:**

$$\mu'_m = E(X-a)^m$$

03:40

m^{th} Moment about the Mean

In the expression $E(X-a)^m$ if we put $a = \mu$, we obtain

$$E(X-\mu)^m = \int_{-\infty}^{\infty} (x-\mu)^m f(x) dx$$

or

$$E(X-\mu)^m = \sum_x (x-\mu)^m p(x)$$

depending on whether X is a continuous or a discrete random variable and it is known as the m^{th} moment about the Mean.

04:59

**Notation
for m^{th} Moment about
The Mean:**

$$\mu_m = E(X - \mu)^m$$

09:14

Another Special Case:

In the m^{th} moment about the Mean, if we put $m = 2$, we obtain

$$\mu_2 = E(X - \mu)^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

or

$$\mu_2 = E(X - \mu)^2 = \sum_x (x - \mu)^2 p(x)$$

implying that the 2nd moment about the mean is the Variance of the distribution.

05:23

Moments about the Mean are also known as **Central Moments**.

00:14

Topic No. 56

**Moment Ratios depicting
Skewness (β_1) and Kurtosis (β_2)**

05:33

m^{th} Moment about the Origin

In the expression $E(X - a)^m$
if we put $a = 0$, we obtain

$$E(X - 0)^m = E(X^m) = \int_{-\infty}^{\infty} x^m f(x) dx$$

or

$$E(X - 0)^m = E(X^m) = \sum_x x^m p(x)$$

depending on whether X is a continuous
or a discrete random variable
and it is known as the m^{th} moment
about the Origin.

01:23

Measures of Skewness

Let X be a random variable with mean μ and variance σ^2
(i.e. standard deviation σ) such that the third moment about μ

i.e. $\mu_3 = E[(X - \mu)^3]$ exists.

The value of the ratio

$$\frac{E[(X - \mu)^3]}{\sigma^3} = \frac{\mu_3}{\sigma^3}$$

is often used as a measure of skewness.

07:35

**Notation
for m^{th} Moment about
The Origin:**

$$\mu'_m = E(X^m)$$

02:35

It is important to note that this measure of skewness is

- negative for distributions that are skewed to the left,
- zero for distributions that are not skewed (symmetric), and
- Positive for distributions that are skewed to the right.

08:14

A Special Case:

In the m^{th} moment about the Origin, if we put $m = 1$, we obtain

$$\mu'_1 = E(X^1) = E(X) = \int_{-\infty}^{\infty} xf(x) dx$$

or

$$\mu'_1 = E(X^1) = E(X) = \sum_x xp(x)$$

implying that the 1st moment about the Origin is the Mean of the distribution.

03:11

The square of this ratio i.e.

$$\frac{\mu_3^2}{(\sigma^3)^2} = \frac{\mu_3^2}{(\sigma^2)^3} = \frac{\mu_3^2}{(\mu_2)^3}$$

is known as the 'first' moment-ratio and is denoted by β_1 i.e.

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

06:38

Measure of Kurtosis

Let X be a random variable with mean μ and variance σ^2 (i.e. standard deviation σ) such that the fourth moment about μ

i.e. $\mu_4 = E[(X - \mu)^4]$ exists.

The value of the ratio

$$\frac{E[(X - \mu)^4]}{\sigma^4} = \frac{\mu_4}{(\sigma^2)^2} = \frac{\mu_4}{(\mu_2)^2}$$

is often used as a measure of kurtosis.

08:10

This ratio is known as the 'second' moment-ratio and is denoted by β_2 i.e.

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

08:30

It is important to note that the numerical value of this measure of kurtosis is

- greater than 3 for distributions that are leptokurtic,
- equal to 3 for distributions that are mesokurtic, and
- less than 3 for distributions that are platykurtic.

00:33

Topic No. 57

**The Concept
of
Moment Generating
Function**

00:51

Definition:

Let X be a random variable such that for some $h > 0$, the expectation of e^{tX} exists for

$$-h < t < h.$$

The **moment generating function** of X is defined to be the function

$$M(t) = E(e^{tX}) \quad -h < t < h.$$

We use the abbreviation **mgf** to denote the moment generating function of a random variable.

03:14

Role of mgf

In general, if m is a positive integer and if $M^{(m)}(t)$ means the m^{th} derivative of $M(t)$, we have, by repeated differentiation with respect to t ,

$$M^{(m)}(0) = E(X^m).$$

Since $M(t)$ generates the values of $E(X^m)$; $m = 1, 2, 3, \dots$, it is called the **moment-generating function** (mgf).

04:20

In the same way, if we take the second derivative of the first derivative, and putting $t = 0$, we will get the second moment about the origin.

05:20

Example

Consider the exponential distribution given by

$$f(x; \lambda) = \lambda e^{-\lambda x}, \quad 0 < x < \infty, \lambda > 0$$

Find the mgf of X and use it to find the mean and variance of the exponential distribution.

06:27

Solution

By definition, we have

$$\begin{aligned} M(t) &= E(e^{tX}) = \lambda \int_0^{\infty} e^{tx} e^{-\lambda x} dx \\ &= \lambda \int_0^{\infty} e^{x(t-\lambda)} dx = \lambda \left(\frac{e^{x(t-\lambda)}}{(t-\lambda)} \right)_0^{\infty} \\ &= \frac{\lambda}{(t-\lambda)} (e^{x(t-\lambda)})_0^{\infty} \end{aligned}$$

07:38

Now, if $\lambda > t$, then $t - \lambda < 0$ and hence

$$\begin{aligned} M(t) &= \frac{\lambda}{(t-\lambda)} (e^{-\infty} - e^0) \\ &= \frac{\lambda}{(t-\lambda)} (0 - 1) = \frac{\lambda}{(\lambda - t)} \end{aligned}$$

for $t < \lambda$.

09:30

Now

$$\begin{aligned}
 M^{(1)}(t) &= M'(t) = \frac{d}{dt}(M(t)) \\
 &= \frac{d}{dt} \left(\frac{\lambda}{(\lambda-t)} \right) = \lambda \frac{d}{dt} ((\lambda-t)^{-1}) \\
 &= -\lambda ((\lambda-t)^{-2} (0-1)) \\
 &= \lambda ((\lambda-t)^{-2}) = \frac{\lambda}{(\lambda-t)^2} \\
 &\text{for } t < \lambda.
 \end{aligned}$$



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10:17

Evaluating the derivative at $t=0$, we obtain

$$M^{(1)}(0) = \frac{\lambda}{(\lambda-0)^2} = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}$$

which is the first moment about zero

i.e. $E(X)$

i.e. the well-known mean of the distribution.

00:23

Topic No. 58

Algebraic expressions of Some well-known MGFs

11:14

Next

$$\begin{aligned}
 M^{(2)}(t) &= M''(t) = \frac{d}{dt} \left(\frac{d}{dt} (M(t)) \right) \\
 &= \frac{d}{dt} \left(\frac{\lambda}{(\lambda-t)^2} \right) = \lambda \frac{d}{dt} ((\lambda-t)^{-2}) \\
 &= (-2)\lambda ((\lambda-t)^{-3} (0-1)) \\
 &= 2\lambda ((\lambda-t)^{-3}) = \frac{2\lambda}{(\lambda-t)^3} \\
 &\text{for } t < \lambda.
 \end{aligned}$$

01:15

Discrete Distributions

12:20

Evaluating the derivative at $t=0$, we obtain

$$M^{(2)}(0) = M''(0) = \frac{2\lambda}{(\lambda-0)^3} = \frac{2\lambda}{\lambda^3} = \frac{2}{\lambda^2}$$

which is the second moment about zero

i.e. $E(X^2)$.

01:20

Distribution	Moment Generating Function (MGF)
Bernoulli $P(X=1) = p$	$1 - p + pe^t$
Geometric $(1-p)^{k-1}p$	$\frac{pe^t}{1-(1-p)e^t} \quad \forall t < -\ln(1-p)$
Binomial $B(n, p)$	$(1 - p + pe^t)^n$
Poisson λ	$e^{\lambda(e^t-1)}$
Negative binomial $NB(r, p)$	$\frac{(1-p)^r}{(1-pe^t)^r}$
Uniform (discrete) $U(a, b)$	$\frac{e^{bt} - e^{at}}{(b-a+1)(1-e^t)}$

13:07

Using the short-cut formula for the variance, we obtain

$$\begin{aligned}
 \text{Var}(X) &= E(X^2) - [E(X)]^2 \\
 &= \frac{2}{\lambda^2} - \left(\frac{1}{\lambda} \right)^2 \\
 &= \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}
 \end{aligned}$$

which is well-known.

06:56

Continuous Distributions

07:03	Distribution	Moment Generating Function (MGF)
	Uniform (continuous) $U(a, b)$	$\frac{e^{tb} - e^{ta}}{t(b-a)}$
	Normal $N(\mu, \sigma^2)$	$e^{i\mu + \frac{1}{2}\sigma^2 t^2}$
	Chi-Squared χ^2_k	$(1 - 2t)^{-\frac{k}{2}}$
	Gamma $\Gamma(k, \theta)$	$(1 - t\theta)^{-k}; \forall t < \frac{1}{\theta}$
	Exponential λ	$(1 - t\lambda^{-1})^{-1}, (t < \lambda)$

09:54

Distribution	Moment Generating Function (MGF)
Laplace $L(\mu, b)$	$\frac{e^{t\mu}}{1 - b^2 t^2}$
Cauchy $N(\mu - \theta)$	Do not exist

00:13

Topic No. 59

Derivation

of

Mean and Variance of a distribution in terms of its MGF

(by repeated differentiation of the mgf)

00:25

- Since a distribution that has an mgf $M(t)$ is **completely determined** by $M(t)$, it would not be surprising if we could obtain some **properties** of this distribution **directly** from $M(t)$.
- For example, the existence of $M(t)$ for $-h < t < h$ implies that **derivatives** of $M(t)$ **of all orders exist** at $t = 0$.

01:38

So here we will apply the method of **successive derivatives** of the MGF in order to prove that, for any pdf the MGF of which exists,

$$\mu = M'(0)$$

and

$$\sigma^2 = M''(0) - [M'(0)]^2$$

03:06

If X is a continuous random variable, then

$$M'(t) = \frac{dM(t)}{dt} = \frac{d}{dt} \int_{-\infty}^{\infty} e^{tx} f(x) dx = \int_{-\infty}^{\infty} \left(\frac{d}{dt} e^{tx} \right) f(x) dx = \int_{-\infty}^{\infty} x e^{tx} f(x) dx$$

Now, $\frac{d}{dt} e^{tx} = x e^{tx}$

But here x is acting as the “constant” and t is the variable.

$$\therefore \frac{d}{dt} e^{tx} = \frac{d}{dt} e^{xt} = x e^{xt} = x e^{tx}$$

Hence, $M'(t) = \frac{dM(t)}{dt} = \int_{-\infty}^{\infty} x e^{tx} f(x) dx$

06:20

Upon setting $t = 0$, we have

$$M'(0) = \left. \frac{dM(t)}{dt} \right|_{t=0} = \int_{-\infty}^{\infty} x e^{0x} f(x) dx = \int_{-\infty}^{\infty} x e^0 f(x) dx$$

$$= \int_{-\infty}^{\infty} x f(x) dx = E(X) = \mu.$$

08:29

Now, the second derivative:

$$M''(t) = \frac{dM'(t)}{dt} = \frac{d}{dt} \int_{-\infty}^{\infty} x e^{tx} f(x) dx = \int_{-\infty}^{\infty} \left(\frac{d}{dt} x e^{tx} \right) f(x) dx = \int_{-\infty}^{\infty} x f(x) \left(\frac{d}{dt} e^{tx} \right) dx$$

Now, $\frac{d}{dt} e^{tx} = x e^{tx}$

But here x is acting as the “constant” and t is the variable

Therefore $\frac{d}{dt} e^{tx} = x e^{tx}$

& hence $M''(t) = \int_{-\infty}^{\infty} x f(x) x e^{tx} dx = \int_{-\infty}^{\infty} x^2 f(x) e^{tx} dx$

10:46

Upon setting $t = 0$, we have

$$M''(0) = \int_{-\infty}^{\infty} x^2 f(x) e^{0x} dx = \int_{-\infty}^{\infty} x^2 f(x) e^0 dx$$

$$= \int_{-\infty}^{\infty} x^2 f(x) dx = E(X^2)$$

12:10

According to the short-cut formula,

$$\sigma^2 = \text{Var}(X) = E(X^2) - \mu^2$$

But since $M'(0) = \mu$ and $M''(0) = E(X^2)$ (as proved above)

Therefore

$$\sigma^2 = M''(0) - [M'(0)]^2$$

13:01

The point to be remembered is that sometimes one way is **easier** than the other.

10:40

- Since $M(t)$ generates the values of $E(X^m)$, $m = 1, 2, 3, \dots$, it is called the moment generating function (mgf).
- m^{th} moment about the Origin of a distribution from its MGF is the same as the m^{th} derivative evaluated at $t = 0$.

00:13

Topic No. 60

Derivation of m^{th} moment about the Origin of a distribution from its MGF (by repeated differentiation of the MGF)

00:30

Theorem

If m is a positive integer and if $M^{(m)}(t)$ means the m^{th} derivative of $M(t)$, we have, by repeated differentiation with respect to t ,

$$M^{(m)}(0) = E(X^m).$$

and

$$E(X^m) = \int_{-\infty}^{\infty} x^m f(x) dx$$

where

$$\text{or} \\ \sum_x x^m p(x),$$

04:24

Proof:

$$M'(t) = \int_{-\infty}^{\infty} x e^{tx} f(x) dx,$$

$$M''(t) = \int_{-\infty}^{\infty} x^2 e^{tx} f(x) dx,$$

$$M'''(t) = \int_{-\infty}^{\infty} x^3 e^{tx} f(x) dx,$$

09:01

$$M^{(m)}(t) = \int_{-\infty}^{\infty} x^m e^{tx} f(x) dx$$

$$\Rightarrow M^{(m)}(0) = \int_{-\infty}^{\infty} x^m e^{0x} f(x) dx$$

$$= \int_{-\infty}^{\infty} x^m f(x) dx = E(X^m) = m^{th} \text{ moment about}$$

about the origin.

00:17

Topic No. 61

Derivation of m^{th} moment about an Arbitrary origin of a distribution from its MGF (by repeated differentiation of the mgf)

02:00

Theorem

Let X be a random variable such that

$$R(t) = E(e^{t(X-a)})$$

exists for t such that $-h < t < h$.

If m is a positive integer, $R^{(m)}(0)$ is equal to the m^{th} moment of the distribution about the point a .

03:38

Proof:

$$R(t) = E[e^{t(X-a)}] = \int_{-\infty}^{\infty} e^{t(x-a)} f(x) dx = \int_{-\infty}^{\infty} e^{t(x-a)} f(x) dx$$

$$R'(t) = \frac{d}{dt} \int_{-\infty}^{\infty} e^{t(x-a)} f(x) dx = \int_{-\infty}^{\infty} \left(\frac{d}{dt} e^{t(x-a)} \right) f(x) dx$$

05:34

Now,

$$\frac{d}{dt} e^{t(x-a)} = (x-a) e^{t(x-a)}$$

$$\therefore R'(t) = \int_{-\infty}^{\infty} (x-a) e^{t(x-a)} f(x) dx$$

$$\Rightarrow R'(0) = \int_{-\infty}^{\infty} (x-a) e^{0(x-a)} f(x) dx = \int_{-\infty}^{\infty} (x-a) e^0 f(x) dx$$

08:05

or

$$R'(0) = \int_{-\infty}^{\infty} (x-a) f(x) dx$$

$$= E(x-a) = 1^{st} \text{ moment about } a$$

or

$$E(x-a)^1 = 1^{st} \text{ moment about } a.$$

09:48 Next,

$$R''(0) = \frac{d}{dt} R'(t) = \frac{d}{dt} \left[\int_{-\infty}^{\infty} (x-a) e^{t(x-a)} f(x) dx \right]$$

$$= \int_{-\infty}^{\infty} (x-a) f(x) \left(\frac{d}{dt} e^{t(x-a)} \right) dx = \int_{-\infty}^{\infty} (x-a)(x-a) e^{t(x-a)} f(x) dx$$

$$= \int_{-\infty}^{\infty} (x-a)^2 e^{t(x-a)} f(x) dx$$

$$\Rightarrow R''(0) = \int_{-\infty}^{\infty} (x-a)^2 e^{0(x-a)} f(x) dx = \int_{-\infty}^{\infty} (x-a)^2 f(x) dx$$

$$= E(X-a)^2 = 2^{nd} \text{ moment about } a.$$

00:17

Topic No. 62

Alternative method for finding
the moments from the MGF using
Maclaurin's series expansion
of the mgf
(through an Example)

00:30

We can differentiate M(t) any number of times to find the moments of X.

However, it is instructive to consider the following **alternative** method --- one that involves the Maclaurin Series expansion of M(t).

03:01

Maclaurin Series

By definition, for any function $f(x)$, a Maclaurin's series is given by:

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \dots + \frac{f^{(n)}(0)}{n!}x^n + \dots$$

05:00

If our function is M(t), the Maclaurin's series will be given by

$$M(t) = M(0) + \frac{M'(0)}{1!}t + \frac{M''(0)}{2!}t^2 + \dots + \frac{M^{(m)}(0)}{m!}t^m + \dots$$

06:08

But we know that

$$M^{(m)}(0) = E(X^m)$$

Also, we know that

$$M(t) = E(e^{tX}) \Rightarrow M(0) = E(e^{0X})$$

$$\Rightarrow M(0) = E(e^0) \Rightarrow M(0) = E(1) = 1$$

09:53

Therefore

$$M(t) = 1 + \frac{E(X)}{1!}t + \frac{E(X^2)}{2!}t^2 + \dots + \frac{E(X^m)}{m!}t^m + \dots$$

or

$$M(t) = 1 + E(X)\frac{t}{1!} + E(X^2)\frac{t^2}{2!} + \dots + E(X^m)\frac{t^m}{m!} + \dots$$

Thus the **coefficient** of $(t^m/m!)$ in the Maclaurin's series representation of M(t) is $E(X^m)$ i.e. the **m^{th} moment about zero**.

00:17

Topic No. 63

Derivation of the

Relationship

between the MGF of the
Standardized Variable and
the MGF of the Original Random
Variable

00:39

Theorem:

Let the random variable X have mean μ , standard deviation σ , and mgf $M_X(t)$, $-h < t < h$

$$\text{Then } M_Z(t) = e^{-\frac{t\mu}{\sigma}} M_X\left(\frac{t}{\sigma}\right), \quad -h\sigma < t < h\sigma.$$

$$\text{where } Z = \frac{X - \mu}{\sigma}.$$

02:50

Proof:

The first point to be noted is that

$$\text{if } -h\sigma < t < h\sigma, \text{ then } -h < \frac{t}{\sigma} < h,$$

which shows that $\frac{t}{\sigma}$ is in the domain of $M_X(t)$.

13:00

Note that the third result i.e.

$$M_{\frac{X+a}{b}}(t) = e^{\frac{at}{b}} M_X\left(\frac{t}{b}\right)$$

is of special importance when $a = -\mu$ and $b = \sigma$,

in which case, we have

$$M_{\frac{X-\mu}{\sigma}}(t) = e^{\frac{-\mu t}{\sigma}} M_X\left(\frac{t}{\sigma}\right)$$

or, in other words,

$$M_Z(t) = e^{\frac{-\mu t}{\sigma}} M_X\left(\frac{t}{\sigma}\right)$$

exactly the same as what we proved.

05:53

Now, using the definition of $M_X(t) = E[e^{tX}]$ and the linear properties of the expected value, we start from the RHS of the equation:

00:28

Topic No. 64

Proof of the fact

that for a PDF that is Symmetric about 0 and for which the MGF exists,

$$M(-t) = M(t)$$

06:51

$$\begin{aligned} RHS &= e^{\frac{-\mu t}{\sigma}} M_X\left(\frac{t}{\sigma}\right) = e^{\frac{-\mu t}{\sigma}} E\left[e^{\frac{t}{\sigma} X}\right] = E\left[e^{\frac{tX}{\sigma} - \frac{\mu t}{\sigma}}\right] \\ &= E\left[e^{t \frac{(X-\mu)}{\sigma}}\right] = E\left[e^{tZ}\right] = M_Z(t) = LHS \end{aligned}$$

which verifies the equation.

01:57

Theorem:

Let X be a random variable with a pdf $f(x)$ and mgf $M(t)$.

Suppose f is symmetric about 0 i.e. $f(-x) = f(x)$.

Then $M(-t) = M(t)$.

09:40

Next:

Three general results

pertaining to

the moment generating function:

02:57

Proof:

By definition

$$M(-t) = E\left(e^{(-t)x}\right) = \int_{-\infty}^{\infty} e^{(-t)x} f(x) dx$$

$$\because \left(\begin{array}{l} \text{if } f(x) \text{ is symmetric about 0,} \\ \text{then } f(x) \text{ itself is } f(-x) \end{array} \right)$$

$$= \int_{-\infty}^{\infty} e^{t(-x)} f(x) dx = \int_{-\infty}^{\infty} e^{t(-x)} f(-x) dx$$

09:47

Three general results pertaining to the moment generating function:

1. $M_{X+a}(t) = E\left[e^{(X+a)t}\right] = e^{at} M_X(t);$
2. $M_{bX}(t) = E\left[e^{tbX}\right] = E\left[e^{(bt)X}\right] = M_X(bt);$
3. $M_{\frac{X+a}{b}}(t) = E\left[e^{t\left(\frac{X+a}{b}\right)}\right] = E\left[e^{t\left(\frac{a}{b}\right) + t\left(\frac{X}{b}\right)}\right] = e^{t\left(\frac{a}{b}\right)} E\left[e^{\left(\frac{t}{b}\right)X}\right] = e^{\frac{at}{b}} M_X\left(\frac{t}{b}\right).$

05:11

Applying the transformation

$$u = -x \Rightarrow du = -dx$$

&

$$x \rightarrow -\infty \Rightarrow u \rightarrow \infty, x \rightarrow \infty \Rightarrow u \rightarrow -\infty$$

06:40

So

$$M(-t) = \int_{-\infty}^{\infty} e^{t(-x)} f(-x) dx = - \int_{\infty}^{-\infty} e^{tu} f(u) du = \int_{-\infty}^{\infty} e^{tu} f(u) du$$

$$= \int_{-\infty}^{\infty} e^{tx} f(x) dx = E(e^{tX}) = M(t).$$

00:21

Topic No. 65

Cumulant Generating Function (CGF) and its role in finding the mean and variance of a probability distribution

00:40

We begin with the Concept of Cumulants

01:09

The First Three Cumulants

- The **first cumulant** is the **first moment about zero** i.e. the **mean** of the distribution.
- The **second cumulant** is the **second central moment** i.e. the **variance**.
- The **third cumulant** is the **third central moment**.

02:54

Higher Order Cumulants

The higher cumulants are neither moments about zero nor central moments, but rather **more complicated polynomial functions of moments**.

In fact, it can be shown that the k^{th} cumulant $\kappa_n(X)$ of a random variable X is the value of **a certain polynomial in the first k moments of X** i.e. a polynomial in $E(X^\ell)$, $\ell=1, \dots, k$

04:44

Next, we consider the concept of the Cumulant Generating Function

05:28

Definition of Cumulant Generating Function:

The natural logarithm of the moment generating function is called the cumulant generating function.

It is denoted by $\Psi(t)$.

In other words:

$$\Psi(t) = \log M(t)$$

05:52

Next, let us consider the Role of the Cumulant Generating Function in finding the mean and variance of a distribution.

06:25

Two simple results:

$$\Psi'(0) = \mu$$

and

$$\Psi''(0) = \sigma^2$$

07:07

Proof:

i) Mean:

$$\Psi'(t) = \frac{d}{dt} (\Psi(t)) = \frac{d}{dt} (\log M(t)) = \frac{1}{M(t)} (M'(t)) = \frac{M'(t)}{M(t)}$$

$$\text{Hence } \Psi'(0) = \left. \frac{M'(t)}{M(t)} \right|_{t=0} = \frac{M'(0)}{M(0)} = \frac{\mu}{M(0)}$$

08:47

However

$$\therefore M(t) = E(e^{tX}) \therefore M(0) = E(e^{0X}) = E(e^0) = E(1) = 1$$

$$\text{So that } \Psi'(0) = \frac{\mu}{1} = \mu. \quad \text{Hence proved.}$$

09:44

Proof:

ii) Variance:

$$\text{We have } \Psi'(t) = \frac{M'(t)}{M(t)}$$

$$\text{Hence } \Psi''(t) = \frac{d}{dt} \left(\frac{M'(t)}{M(t)} \right) = \frac{M(t)M''(t) - (M'(t))^2}{(M(t))^2}$$

11:00

$$\text{so that } \Psi''(0) = \frac{M(0)M''(0) - (M'(0))^2}{(M(0))^2} = \frac{(1)M''(0) - (\mu)^2}{(1)^2}$$

$$= M''(0) - (\mu)^2$$

$$\text{However, we know that } M''(0) = E(X^2)$$

$$\therefore \Psi''(0) = E(X^2) - (\mu)^2 = \text{Var}(X) = \sigma^2. \quad \text{Hence proved.}$$

00:14

Topic No. 66

This is known as the

Additivity Property of the CGF and the Cumulants

00:31

The CGF of the sum
of two independent random variables
equals the sum of their CGFs

so that

each cumulant of a sum
of independent random variables
is the sum of
the corresponding cumulants of the addends

01:10

1. Additivity Property of the CGF:

If X and Y are independent random variables,
then,

the cumulant generating function of $X+Y$ is related to the
cumulant generating function of X and the cumulant
generating function of Y by the relation

$$K_{X+Y}(t) = K_X(t) + K_Y(t).$$

02:08

Proof:

If X and Y are independent random variables,
then

$$\begin{aligned} K_{X+Y}(t) &= \log E[e^{t(X+Y)}] \\ &= \log E[e^{t(X)}e^{t(Y)}] \\ &= \log (E[e^{tX}]E[e^{tY}]) \\ &= \log E[e^{tX}] + \log E[e^{tY}] \\ &= K_X(t) + K_Y(t). \end{aligned}$$

07:05

2. Additivity Property of the Cumulants:

If X and Y are independent random variables,
then,

the n^{th} cumulant of $X+Y$ is related to the n^{th} cumulant of X
and the n^{th} cumulant of Y by the relation

$$\kappa_n(X+Y) = \kappa_n(X) + \kappa_n(Y).$$

The proof of this result is somewhat advanced.

00:09

Topic No. 67

**Relation between Central MGF and CGF
AND**

**Relation between Moments about the Mean
and Cumulants**

01:47

- The central moment generating function is given denoted by $C(t)$

• Definition

It is shown as the

$$C(t) = E(e^{t(x-\mu)})$$

02:37

When we consider the mgf for moments about '0'

also

we say $E\left(e^{tx}\right)$ can also be written as $E\left(e^{t(x-0)}\right)$

03:48

The central moment generating function is given

$$\begin{aligned} C(t) &= E\left[e^{t(x-\mu)}\right] = E\left[e^{tx-\mu t}\right] = E\left[e^{tx}e^{-\mu t}\right] \\ &= e^{-\mu t} E\left[e^{tx}\right] = e^{-\mu t} M(t) \end{aligned}$$

05:23

Now, by definition, we have

$$\begin{aligned} K(t) &= \log M(t) \\ \therefore e^{K(t)} &= e^{\log M(t)} \Rightarrow e^{K(t)} = M(t) \\ \therefore C(t) &= e^{K(t)} e^{-\mu t} = e^{(K(t)-\mu t)}. \end{aligned}$$

09:25

- To express the *central* moments as functions of the cumulants, just drop from these polynomials all terms in which κ_1 appears as a factor:

09:46

$$\mu_1 = 0$$

$$\mu_2 = \kappa_2$$

$$\mu_3 = \kappa_3$$

$$\mu_4 = \kappa_4 + 3\kappa_2^2$$

$$\mu_5 = \kappa_5 + 10\kappa_3\kappa_2$$

$$\mu_6 = \kappa_6 + 15\kappa_4\kappa_2 + 10\kappa_3^2 + 15\kappa_2^3.$$

11:52

To express the cumulants κ_n for $n > 1$ as functions of the central moments, drop from these polynomials all terms in which μ'_1 appears as a factor:

12:06

$$\kappa_2 = \mu_2$$

$$\kappa_3 = \mu_3$$

$$\kappa_4 = \mu_4 - 3\mu_2^2$$

$$\kappa_5 = \mu_5 - 10\mu_3\mu_2$$

$$\kappa_6 = \mu_6 - 15\mu_4\mu_2 - 10\mu_3^2 + 30\mu_2^3.$$

00:15

Topic No. 68

Characteristic Function

00:43

The Characteristic function of a probability distribution is denoted by $\varphi_X(t)$ and is defined as the expected value of e^{itX} i.e.

$$\varphi_X(t) = E\left(e^{itX}\right)$$

where t is an arbitrary real number and i is the imaginary number given by $i = \sqrt{-1}$.

01:53

If the PDF of X is given by $f(x)$, then

$$\varphi_X(t) = E\left(e^{itX}\right) = \int_{-\infty}^{\infty} e^{itx} f(x) dx.$$

02:36

The characteristic function exists for every distribution and it possesses the uniqueness property. In other words, every distribution has unique characteristic function.

In other words, the characteristic function uniquely identifies the probability distribution.

In fact, it completely determines the behavior and properties of the distribution.

04:13

Algebraic expressions of Some well-known
Characteristic Functions

05:09

Characteristic Functions of some well-known
discrete distributions:

Distribution	Cumulative function $\phi(t)$
Bernoulli $P(X = 1) = p$	$1 - p + pe^{it}$

05:32

Characteristic Functions of some well-known
discrete distributions:

Distribution	Cumulative function $\phi(t)$
Binomial $B(n, p)$	$(1 - p + pe^{it})^n$

06:26

Characteristic Functions of some well-known
discrete distributions:

Distribution	Cumulative function $\phi(t)$
Poisson $Pois(\lambda)$	$e^{\lambda(e^{it} - 1)}$

06:51

Characteristic Functions of some well-known
discrete distributions:

Distribution	Cumulative function $\phi(t)$
Geometric $(1 - p)^{k-1} p$	$\frac{pe^{it}}{1 - (1 - p)e^{it}}$

06:56

Characteristic Functions of some well-known
discrete distributions:

Distribution	Cumulative function $\phi(t)$
Uniform (discrete) $U(a, b)$	$\frac{e^{ait} - e^{(b+1)it}}{(b - a + 1)(1 - e^{it})}$

07:04

Characteristic Functions of some well-known
continuous distributions:

- Exponential
- Normal
- Chi-squared
- Gamma
- Cauchy
- Laplace

07:09

Distribution	Characteristic Function $\phi(t)$
Exponential $Exp(\lambda)$	$(1 - it\lambda^{-1})^{-1}$
Normal $N(\mu, \sigma^2)$	$e^{it\mu} - \frac{1}{2}\sigma^2 t^2$

08:32

Distribution	Characteristic Function $\phi(t)$
Chi-squared X^2_k	$(1 - 2it)^{-\frac{k}{2}}$
Gamma $\Gamma(k, \theta)$	$(1 - it\theta)^{-k}$
Cauchy $Cauchy(\mu, \theta)$	$e^{it\mu - \theta t }$
Laplace $L(\mu, b)$	$\frac{e^{it\mu}}{1 + b^2 t^2}$

00:27

Topic No. 69

Derivation of mean and variance of a distribution from its

Characteristic Function

(by repeated differentiation of the Characteristic Function)

00:55

If X has a distribution with characteristic function $\varphi(t)$, then, if $E(X)$ and $E(X^2)$ exist, they are given respectively, by

$$iE(X) = \varphi'(0) \text{ and } i^2E(X^2) = \varphi''(0).$$

01:35

Example:

- Use the characteristic function of the Exponential distribution in order to find the mean and variance.

01:48

Solution:

CF of exponential distribution is given by

$$\varphi(t) = \frac{\lambda}{\lambda - it}$$

Now, taking the first derivative w.r.t 't', we get

$$\begin{aligned}\varphi'(t) &= \frac{d}{dt} \left(\frac{\lambda}{\lambda - it} \right) = \lambda \frac{d}{dt} (\lambda - it)^{-1} = \lambda (-1) (\lambda - it)^{-1-1} \frac{d}{dt} (\lambda - it) \\ &= -\lambda (\lambda - it)^{-2} \frac{d}{dt} (0 - i(1)) = -\lambda (\lambda - it)^{-2} (-i) = \frac{i\lambda}{(\lambda - it)^2}\end{aligned}$$

04:33

Now putting $t=0$, we have

$$\varphi'(0) = \frac{i\lambda}{(\lambda - i(0))^2} = \frac{i\lambda}{(\lambda - 0)^2} = \frac{i\lambda}{\lambda^2} = \frac{i}{\lambda}$$

Therefore,

$$\varphi'(0) = \frac{i}{\lambda} = i \left(\frac{1}{\lambda} \right) = iE(X).$$

07:10

Now, taking the second derivative w.r.t 't', we get

$$\begin{aligned}\varphi''(t) &= \frac{d}{dt} \left(\frac{i\lambda}{(\lambda - it)^2} \right) = i\lambda \frac{d}{dt} (\lambda - it)^{-2} = \lambda (-2) (\lambda - it)^{-2-1} \frac{d}{dt} (\lambda - it) \\ &= -2\lambda (\lambda - it)^{-3} \frac{d}{dt} (0 - i(1)) = \frac{2i^2\lambda}{(\lambda - it)^3} = \frac{-2\lambda}{(\lambda - it)^3} \quad (\because i = \sqrt{-1} \Rightarrow i^2 = -1)\end{aligned}$$

08:57

Now putting $t=0$, we have

$$\varphi''(0) = \frac{-2\lambda}{(\lambda - 0)^3} = \frac{-2\lambda}{\lambda^3} = \frac{-2}{\lambda^2} = \frac{-2}{\lambda^2}$$

Therefore,

$$\varphi'(0) = -1 \left(\frac{2}{\lambda^2} \right) = i^2 \left(\frac{2}{\lambda^2} \right) = i^2 E(X^2). \quad (\because i = \sqrt{-1}, i^2 = -1)$$

11:24

As we know that,

$$V(X) = E(X^2) - [E(X)]^2$$

Therefore, putting the values in the above

$$\frac{1}{\lambda^2} = E(X^2) - \left[\frac{1}{\lambda} \right]^2 \Rightarrow E(X^2) = \frac{1}{\lambda^2} + \frac{1}{\lambda^2} = \frac{2}{\lambda^2}$$

12:02

For any distribution, if a higher moment exists, then all lower moments necessarily exist.

12:29

It is worthwhile to note that

$$\varphi(t) = M(it)$$

00:09

Topic No. 70

Chebyshev's Inequality and its proof & an alternative form of the inequality

00:22

- **Chebyshev's inequality** guarantees that, for a wide class of probability distributions, **no more than a certain fraction of values can be more than a certain distance from the mean.**

00:59

According to the Chebychev Theorem, no more than $1/k^2$ of the distribution's values can be more than k standard deviations away from the mean

where k is any positive real number

00:59

or equivalently,

at least $1-1/k^2$ of the distribution's values lie within k standard deviations of the mean.

02:02

When a random variable is not normally distributed, then we can make use of **Chebyshev's inequality** in order to find out the **minimum** amount of data that is **within k standard deviations** of the mean --- in percentage form.

02:40

In reference to normal distribution, we know that

68.26% are under the curve lies between $\mu \pm \sigma$
95.45% are under the curve lies between $\mu \pm 2\sigma$
and
99.73% are under the curve lies between $\mu \pm 3\sigma$

04:42

If our data is **not** normally distributed, then either we will go to the exact calculations through some amount of labor or we can make use of **Chebyshev's Inequality.**

05:23

Theorem: (Chebyshev's Inequality)

Let the random variable X have a distribution probability about which we assume only that there is a finite variance σ^2 . Then for every $k > 0$,

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2},$$

or, equivalently

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}.$$

06:54

Note

If the variance of a distribution exists, then its mean necessarily exist.

Short-cut formula

07:42

Again see the statement as

Let the random variable X have a distribution probability about which we assume only that there is a finite variance σ^2 .

08:29 Example:

Let $k=2$

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2} \Rightarrow P(|X - \mu| < 2\sigma) \geq 1 - \frac{1}{2^2}$$

$$\Rightarrow P(|X - \mu| < 2\sigma) \geq 1 - \frac{1}{4} \Rightarrow P(|X - \mu| < 2\sigma) \geq \frac{4-1}{4}$$

$$\Rightarrow P(|X - \mu| < 2\sigma) \geq \frac{3}{4}$$

Area that is within $\mu-2\sigma$ and $\mu+2\sigma$, this area is greater than or equal to $\frac{3}{4}$ or 75%

11:09

Hence, the number $1/k^2$ or Chebyshev inequality give us an **upper bound** for the **probability** $P(|X - \mu| \geq k\sigma)$.

00:20

Topic No. 71

Application of Chebyshev's Inequality

00:30

Example:

Let X have the pdf

$$f(x) = \begin{cases} \frac{1}{2\sqrt{3}} & -\sqrt{3} < x < \sqrt{3} \\ 0 & \text{elsewhere.} \end{cases}$$

Here $\mu=0$ and $\sigma^2=1$.

01:58

If $k=3/2$, we have the exact probability

$$P(|X - \mu| \geq k\sigma) = P\left(|X - 0| \geq \left(\frac{3}{2}\right)(1)\right)$$

$$= P\left(|X| \geq \frac{3}{2}\right)$$

$$= 1 - \int_{-3/2}^{3/2} \frac{1}{2\sqrt{3}} dx = 1 - \left(\frac{1}{2\sqrt{3}} (x)^{3/2} \Big|_{-3/2}^{3/2}\right)$$

05:24

$$P\left(|X| \geq \frac{3}{2}\right) = 1 - \left[\frac{1}{2\sqrt{3}} \left(\frac{3}{2} + \frac{3}{2}\right)\right]$$

$$= 1 - \frac{1}{\sqrt{3}} \left(\frac{3}{2}\right) = 1 - \frac{3 \cdot 3^{-\frac{1}{2}}}{2} = 1 - \frac{\sqrt{3}}{2}.$$

07:19

- By Chebyshev's inequality, this probability has the upper bound $1/k^2=4/9$.
- Since $1-\sqrt{3}/2=0.134$, approximately, the exact probability in this case is considerably less than the upper bound $4/9$.

00:12

Topic No. 72

Another application Of Chebyshev's Inequality

00:48

Alternative form

By Chebyshev's inequality, putting $k\sigma = m$, we have

$$P[|X - \mu| \geq m] \leq \frac{\sigma^2}{m^2}, \text{ for all } m > 0$$

$$\text{or } P[|X - \mu| \geq m] \leq \left(\frac{\text{Var}(X)}{m^2}\right)$$

$$\text{Also, } \text{Var}(X) = E(X)^2 - [E(X)]^2 = 13 - 9 = 4.$$

02:42

Example:

If X is a random variable such that $E(X) = 3$ and $E(X^2) = 13$.

Use Chebyshev's inequality to determine a **lower bound** for the probability $P(-2 < X < 8)$.

04:48 Solution

By Chebyshev's inequality, we have

$$P[|X - \mu| \geq m] \leq \frac{\text{Var}(X)}{m^2}, \text{ for all } m > 0$$

In order to obtain a lower bound, we flip the inequalities:

$$P[|X - \mu| < m] > 1 - \left(\frac{\text{Var}(X)}{m^2} \right)$$

06:08

$$\text{or } P[-m < (X - \mu) < m] > 1 - \left(\frac{\text{Var}(X)}{m^2} \right)$$

$$\text{or } P[\mu - m < X < \mu + m] > 1 - \left(\frac{\text{Var}(X)}{m^2} \right)$$

08:07

So,

$$\begin{aligned} \text{Var}(X) &= E(X)^2 - [E(X)]^2 \\ &= 13 - 9 = 4. \end{aligned}$$

08:38

Now, we note that if we put $k = 5$,

then $3 - k = -2$ and $3 + k = 8$,

Therefore, the required probability

$$P[-2 < X < 8] = P[3 - k < X < k + 3]$$

09:47

But, according to Chebyshev's Inequality,

$$P[3 - m < X < m + 3] > 1 - \frac{4}{m^2}$$

So

$$P[-2 < X < 8] > 1 - 4/25 \text{ or } P[-2 < X < 8] > (25 - 4)/25$$

$$\text{or } P[-2 < X < 8] > 21/25 \text{ or } P[-2 < X < 8] > 0.84$$

00:12

Topic No. 73

H.M. $\leq$ G.M. $\leq$ A.M.

00:30

Example: (Harmonic and Geometric Means)

Let $\{a_1, \dots, a_n\}$ be a set of positive numbers.

Create a distribution for a random variable X by placing weight $1/n$ on each of the numbers $a_1, \dots, a_n$.

01:15

Then the mean of X is the arithmetic mean, (AM),

$$E(X) = n^{-1} \sum_{i=1}^n a_i.$$

01:15

Then the mean of X is the arithmetic mean, (AM),

$$\begin{aligned} E(X) &= \sum xp(x) = \frac{a_1}{n} + \frac{a_2}{n} + \dots + \frac{a_n}{n} \\ &= \frac{1}{n}(a_1 + a_2 + \dots + a_n) = \frac{1}{n} \sum a_i \end{aligned}$$

03:44

Theorem (Jensen's Inequality)

If ϕ is convex on an open interval I and X is a random variable whose support I contained in I and has finite expectation, then

$$\phi[E(X)] \leq E[\phi(X)].$$

If ϕ is strictly convex, then the inequality is strict unless X is a constant random variable.

04:52

Then, since $-\log x$ is a convex function, we have by Jensen's inequality that

$$-\log\left(\frac{1}{n}\sum_{i=1}^n a_i\right) \leq E(-\log X) = -\frac{1}{n}\sum_{i=1}^n \log a_i = -\log(a_1 a_2 \dots a_n)^{1/n}$$

08:52
or, equivalent,

$$\log\left(\frac{1}{n}\sum_{i=1}^n a_i\right) \geq \log(a_1 a_2 \dots a_n)^{1/n},$$

and, hence

$$(a_1 a_2 \dots a_n)^{1/n} \leq \frac{1}{n}\sum_{i=1}^n a_i \tag{1}$$

The quantity on the left side of this inequality is called the geometric mean (G.M).
So, equation 1 is equivalent to saying that $GM \leq AM$ for any finite set of positive numbers.

10:52

Now in eq. (1) replace a_i by $1/a_i$ (which is positive).

We then obtain

$$\frac{1}{n}\sum_{i=1}^n \frac{1}{a_i} \geq \left(\frac{1}{a_1} \frac{1}{a_2} \dots \frac{1}{a_n}\right)^{1/n}$$

Or, equivalently,

$$\frac{1}{\frac{1}{n}\sum_{i=1}^n \frac{1}{a_i}} \leq (a_1 a_2 \dots a_n)^{1/n} \tag{2}$$

The left member of this inequality is called harmonic mean HM.

14:00

Putting the equations together, we have shown the relationship

$$HM \leq GM \leq AM, \tag{3}$$

for any finite set of positive numbers.

00:12

Topic No. 74

Concept of a Random Vector
(explained through an example)

01:07

Let us begin the discussion of a **pair of random variables** with the following example.

A coin is tossed three times and our interests is in the ordered number pair (number of H's on first two tosses, number of H's on all three tosses),

where H and T represent respectively, heads and tails.

02:04

Let $C = \{TTT, TTH, THT, HTT, THH, HTH, HHT, HHH\}$ denote the sample space.

Let X_1 denote the number of H's on the first two tosses and X_2 denotes the number of HH's on all three flips.

Then our interest can be represented by the pair of random variables (X_1, X_2) .

03:52

For example, $(X_1(HTH), X_2(HTH))$ represents the outcome (1,2).

Continuing in this way, X_1 and X_2 are real-valued functions defined on the sample space C , which take us from the **sample space** to the space of ordered number pairs.

$$D = \{(0,0), (0,1), (1,1), (1,2), (2,2), (2,3)\}.$$

06:35

Thus X_1 and X_2 are two random variables defined on the space C ,

and,

(in this example), the space of these random variables is the **two-dimensional** set D , which is a subset of the two-dimensional Euclidean Space R^2 .

Hence (X_1, X_2) is a **vector function** from C to D .

07:23

In simple words, the **pair** of random variables (X_1, X_2) is a **random vector**.

07:39

So
Formal Definition of a Random Vector in the case of Two random variables:

07:50

Definition: (Random Vector)

From a random experiment with a sample space C , consider two random variables X_1 and X_2 , which assign to each element c of C one and only one ordered pair of numbers

$$X_1(c) = x_1, \quad X_2(c) = x_2.$$

Then we say that (X_1, X_2) is a random vector.

09:25

The space of (X_1, X_2) is the set of ordered pairs

$$\mathcal{D} = \{(x_1, x_2) : x_1 = X_1(c), x_2 = X_2(c), c \in C\}$$

We often denote random vector notation

$$X = (X_1, X_2)',$$

where the $'$ denotes the transpose of the row vector (X_1, X_2) .

10:23

Conversely,
If (X_1, X_2) is random vector then both X_1 and X_2 are random variables.

00:12

Topic No. 75

**Concept of an event
in a case of a two-dimensional space
(i.e. a set of ordered pairs)**

00:31

- Let $\mathcal{D}$ be the space associated with the random vector (X_1, X_2) .
- Let A be a subset of $\mathcal{D}$. As in the case of one random variable, we speak of the event A .
- We wish to define the probability of the event A , which we denote by

$$P_{X_1, X_2}[A]$$

01:48

- A coin is tossed three times and our interests is in the ordered number pair (number of H's on first two tosses, number of H's on all three tosses), where H and T represent respectively, heads and tails.

02:19

Let $C = \{TTT, TTH, THT, HTT, THH, HTH, HHT, HHH\}$ denote the sample space.

Let X_1 denote the number of H's on the first two tosses and X_2 denotes the number of HH's on all three flips.

Then our interest can be represented by the pair of random variables (X_1, X_2) .

For example, $(X_1(HTH), X_2(HTH))$ represents the outcome $(1, 2)$.

03:28

Continuing in this way, X_1 and X_2 are real-valued functions defined on the sample space C , which take us from the sample space to the space of ordered number pairs.

$$\mathcal{D} = \{(0,0), (0,1), (1,1), (1,2), (2,2), (2,3)\}.$$

04:45

Thus X_1 and X_2 are two random variables defined on the space C , and, in this example, the space of these random variables is the two-dimensional set $\mathcal{D}$, which is a subset of two-dimensional Euclidean R^2 .

Hence (X_1, X_2) is a vector function from C to $\mathcal{D}$.

05:10

Now we can define various events:

e.g.

‘Number of Heads on the first two tosses greater than zero and number of Heads on all three tosses less than 2’

Then Event $A = \{(1,1)\}$.

00:12

Topic No. 76

Joint Cumulative Distribution Function

00:36

- Given random variables $X_1, X_2, \dots$, that are defined on a probability space, the **joint probability distribution** for $X_1, X_2, \dots$ is a probability distribution that gives the probability that each of $X_1, X_2, \dots$ falls in any particular range or discrete set of values specified for that variable.
- In the case of only two random variables, this is called a **bivariate distribution**, but the concept generalizes to any number of random variables, giving a **multivariate distribution**.

01:45

- The joint probability distribution can be expressed either in terms of a **joint cumulative distribution function** or in terms of a joint probability density function (in the case of continuous variables) or joint probability mass function (in the case of discrete variables).

02:25

So, let us focus on

the Joint Cumulative Distribution Function

02:38

We can uniquely define P_{X_1, X_2} in terms of the **cumulative distribution (cdf)**, which is given by

$$F_{X_1, X_2}(x_1, x_2) = P[\{X_1 \leq x_1\} \cap \{X_2 \leq x_2\}], \quad (1)$$

for all $(x_1, x_2) \in \mathbb{R}^2$

because

$X_1 \leq x_1$ and $X_2 \leq x_2$ are events with reference to the random variables X_1 and X_2 separately --- AND the intersection of these two events is the ‘joint event’.

05:40

Thus the expression is well defined.

As with random variables, we write

$$P[\{X_1 \leq x_1\} \cap \{X_2 \leq x_2\}] \text{ as } P[X_1 \leq x_1, X_2 \leq x_2]$$

06:12

Also, we can write,

$$P[a_1 < X_1 \leq b_1, a_2 < X_2 \leq b_2] = F_{X_1, X_2}(b_1, b_2) - F_{X_1, X_2}(a_1, b_2) - F_{X_1, X_2}(b_1, a_2) + F_{X_1, X_2}(a_1, a_2) \quad (2)$$

Hence, all induced probabilities of sets of the form can be formulated in terms of the cdf. $(a_1, b_1] \times (a_2, b_2]$

We often call this cdf the **joint cumulative distribution function** of (X_1, X_2) .

00:12

Topic No. 77

Discrete Random Vector and Joint Probability Mass Function

00:35

Definition: A random vector (X_1, X_2) is a discrete random vector if its space $\mathcal{D}$ is finite or countable.

Hence, X_1 and X_2 are both discrete also.

01:36

The **joint probability mass function (pmf)** of (X_1, X_2) is defined by

$$p_{X_1, X_2}(x_1, x_2) = P[X_1 = x_1, X_2 = x_2],$$

for all $(x_1, x_2) \in \mathcal{D}$

02:32

Properties

It also is characterized by the two properties

$$\begin{aligned} (i) \quad & 0 \leq p_{X_1, X_2}(x_1, x_2) \leq 1 \quad \text{and} \\ (ii) \quad & \sum_D \sum p_{X_1, X_2}(x_1, x_2) = 1. \end{aligned} \quad (1)$$

04:25

For an event $B \in \mathcal{D}$, we have

$$P[(X_1, X_2) \in B] = \sum_B \sum p_{X_1, X_2}(x_1, x_2).$$

05:49

Likewise we may extend the pmf $p_{X_1, X_2}(x_1, x_2)$ over a convenient set by using zero elsewhere.

Hence, we replace

$$\sum_D \sum p_{X_1, X_2}(x_1, x_2) \quad \text{by} \quad \sum_{x_2} \sum_{x_1} p(x_1, x_2).$$

00:12

Topic No. 78

Concept of the Support of a Discrete Random Vector (explained through an example)

00:19

- We begin with the concept of the Support of a random variable:
- The support of a random variable is the set of values that the random variable can take.

01:45

- Concentrating on the Support of a discrete variable:
- For discrete random variables, it is the set of all the realizations that have a strictly **positive** probability of being observed.

03:01

The above ideas can be extended to develop the Concept of the Support of a Discrete Random **Vector**.

03:12

Let us attempt this with the help of the following example.

A coin is tossed three times and our interest is in the ordered number pair (number of heads on first two tosses, number of heads on all three tosses),

03:45

Let H and T represent respectively, heads and tails. Then the sample space C is given by

$$C = \{TTT, TTH, THT, HTT, THH, HTH, HHT, HHH\}.$$

Let

X_1 denote the number of H's on the first two tosses

and

X_2 denotes the number of H's on all three flips.

Then our interest can be represented by the **pair** of random variables (X_1, X_2) .

05:05

Now,
 $X_1(TTT)=0$ and $X_2(TTT)=0$
 $X_1(TTH)=0$ and $X_2(TTH)=1$
 $X_1(THT)=1$ and $X_2(THT)=1$
 $X_1(HTT)=1$ and $X_2(HTT)=1$
 $X_1(THH)=1$ and $X_2(THH)=2$
 $X_1(HTH)=1$ and $X_2(HTH)=2$
 $X_1(HHT)=2$ and $X_2(HHT)=2$
 $X_1(HHH)=2$ and $X_2(HHH)=3$

06:17

Now, our interest can be represented by the **pair** of random variables (X_1, X_2) .
So, the eight possible pairs are
 $(X_1, X_2)=(0,0),$
 $(X_1, X_2)=(0,1),$
 $(X_1, X_2)=(1,1),$
 $(X_1, X_2)=(1,1),$
 $(X_1, X_2)=(1,2),$
 $(X_1, X_2)=(1,2),$
 $(X_1, X_2)=(2,2),$
 $(X_1, X_2)=(2,3),$

07:03

So, we write
 $D = \{(0,0), (0,1), (1,1), (1,2), (2,2), (2,3)\}.$
with **probabilities**
 $P[(X_1, X_2)=(0,0)] = 1/8$
 $P[(X_1, X_2)=(0,1)] = 1/8$
 $P[(X_1, X_2)=(1,1)] = 2/8$
 $P[(X_1, X_2)=(1,2)] = 2/8$
 $P[(X_1, X_2)=(2,2)] = 1/8$
 $P[(X_1, X_2)=(2,3)] = 1/8$

08:36

We can conveniently table the pmf of the random vector (X_1, X_2) as

10:00

		Support of X_2			
		0	1	2	3
Support of X_1	0	1/8	1/8	0	0
	1	0	2/8	2/8	0
	2	0	0	1/8	1/8

00:20

Topic No. 79

Continuous Random Vector
and Joint Probability Density
Function

00:36

- A **random vector** (X_1, X_2) with space D is of the continuous type if its cdf $F_{X_1, X_2}(x_1, x_2)$ is continuous .
- That is $F_{X_1, X_2}(x_1, x_2)$ can be expressed as

$$F_{X_1, X_2}(x_1, x_2) = \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} f_{X_1, X_2}(w_1, w_2) dw_1 dw_2$$

for all $(x_1, x_2) \in R^2$.

04:15

Joint Probability Density Function

We call the integrand

$$F_{X_1, X_2}(x_1, x_2) = \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} f_{X_1, X_2}(w_1, w_2) dw_1 dw_2$$

the **joint probability density function (PDF)** of (X_1, X_2) .

06:10

Then

$$\frac{\partial^2 F_{X_1, X_2}(x_1, x_2)}{\partial x_1 \partial x_2} = f_{X_1, X_2}(x_1, x_2)$$

except possibly on events which have probability zero.

07:40

Properties

A pdf is essentially characterized by the two properties

$$(i) f_{X_1, X_2}(x_1, x_2) \geq 0 \quad \text{and}$$

$$(ii) \int_D \int f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 = 1. \tag{1}$$

09:12

We may extend the definition of a pdf

$$f_{X_1, X_2}(x_1, x_2) \text{ over } R^2$$

by using zero elsewhere.

10:50

- We do this consistently so that tedious, repetitious references to the space D can be avoided. Once this is done, we replace

$$f_{X_1, X_2}(x_1, x_2) \text{ over } R^2$$

$$\iint_D f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 \text{ by } \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x_1, x_2) dx_1 dx_2.$$

12:27

- For an event $A \in \mathcal{D}$, we have

$$P[(X_1, X_2) \in A] = \iint_A f_{X_1, X_2}(x_1, x_2) dx_1 dx_2$$

- Note that the $P[(X_1, X_2) \in A]$ is just the volume under the surface $z = f_{X_1, X_2}(x_1, x_2)$ over the set A .

00:32

Topic No. 80

Determination of the probability of an event in the case of a continuous random vector (explained through an example)

00:43 **Example:** Let

$$f(x_1, x_2) = \begin{cases} 6x_1^2 x_2 & 0 < x_1 < 1, 0 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

be the pdf of the random vector (X_1, X_2) where both X_1 and X_2 are random variables of the continuous type.

Suppose, we wish to compute the probability

$$P\left(0 < X_1 < \frac{3}{4}, \frac{1}{3} < X_2 < 2\right)$$

02:28

Solution:

Since (X_1, X_2) is a continuous random vector, therefore

$$\begin{aligned} P\left(0 < X_1 < \frac{3}{4}, \frac{1}{3} < X_2 < 2\right) &= \int_{1/3}^2 \int_0^{3/4} f(x_1, x_2) dx_1 dx_2 \\ &= \int_{1/3}^2 \int_0^{3/4} 6x_1^2 x_2 dx_1 dx_2 + \int_1^2 \int_0^{3/4} 0 dx_1 dx_2 \\ &= 6x_2 \int_{1/3}^2 \left(\int_0^{3/4} x_1^2 dx_1 \right) dx_2 + 0 \end{aligned}$$

10:10

$$\begin{aligned} P\left(0 < X_1 < \frac{3}{4}, \frac{1}{3} < X_2 < 2\right) &= 6x_2 \int_{1/3}^2 \left(\frac{x_1^{2+1}}{2+1} \Big|_0^{3/4} \right) dx_2 + 0 = 6x_2 \int_{1/3}^2 \left(\frac{\left(\frac{3}{4}\right)^3}{3} - 0 \right) dx_2 + 0 \\ &= 6x_2 \int_{1/3}^2 \left(\frac{27}{64 \times 3} \right) dx_2 + 0 = 3x_2 \int_{1/3}^2 \left(\frac{9}{32} \right) dx_2 + 0 = \frac{27}{32} \int_{1/3}^2 x_2 dx_2 + 0 \end{aligned}$$

11:20

or

$$\begin{aligned} P\left(0 < X_1 < \frac{3}{4}, \frac{1}{3} < X_2 < 2\right) &= \frac{27}{32} \int_{1/3}^2 x_2 dx_2 \\ &= \frac{27}{32} \left(\frac{x_2^{1+1}}{1+1} \Big|_{1/3}^2 \right) = \frac{27}{32} \left(\frac{x_2^2}{2} \Big|_{1/3}^2 \right) = \frac{27}{32} \left(\frac{(2)^2}{2} - \frac{(1/3)^2}{2} \right) \\ &= \frac{27}{32} \left(\frac{1}{2} - \frac{1}{9 \times 2} \right) = \frac{27}{32} \left(\frac{1}{2} - \frac{1}{18} \right) = \frac{27}{32} \left(\frac{9-1}{18} \right) \\ &= \frac{27}{32} \left(\frac{8}{18} \right) = \frac{3}{4} \left(\frac{1}{2} \right) = \frac{3}{8}. \end{aligned}$$

12:35

Note that this probability is the **volume** under the surface given by $f(x_1, x_2) = 6x_1^2 x_2$ above the **rectangular** set

$$\left\{ (x_1, x_2) : 0 < x_1 < \frac{3}{4}, \frac{1}{3} < x_2 < 1 \right\} \in R^2$$

(i.e. the rectangular area given by

$$\left\{ (x_1, x_2) : 0 < x_1 < \frac{3}{4}, \frac{1}{3} < x_2 < 1 \right\} \in R^2).$$

00:12

Topic No. 81

Concept of the Support of a Continuous Random Vector (explained through an example)

00:20

- For a continuous random vector (X_1, X_2) , the support of (X_1, X_2) contains all points (x_1, x_2) for which $f(x_1, x_2) > 0$.

02:10

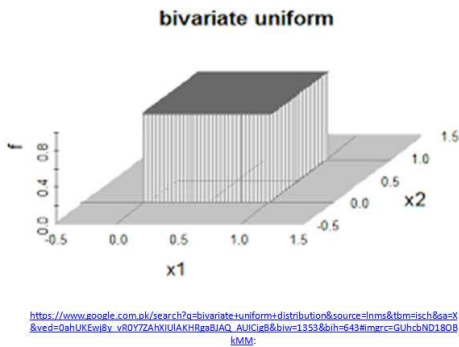
Example:

Let us consider the bivariate uniform distribution given by

$$f(x_1, x_2) = \begin{cases} 1 & 0 < x_1 < 1, 0 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

03:12

Then:



06:50

And

The support of the random vector is

$$\{0 < x_1 < 1, 0 < x_2 < 1\}.$$

00:10

Topic No. 82

**Properties of the
Joint Cumulative Distribution Function**

00:32

Definition

- The joint cumulative function of two random variables X and Y is defined as

$$F_{XY}(x, y) = P(X \leq x, Y \leq y).$$

01:14

Properties:

The joint CDF satisfies the following properties:

- 01:19
- $F_X(x) = F_{XY}(x, \infty)$, for any x (marginal CDF of X);
 - $F_Y(y) = F_{XY}(\infty, y)$, for any y (marginal CDF of Y);
 - $F_{XY}(\infty, \infty) = 1$;
 - $F_{XY}(-\infty, y) = F_{XY}(x, -\infty) = 0$;
 - $P(x_1 < X \leq x_2, y_1 < Y \leq y_2) = F_{XY}(x_2, y_2) - F_{XY}(x_1, y_2) - F_{XY}(x_2, y_1) + F_{XY}(x_1, y_1)$;
 - If X and Y are independent, then $F_{XY}(x, y) = F_X(x) F_Y(y)$.

01:39

- $F_X(x) = F_{XY}(x, \infty)$, for any x (marginal CDF of X);

02:04

- $F_Y(y) = F_{XY}(\infty, y)$, for any y (marginal CDF of Y);

02:15

Explanation

1. $F_X(x) = F_{XY}(x, \infty)$, for any x (marginal CDF of X);
2. $F_Y(y) = F_{XY}(\infty, y)$, for any y (marginal CDF of Y);

05:27

$$3. F_{XY}(\infty, \infty) = 1;$$

Virtual University of Pakistan
Probability Distributions

by

Dr. Saleha Naghmi Habibullah

00:15

Topic No. 83

Properties of the Joint Cumulative Distribution Function

00:23

Definition

- The joint cumulative function of two random variables X and Y is defined as

$$F_{XY}(x, y) = P(X \leq x, Y \leq y).$$

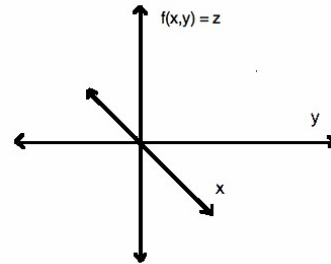
01:03

Properties:

The joint CDF satisfies the following properties:

1. $F_X(x) = F_{XY}(x, \infty)$, for any x (marginal CDF of X);
2. $F_Y(y) = F_{XY}(\infty, y)$, for any y (marginal CDF of Y);
3. $F_{XY}(\infty, \infty) = 1$;
4. $F_{XY}(-\infty, y) = F_{XY}(x, -\infty) = 0$;
5. $P(x_1 < X \leq x_2, y_1 < Y \leq y_2) = F_{XY}(x_2, y_2) - F_{XY}(x_1, y_2) - F_{XY}(x_2, y_1) + F_{XY}(x_1, y_1)$;
6. If X and Y are independent, then $F_{XY}(x, y) = F_X(x) F_Y(y)$.

4. $F_{XY}(-\infty, y) = F_{XY}(x, -\infty) = 0$;



04:52

5. $P(x_1 < X \leq x_2, y_1 < Y \leq y_2) = F_{XY}(x_2, y_2) - F_{XY}(x_1, y_2) - F_{XY}(x_2, y_1) + F_{XY}(x_1, y_1)$;

06:30

6. If X and Y are independent, then

$$F_{XY}(x, y) = F_X(x) F_Y(y).$$

00:12

Topic No. 84

Marginal
Probability Mass Functions
(explained through an example)

00:20

- Consider a discrete random vector, that is, a vector whose entries are discrete random variables.
- When one of these entries is taken in **isolation**, its distribution can be characterized in terms of its probability mass function.

01:02

- This is called **marginal probability mass function**, in order to distinguish it from the joint probability mass function,
- which is instead used to characterize the joint distribution of all the entries of the random vector considered **together**.

01:52

Why the word ‘marginal’?

02:04

- In terms of a **tabled** joint pmf with rows comprised of X_1 support values and columns comprised of X_2 support values, this says that the distribution of X_1 can be obtained by **marginal sums of the rows**.

02:20

		Support of X_2				
		0	1	2	3	
Support of X_1	0	Probabilities				Total
	1					
	2					

→ Marginal of X_1

04:20

		Support of Y				
		0	1	2	3	
Support of X	0	Probabilities				Total
	1					
	2					
		Total				

→ Marginal of X

↓
Marginal of Y

05:20

- To find the probability that X_1 is equal to x_1 , **keep x_1 fixed**, in continuous case, instead of summing we are integrating one variable over the other.

Virtual University of Pakistan
Probability Distributions

by

Dr. Saleha Naghmi Habibullah

00:12

Topic No. 85

Marginal
Probability Mass Functions
(explained through an example)

00:34 **Example:**

- Let us attempt to understand this with the help of the following **example**.

00:36

A coin is tossed three times and our interest is in the ordered number pair (number of heads on first two tosses, number of heads on all three tosses),

00:55

Let H and T represent respectively, heads and tails. Then the sample space C is given by

$$C=\{TTT,TTH,THT,HTT,THH, HTH,HHT,HHH\}.$$

Let
 X_1 denote the **number of H's on the first two tosses**
and
 X_2 denotes the **number of H's on all three flips**.

01:58

Then the space of the discrete random vector (X_1,X_2) is given by

$$D = \{(0,0), (0,1),(1,1), (1,2),(2,2),(2,3)\}.$$

with **probabilities**

$$P[(X_1,X_2)=(0,0)]= 1/8$$

$$P[(X_1,X_2)=(0,1)]= 1/8$$

$$P[(X_1,X_2)=(1,1)]= 2/8$$

$$P[(X_1,X_2)=(1,2)]= 2/8$$

$$P[(X_1,X_2)=(2,2)]= 1/8$$

$$P[(X_1,X_2)=(2,3)]= 1/8$$

04:10

We can conveniently table the pmf of the random vector (X_1, X_2) as:

		Support of X_2			
		0	1	2	3
Support of X_1	0	1/8	1/8	0	0
	1	0	2/8	2/8	0
	2	0	0	1/8	1/8

08:20

Now, to find the marginal probabilities, we have:

08:24

Support of X_2						
		0	1	2	3	$p_{X_1}(x_1)$
Support of X_1	0	1/8	1/8	0	0	2/8
	1	0	2/8	2/8	0	4/8
	2	0	0	1/8	1/8	2/8
	$p_{X_2}(x_2)$	1/8	3/8	3/8	1/8	8/8=1

10:45

Note that it is not **necessary** to have a **formula** for $p(x_1,x_2)$ in order to obtain the marginal PMFs

But

If we can develop a formula, all the better !

00:12

Topic No. 86

Marginal Probability Density Functions
(explained through and example)

00:27

- In case of two continuous random variables X_1 and X_2 , the joint probability density function is given by

$$f_{X_1,X_2}(x_1,x_2)$$

Obviously,

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X_1,X_2}(x_1,x_2) dx_1 dx_2 = 1$$

01:23

- In the continuous case, the **marginal** pdf of X_1 is found by integrating out x_2 i.e.,

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} f_{X_1, X_2}(x_1, x_2) dx_2 \quad (1)$$

- Similarly, the marginal pdf of X_2 is found by integrating out x_1 .

$$f_{X_2}(x_2) = \int_{-\infty}^{\infty} f_{X_1, X_2}(x_1, x_2) dx_1 \quad (2)$$

03:21

Example:

Let X_1 and X_2 have the joint pdf

$$f(x_1, x_2) = \begin{cases} x_1 + x_2 & 0 < x_1 < 1, 0 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

04:00

It is easy to verify that

$$\int_0^1 \int_0^1 (x_1 + x_2) dx_1 dx_2 = 1$$

i.e.

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 = 1$$

04:39

The marginal pdf of X_1 is

$$f_1(x_1) = \int_0^1 (x_1 + x_2) dx_2 = x_1 \int_0^1 1 dx_2 + \int_0^1 x_2 dx_2 = x_1 \left(x_2 \Big|_0^1 + \left(\frac{x_2^2}{2} \Big|_0^1 \right) \right)$$

$$= x_1(1-0) + \left(\frac{1^2}{2} - 0 \right) = x_1(1) + \frac{1}{2}$$

$$f_1(x_1) = x_1 + \frac{1}{2}, \quad 0 < x_1 < 1$$

zero elsewhere,

06:29

and the **marginal** pdf of X_2 is

$$f_2(x_2) = \int_0^1 (x_1 + x_2) dx_1 = \int_0^1 x_1 dx_1 + x_2 \int_0^1 1 dx_1 = \left(\frac{x_1^2}{2} \Big|_0^1 + x_2 \left(x_1 \Big|_0^1 \right) \right)$$

$$= \left(\frac{1^2}{2} - 0 \right) + x_2(1-0) = \frac{1}{2} + x_2(1)$$

$$f_2(x_2) = \frac{1}{2} + x_2, \quad 0 < x_2 < 1,$$

and zero elsewhere.

00:16

Topic No. 87

Another example of Computation of probabilities that can not be found through Marginal PDFs

00:34

Example:

Let $f(x_1, x_2) = \begin{cases} 4x_1x_2, & 0 < x_1 < 1, 0 < x_2 < 1, \\ \text{zero elsewhere,} \end{cases}$

be the joint pdf of X_1 and X_2 .

Find

i) $P(0 < X_1 < \frac{1}{2}, \frac{1}{4} < X_2 < 1),$

ii) $P(X_1 = X_2),$

iii) $P(X_1 < X_2)$

and

iv) $P(X_1 \leq X_2).$

01:20

Solution:

The probability density function is given by

$$f(x_1, x_2) = \begin{cases} 4x_1x_2, & 0 < x_1 < 1, 0 < x_2 < 1, \\ \text{zero elsewhere,} \end{cases}$$

As such:

$$\begin{aligned} \text{i) } P(0 < X_1 < \frac{1}{2}, \frac{1}{4} < X_2 < 1) &= \int_{\frac{1}{4}}^{\frac{1}{2}} \int_0^{\frac{1}{2}} f(x_1, x_2) dx_1 dx_2 \\ &= \int_{\frac{1}{4}}^{\frac{1}{2}} \int_0^{\frac{1}{2}} 4x_1x_2 dx_1 dx_2 = 4 \left(\int_{\frac{1}{4}}^{\frac{1}{2}} \left(\int_0^{\frac{1}{2}} x_1x_2 dx_1 \right) dx_2 \right) \end{aligned}$$

02:47

or

$$\begin{aligned} P(0 < X_1 < \frac{1}{2}, \frac{1}{4} < X_2 < 1) &= 4 \int_{\frac{1}{4}}^{\frac{1}{2}} x_2 \left(\frac{x_1^2}{2} \Big|_0^{\frac{1}{2}} \right) dx_2 \\ &= 4 \int_{\frac{1}{4}}^{\frac{1}{2}} x_2 \left(\frac{(1/2)^2}{2} - 0 \right) dx_2 = 4 \int_{\frac{1}{4}}^{\frac{1}{2}} x_2 \left(\frac{1}{4 \times 2} \right) dx_2 \\ &= \frac{1}{2} \int_{\frac{1}{4}}^{\frac{1}{2}} x_2 dx_2 = \frac{1}{2} \left(\frac{x_2^2}{2} \Big|_{1/4}^{\frac{1}{2}} \right) = \frac{1}{2} \left(\frac{(1)^2}{2} - \frac{(1/4)^2}{2} \right) \\ &= \frac{1}{2} \left(\frac{1}{2} - \frac{1}{16 \times 2} \right) = \frac{15}{64} \end{aligned}$$

03:03

ii) Now to find $P(X_1=X_2)$:

$P(X_1=X_2)$ can be re-written as $P(X_1 - X_2 = 0)$

As the random variables, X_1 and X_2 are **continuous random variables**, the **difference** of these random variables is also a **continuous random variable**.

04:05

Now, zero is a constant

and

we know that the probability that a continuous random variable assumes a constant value is zero.

Therefore, $P(X_1 - X_2 = 0)$

$$P(A_1) = \int_{-\infty}^2 \int_{-\infty}^4 f(x, y) dy dx = \frac{7}{8} = \int_{-\infty}^2 \int_{-\infty}^1 f(x, y) dy dx + \int_{-\infty}^2 \int_1^4 f(x, y) dy dx$$

$$P(A_2) = \int_{-\infty}^2 \int_1^4 f(x, y) dy dx = \frac{4}{8}$$

So,

$$\frac{7}{8} = P(A_1) = \frac{4}{8} = P(A_2) + \int_{-\infty}^2 \int_1^4 f(x, y) dy dx$$

$$\frac{7}{8} = \frac{4}{8} + \int_{-\infty}^2 \int_1^4 f(x, y) dy dx$$

05:42

iii) Find $P(X_1 < X_2)$.

According to the definition,

$$\begin{aligned} P(X_1 < X_2) &= \int_0^1 \int_0^{x_2} f(x_1, x_2) dx_1 dx_2 = \int_0^1 \int_0^{x_2} 4x_1 x_2 dx_1 dx_2 \\ &= 4 \int_0^1 x_2 \left[\left. \frac{x_1^2}{2} \right|_0^{x_2} \right] dx_2 = 4 \int_0^1 x_2 \left[\left(\frac{x_2^2}{2} - \frac{0}{2} \right) \right] dx_2 = 4 \int_0^1 x_2 \cdot \frac{x_2^2}{2} dx_2 \\ &= 2 \int_0^1 x_2^3 dx_2 = 2 \left[\left. \frac{x_2^4}{4} \right|_0^1 \right] = 2 \left(\frac{1}{4} - \frac{0}{4} \right) = \frac{1}{2} \end{aligned}$$

Topic No. 88

**Yet another example of
Computation of probabilities that can
not be found through Marginal PDFs**

Example:

Let $A_1 = \{(x, y): x \leq 2, y \leq 4\}$,

$A_2 = \{(x, y): x \leq 2, y \leq 1\}$,

$A_3 = \{(x, y): x \leq 0, y \leq 4\}$,

and

$A_4 = \{(x, y): x \leq 0, y \leq 1\}$

be subsets of the space A of two random variables X and Y which is the entire two-dimensional plane.

If $P(A_1)=7/8$, $P(A_2)=4/8$, $P(A_3)=3/8$, and $P(A_4)=2/8$, find $P(A_5)$, where $A_5 = \{(x, y): 0 < x \leq 2, 1 < y \leq 4\}$.

HINT:

The required probability will be obtained by splitting the integrals representing the various probabilities.

$$P(A_3) = \int_{-\infty}^0 \int_1^4 f(x, y) dy dx$$

So,

$$P(A_1) = P(A_3) + \int_0^2 \int_{-\infty}^4 f(x, y) dy dx$$

$$\frac{7}{8} = \frac{3}{8} + \int_0^2 \int_{-\infty}^4 dy dx$$

$$P(A_4) = \int_{-\infty}^0 \int_1^1 f(x, y) dy dx$$

So,

$$P(A_3) = P(A_4) + \int_{-\infty}^0 \int_1^4 f(x, y) dy dx$$

$$\frac{3}{8} = \frac{2}{8} + \int_{-\infty}^0 \int_1^4 f(x, y) dy dx$$

So,

$$\int_{-\infty}^2 \int_1^4 f(x, y) dy dx = \frac{7}{8} - \frac{4}{8} = \frac{3}{8} = P(A_6)$$

$$\int_0^2 \int_{-\infty}^4 f(x, y) dy dx = \frac{7}{8} - \frac{3}{8} = \frac{4}{8} = P(A_7)$$

and

$$\int_{-\infty}^0 \int_1^4 f(x, y) dy dx = \frac{3}{8} - \frac{2}{8} = \frac{1}{8} = P(A_8)$$

Hence,

$$P(A_5) = \int_0^2 \int_1^4 f(x, y) dy dx$$

So,

$$\begin{aligned} &\int_{-\infty}^2 \int_1^4 f(x, y) dy dx - \int_{-\infty}^0 \int_1^4 f(x, y) dy dx \\ &= \frac{3}{8} - \frac{1}{8} = \frac{2}{8} = \frac{1}{4} \end{aligned}$$

00:12

Topic No. 89

Expected Value of a Real-Valued Function of a Random Vector

00:38

It is a straightforward extension of the concept of the expected value of a function of a random variable.

01:01

Let (X_1, X_2) be a random vector and let $Y = g(X_1, X_2)$ where $g : R^2 \rightarrow R$ is some real-valued function of X_1 and X_2 .

For example:

- $Y = X_1 + X_2$,
- $Y = X_1^2 - e^{X_2}$

02:03

- Then Y is a **random variable** and we can determine its expectation by obtaining the **distribution** of Y .

02:35

First and foremost, let us determine the **conditions** under which the expectation (or expected value) of Y will **exist**.

Let us commence this discussion with the conditions that are required in the case of a **single** random variable.

03:14

A random variable X will be said to have a finite or infinite expectation (or expected value) **according as** $E(X)$ is a finite number or not.

If it is finite, then the expectation exists and if it is **infinite** (i.e. if it is **not finite**), we shall say that the expectation of X does **not** exist.

04:12

So, naturally, we would like to determine the conditions under which $E(X)$ will be **finite**.

By definition, In the discrete case

$$E(X) = \sum_{i=1}^{\infty} x_i p(x_i)$$

In the continuous case

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

04:41

$E(X)$ will be **finite** if our summation or integral **converges absolutely**.

Hence, naturally, the next question is: What is meant by **absolute convergence**?

05:24

The Concept of Absolute Convergence:

The term 'Absolutely Convergent' describes a series that converges **when all terms are replaced by their absolute values**.

Stated a little differently,

The term 'Absolutely Convergent' describes a series for which **the sum of all its terms remains finite** when all terms are replaced by their **absolute values**.

06:32

Let X be a random variable of the discrete type with probability mass function $p(x_k) = P\{X = x_k\}$, $k = 1, 2, \dots$

If

$$\sum_{k=1}^{\infty} |x_k| p(x_k) < \infty,$$

then, we say that the expected value of X **exists** and we write

$$\mu = E(X) = \sum_{k=1}^{\infty} x_k p(x_k).$$

08:07

Similarly, if X is a random variable of the continuous type with probability density function $f(x)$

Then, if

$$\int_{-\infty}^{\infty} |x|f(x)dx < \infty,$$

then, we say that the expected value of X **exists** and we write

$$\mu = E(X) = \int_{-\infty}^{\infty} xf(x)dx.$$

09:09

A similar definition is available for a function $g(X)$ of X .

Thus, if X is of continuous type and has probability density function $f(x)$, we will say that $E[g(X)]$ exists and equals

$$\int_{-\infty}^{\infty} g(x)f(x)dx$$

provided that

$$\int_{-\infty}^{\infty} |g(x)|f(x)dx < \infty.$$

10:14

A similar definition is available for a function $g(X_1, X_2)$ of two random variables X_1 and X_2 .

If both X_1 and X_2 are of continuous type and have a joint probability density function $f(x_1, x_2)$, we will say that $E[g(x_1, x_2)]$ exists and equals

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x_1, x_2)f_{X_1, X_2}(x_1, x_2)dx_1dx_2$$

provided that

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |g(x_1, x_2)|f_{X_1, X_2}(x_1, x_2)dx_1dx_2 < \infty.$$

11:26

Likewise, if the random vector (X_1, X_2) is discrete and we let $Y = g(X_1, X_2)$, then $E(Y)$ exists if

$$\sum_{x_1} \sum_{x_2} |g(x_1, x_2)|p_{X_1, X_2}(x_1, x_2) < \infty.$$

and is given by

$$E(Y) = \sum_{x_1} \sum_{x_2} g(x_1, x_2)p_{X_1, X_2}(x_1, x_2).$$

00:42

Topic No. 90

Linear Combination of Expected Values of Real-Valued Functions of a Random Vector (explained through an example)

01:00

Theorem:

Let (X_1, X_2) be a random vector, and,

let $Y_1 = g_1(X_1, X_2)$ and $Y_2 = g_2(X_1, X_2)$ be random variables the expectations of which **exist**.

Then, for all real numbers k_1 and k_2 ,

$$E(k_1Y_1 + k_2Y_2) = k_1E(Y_1) + k_2E(Y_2) \quad (1)$$

02:30

Before we begin the proof, let us remind ourselves what is meant by the **existence** of a function of two random variables.

02:43

If $Y = g(X_1, X_2)$ is a function of two random variables X_1 and X_2 both of which are of the **continuous** type and have a joint probability density function $f(x_1, x_2)$, we will say that $E[g(x_1, x_2)]$ exists and equals

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x_1, x_2)f_{X_1, X_2}(x_1, x_2)dx_1dx_2$$

provided that

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |g(x_1, x_2)|f_{X_1, X_2}(x_1, x_2)dx_1dx_2 < \infty.$$

03:50

In other words, if $Y = g(X_1, X_2)$,

then

$$E(Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x_1, x_2)f_{X_1, X_2}(x_1, x_2)dx_1dx_2$$

provided that

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |g(x_1, x_2)|f_{X_1, X_2}(x_1, x_2)dx_1dx_2 < \infty.$$

03:46 Focusing on the theorem again:

Theorem: Let (X_1, X_2) be a random vector, and let $Y_1 = g_1(X_1, X_2)$ and $Y_2 = g_2(X_1, X_2)$ be random variables the expectations of which exist.

Then, for all real numbers k_1 and k_2 ,

$$E(k_1Y_1 + k_2Y_2) = k_1E(Y_1) + k_2E(Y_2) \quad (1)$$

04:40

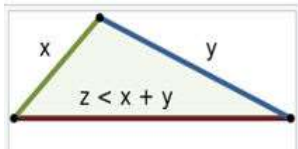
Proof:

For the continuous case, the **existence** of the expected value of $k_1 Y_1 + k_2 Y_2$ follows directly from the **triangle inequality** and **linearity of integrals**.

05:10

So, let us first see what is meant by the Triangle Inequality:

- For any triangle the sum of the length of any two sides must be greater than or equal to the length of the third side.



05:33

So now according to Euclidian geometry:

- If x and y are two real numbers then the absolute value (the modulus) of $|x+y|$ is less than or equal to the modulus of $|x|$ and modulus of $|y|$

$$\|x + y\| \leq \|x\| + \|y\|$$

06:01

Reverting back to the Proof:

For the continuous case, the **existence** of the expected value of $k_1 Y_1 + k_2 Y_2$ follows directly from the **triangle inequality** and **linearity of integrals**.

06:14

Now, according to the triangle inequality, for any two real numbers x and y ,

$$|x + y| \leq |x| + |y|.$$

This implies that

$$\begin{aligned} |k_1 g_1(x_1, x_2) + k_2 g_1(x_1, x_2)| &\leq |k_1 g_1(x_1, x_2)| + |k_2 g_1(x_1, x_2)| \\ \Rightarrow |k_1 g_1(x_1, x_2) + k_2 g_1(x_1, x_2)| &\leq |k_1| |g_1(x_1, x_2)| + |k_2| |g_2(x_1, x_2)| \\ \Rightarrow |k_1 g_1(x_1, x_2) + k_2 g_1(x_1, x_2)| f_{x_1, x_2}(x_1, x_2) &\leq (|k_1| |g_1(x_1, x_2)| + |k_2| |g_2(x_1, x_2)|) f_{x_1, x_2}(x_1, x_2) \end{aligned}$$

or

$$\begin{aligned} |k_1 g_1(x_1, x_2) + k_2 g_1(x_1, x_2)| f_{x_1, x_2}(x_1, x_2) &\leq (|k_1| |g_1(x_1, x_2)| + |k_2| |g_2(x_1, x_2)|) f_{x_1, x_2}(x_1, x_2) \end{aligned}$$

08:19

Now $|k_1 g_1(x_1, x_2) + k_2 g_1(x_1, x_2)| f_{x_1, x_2}(x_1, x_2)$

$$\leq (|k_1| |g_1(x_1, x_2)| + |k_2| |g_2(x_1, x_2)|) f_{x_1, x_2}(x_1, x_2)$$

can be re-written as

$$\begin{aligned} |k_1 g_1(x_1, x_2) + k_2 g_1(x_1, x_2)| f_{x_1, x_2}(x_1, x_2) &\leq |k_1| |g_1(x_1, x_2)| f_{x_1, x_2}(x_1, x_2) + |k_2| |g_2(x_1, x_2)| f_{x_1, x_2}(x_1, x_2) \\ &\dots(1) \end{aligned}$$

08:35

Now, let us consider the concept of Linearity of Integration:

- In calculus, the integral of any linear combination of functions equals the same linear combination of the integrals of functions.; this property is known as linearity of integration.
- Linearity of integration is related to the linearity of summation, since integrals are thought of as infinite sums.

09:05

As such, applying integrals on both sides of inequality (1), we have

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |k_1 g_1(x_1, x_2) + k_2 g_1(x_1, x_2)| f_{x_1, x_2}(x_1, x_2) dx_1 dx_2 &\leq |k_1| \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |g_1(x_1, x_2)| f_{x_1, x_2}(x_1, x_2) dx_1 dx_2 \\ &+ |k_2| \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |g_2(x_1, x_2)| f_{x_1, x_2}(x_1, x_2) dx_1 dx_2 \end{aligned}$$

09:18

Now, according to the statement of the Theorem,

$$Y_1 = g_1(X_1, X_2) \text{ and } Y_2 = g_2(X_1, X_2)$$

are random variables the expectations of which exist i.e.

$$E(Y_1) = E[g_1(X_1, X_2)] \text{ exists} \Rightarrow \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |g_1(x_1, x_2)| f_{x_1, x_2}(x_1, x_2) dx_1 dx_2 < \infty.$$

$$E(Y_2) = E[g_2(X_1, X_2)] \text{ exists} \Rightarrow \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |g_2(x_1, x_2)| f_{x_1, x_2}(x_1, x_2) dx_1 dx_2 < \infty$$

10:20

$$\text{Hence } \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |k_1 g_1(x_1, x_2) + k_2 g_1(x_1, x_2)| f_{x_1, x_2}(x_1, x_2) dx_1 dx_2$$

$$\begin{aligned} &\leq |k_1| \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |g_1(x_1, x_2)| f_{x_1, x_2}(x_1, x_2) dx_1 dx_2 \\ &+ |k_2| \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |g_2(x_1, x_2)| f_{x_1, x_2}(x_1, x_2) dx_1 dx_2 < \infty \end{aligned}$$

$$\text{Therefore } \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} [k_1 g_1(x_1, x_2) + k_2 g_1(x_1, x_2)] f_{x_1, x_2}(x_1, x_2) dx_1 dx_2$$

$$= E[k_1 g_1(x_1, x_2) + k_2 g_1(x_1, x_2)] \text{ exists.}$$

11:33

By **once again** using linearity of the integral, we have

$$\begin{aligned} E(k_1 Y_1 + k_2 Y_2) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} [k_1 g_1(x_1, x_2) + k_2 g_2(x_1, x_2)] f_{x_1, x_2}(x_1, x_2) dx_1 dx_2 \\ &= k_1 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g_1(x_1, x_2) f_{x_1, x_2}(x_1, x_2) dx_1 dx_2 + k_2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g_2(x_1, x_2) f_{x_1, x_2}(x_1, x_2) dx_1 dx_2 \\ &= k_1 E(Y_1) + k_2 E(Y_2), \end{aligned}$$

i.e., the desired result.

Topic No. 91

Determination of the Expectation of the Product of two discrete random variables (explained through an example)

Example:

Let X_1, X_2 be two random variables with the joint pmf.

$$p(x_1, x_2) = \begin{cases} \frac{(x_1 + x_2)}{12} & , \text{ for } x_1 = 1, 2, x_2 = 1, 2, \\ 0 & \text{elsewhere} \end{cases}$$

Compute

$$E(X_1), E(X_2) \text{ and}$$

$$E(X_1 X_2). \text{ Is } E(X_1 X_2) = E(X_1) E(X_2)?$$

Solution:

First we know that,

$$p(x_1) = \sum_{x_2=1,2} \frac{x_1 + x_2}{12} = \frac{x_1 + 1}{12} + \frac{x_1 + 2}{12} = \frac{x_1 + 1 + x_1 + 2}{12} = \frac{2x_1 + 3}{12}, \quad x_1 = 1, 2$$

In the same way,

$$p(x_2) = \sum_{x_1=1,2} \frac{x_1 + x_2}{12} = \frac{1 + x_2}{12} + \frac{2 + x_2}{12} = \frac{2x_2 + 3}{12}, \quad x_2 = 1, 2$$

Therefore, as X_1 and X_2 are symmetric, they have the same pmfs and

Hence, their mathematical expectations will be equal.

Now computing

$$E(X_1) = \sum_{x_1=1,2} x_1 \frac{2x_1 + 3}{12} = (1) \frac{2(1) + 3}{12} + (2) \frac{2(2) + 3}{12} = \frac{5}{12} + \frac{14}{12} = \frac{19}{12}$$

$$E(X_2) = E(X_1) = \sum_{x_2=1,2} x_2 \frac{2x_2 + 3}{12} = (1) \frac{2(1) + 3}{12} + (2) \frac{2(2) + 3}{12} = \frac{5}{12} + \frac{14}{12} = \frac{19}{12}$$

Now

$$\begin{aligned} E(X_1 X_2) &= \sum_{x_1, x_2} x_1 \cdot x_2 \frac{x_1 + x_2}{12} \\ &= (1 \times 1) \frac{1+1}{12} + (1 \times 2) \frac{1+2}{12} + (2 \times 1) \frac{2+1}{12} + (2 \times 2) \frac{2+2}{12} \\ &= \frac{2}{12} + \frac{6}{12} + \frac{6}{12} + \frac{16}{12} = \frac{30}{12} = 2.5 \end{aligned}$$

On the other hand,

$$E(X_1) E(X_2) = \frac{19}{12} \times \frac{19}{12} = 2.5069$$

Hence

$$E(X_1 X_2) \neq E(X_1) E(X_2)$$

00:09

Topic No. 92

Determination of the Expectation of the Product of two continuous random variables (explained through an example)

00:25 Example:

Let X_1, X_2 be two random variables with the joint pdf $f(x_1, x_2) = 4x_1 x_2$, $0 < x_1 < 1$, $0 < x_2 < 1$, zero elsewhere. Compute $E(X_1)$, $E(X_2)$

$$E(X_1), E(X_1^2), E(X_2), E(X_2^2), \text{ and } E(X_1 X_2).$$

$$\text{Is } E(X_1 X_2) = E(X_1) E(X_2)?$$

$$\text{Find } E(3X_1 - 2X_1^2 + 6X_1 X_2).$$

01:30

Firstly, $E(X_1)$

As a rule:

$$\begin{aligned} E(X_1) &= \int_{-\infty}^{\infty} x_1 f_{x_1}(x_1) dx_1 = \int_{-\infty}^{\infty} x_1 f_{x_1}(x_1) dx_1 = \int_{-\infty}^{\infty} x_1 \left(\int_{-\infty}^{\infty} f(x_1, x_2) dx_2 \right) dx_1 \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} x_1 f(x_1, x_2) dx_2 \right) dx_1 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 f(x_1, x_2) dx_2 dx_1 \end{aligned}$$

05:00

Similarly, $E(X_2)$

As a rule:

$$E(X_2) = \int_{-\infty}^{\infty} x_2 f_{X_2}(x_2) dx_2 = \int_{-\infty}^{\infty} x_2 f_{X_2}(x_2) dx_2 = \int_{-\infty}^{\infty} x_2 \left(\int_{-\infty}^{\infty} f(x_1, x_2) dx_1 \right) dx_2$$

$$= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} x_2 f(x_1, x_2) dx_2 \right) dx_1 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_2 f(x_1, x_2) dx_2 dx_1$$

05:55

Firstly, $E(X_1)$

$$E(X_1) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 f(x_1, x_2) dx_2 dx_1 = \int_0^1 \int_0^1 x_1 \cdot 4x_1 x_2 dx_2 dx_1 = \int_0^1 4x_1^2 x_2 dx_2 dx_1$$

$$= \int_0^1 \left(\int_0^1 4x_1^2 x_2 dx_2 \right) dx_1 = \int_0^1 \left(4x_1^2 \cdot \left(\frac{x_2^2}{2} \right) \Big|_0^1 \right) dx_1 = \int_0^1 \left(4x_1^2 \cdot \left(\frac{1}{2} - \frac{0}{2} \right) \right) dx_1$$

$$= \int_0^1 (2x_1^2) dx_1 = 2 \left(\frac{x_1^3}{3} \right) \Big|_0^1 = 2 \left(\frac{1}{3} - \frac{0}{3} \right)$$

$$\therefore E(X_1) = \frac{2}{3}$$

06:27

Then $E(X_2)$

$$E(X_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_2 f(x_1, x_2) dx_1 dx_2 = \int_0^1 \int_0^1 x_2 \cdot 4x_1 x_2 dx_1 dx_2 = \int_0^1 4x_1 x_2^2 dx_1 dx_2$$

$$= \int_0^1 \left(\int_0^1 4x_1 x_2^2 dx_1 \right) dx_2 = \int_0^1 \left(4x_2^2 \cdot \left(\frac{x_1^2}{2} \right) \Big|_0^1 \right) dx_2 = \int_0^1 \left(4x_2^2 \cdot \left(\frac{1}{2} - \frac{0}{2} \right) \right) dx_2$$

$$= \int_0^1 (2x_2^2) dx_2 = 2 \left(\frac{x_2^3}{3} \right) \Big|_0^1 = 2 \left(\frac{1}{3} - \frac{0}{3} \right)$$

$$\therefore E(X_2) = \frac{2}{3}$$

06:51

Similarly $E(X_1 X_2)$

As a rule:

$$E(X_1 X_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f_{X_1, X_2}(x_1, x_2) dx_2 dx_1 = \int_{-\infty}^{\infty} x_1 x_2 \left(\int_{-\infty}^{\infty} f(x_1, x_2) dx_2 \right) dx_1$$

$$= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} x_1 x_2 f(x_1, x_2) dx_2 \right) dx_1 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f(x_1, x_2) dx_2 dx_1$$

07:11

Now $E(X_1 X_2)$

$$E(X_1 X_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f(x_1, x_2) dx_1 dx_2 = \int_0^1 \int_0^1 x_1 x_2 \cdot 4x_1 x_2 dx_1 dx_2 = \int_0^1 4x_1^2 x_2^2 dx_1 dx_2$$

$$= \int_0^1 \left(\int_0^1 4x_1^2 x_2^2 dx_1 \right) dx_2 = \int_0^1 \left(4x_2^2 \cdot \left(\frac{x_1^3}{3} \right) \Big|_0^1 \right) dx_2 = \int_0^1 \left(4x_2^2 \cdot \left(\frac{1}{3} - \frac{0}{3} \right) \right) dx_2$$

$$= \frac{4}{3} \int_0^1 x_2^2 dx_2 = \frac{4}{3} \left(\frac{x_2^3}{3} \right) \Big|_0^1 = \frac{4}{3} \left(\frac{1}{3} - \frac{0}{3} \right) = \frac{4}{9}$$

$$\therefore E(X_1 X_2) = E(X_1) E(X_2) = \left(\frac{2}{3} \right) \left(\frac{2}{3} \right) = \frac{4}{9}$$

00:12

Topic No. 93

Determination of the Expectation of the Ratio of two continuous random variables (explained through an example)

00:22

Example:

Let X_1 and X_2 have the pdf

$$f(x_1, x_2) = \begin{cases} 8x_1 x_2 & 0 < x_1 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Suppose the random variable Y is defined by $Y = X_1/X_2$ and we are interested to determine the $E(Y)$.

00:22

Solution:

We can determine $E(Y)$ in two ways.

The first way is **by definition**, i.e., find the **distribution** of Y and then determine its expectation.

02:24

The cdf of Y , for $0 < y < 1$, is derived as follows:

$$F_Y(y) = P(Y \leq y) = P(X_1 \leq yX_2) = \int_0^1 \int_0^{yx_2} 8x_1 x_2 dx_1 dx_2$$

$$= \int_0^1 \left(\int_0^{yx_2} 8x_1 x_2 dx_1 \right) dx_2 = \int_0^1 \left(8x_2 \cdot \left(\frac{x_1^2}{2} \right) \Big|_0^{yx_2} \right) dx_2 = \int_0^1 \left(8x_2 \cdot \left(\frac{y^2 x_2^2}{2} - \frac{0}{2} \right) \right) dx_2$$

$$= \int_0^1 \left(8x_2 \cdot \left(\frac{y^2 x_2^2}{2} \right) \right) dx_2 = \int_0^1 4y^2 x_2^3 dx_2 = 4y^2 \left(\frac{x_2^4}{4} \right) \Big|_0^1 = 4y^2 \left(\frac{1}{4} - \frac{0}{4} \right)$$

$$= 4y^2 \left(\frac{1}{4} \right) = y^2.$$

08:29

Hence

$$F_Y(y) = \begin{cases} y^2 & 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

And hence, the pdf of Y is

$$f_Y(y) = F'_Y(y) = \begin{cases} 2y & 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

09:14

which leads to

$$\begin{aligned}
 E(Y) &= \int_0^1 y(2y)dy \\
 &= 2 \int_0^1 y^2 dy = 2 \left(\frac{y^3}{3} \right)_0^1 = 2 \left(\frac{1}{3} - \frac{0}{3} \right) = 2 \left(\frac{1}{3} \right) \\
 \therefore E(Y) &= \frac{2}{3},
 \end{aligned}$$

09:55

$$\begin{aligned}
 E(Y) &= E\left(\frac{X_1}{X_2}\right) = \int_0^1 \left\{ \int_0^{x_2} \left(\frac{x_1}{x_2}\right) 8x_1 x_2 dx_1 \right\} dx_2 = \int_0^1 \left\{ \int_0^{x_2} 8x_1^2 dx_1 \right\} dx_2 \\
 &= \int_0^1 \left\{ 8 \left(\frac{x_1^3}{3} \right)_0^{x_2} \right\} dx_2 = \int_0^1 \left\{ 8 \left(\frac{x_2^3}{3} - \frac{0}{3} \right) \right\} dx_2 = \int_0^1 \left\{ 8 \left(\frac{x_2^3}{3} \right) \right\} dx_2 \\
 &= \int_0^1 \frac{8}{3} x_2^3 dx_2 = \frac{8}{3} \left(\frac{x_2^4}{4} \right)_0^1 = \frac{8}{3} \left(\frac{1}{4} - \frac{0}{4} \right) = \frac{8}{3} \times \frac{1}{4} = \frac{2}{3}.
 \end{aligned}$$

00:13

Topic No. 94

**How to obtain product moments
and simple moments from the
MGF of a random vector?**

02:17

In a simplified notation

$$\begin{aligned}
 E(X) &= \frac{\partial M(0,0)}{\partial t_1}, & E(Y) &= \frac{\partial M(0,0)}{\partial t_2} \\
 E(X^2) &= \frac{\partial^2 M(0,0)}{\partial t_1^2}, & E(Y^2) &= \frac{\partial^2 M(0,0)}{\partial t_2^2}
 \end{aligned}$$

04:41

and

$$E(XY) = \frac{\partial^2 M(0,0)}{\partial t_1 \partial t_2}$$

06:17

Therefore

$$\mu_1 = E(X) = \frac{\partial M(0,0)}{\partial t_1},$$

$$\mu_2 = E(Y) = \frac{\partial M(0,0)}{\partial t_2}$$

and

$$\sigma_1^2 = E(X^2) - [E(X)]^2 = \left(\frac{\partial^2 M(0,0)}{\partial t_1^2} \right) - \left(\frac{\partial M(0,0)}{\partial t_1} \right)^2$$

$$\sigma_2^2 = E(Y^2) - [E(Y)]^2 = \left(\frac{\partial^2 M(0,0)}{\partial t_2^2} \right) - \left(\frac{\partial M(0,0)}{\partial t_2} \right)^2$$

07:20

And, as far as the Co variance is concerned
we have

$$\begin{aligned}
 E[(X - \mu_1)(Y - \mu_2)] &= E(XY) - E(X)E(Y) \\
 &= \frac{\partial^2 M(0,0)}{\partial t_1 \partial t_2} - \left(\frac{\partial M(0,0)}{\partial t_1} \right) \left(\frac{\partial M(0,0)}{\partial t_2} \right)
 \end{aligned}$$

And from the above we can compute
the correlation coefficient ρ .

09:10

- Thus the **correlation coefficient** may be computed by using the **mgf** of the joint distribution if that function is readily available.

00:13

Topic No. 95

**Expected Value of a Random
Vector (defined in terms of
Component wise expectation)**

00:20

Definition
(Expected Value of a Random Vector)

Let $X = (X_1, X_2)'$ be a random vector.

Then the expected value of X exists if the expectations of X_1 and X_2 exist.

01:29

If $E(X_1)$ and $E(X_2)$ exist, then the expected value of X is given by

$$E(X) = \begin{bmatrix} E(X_1) \\ E(X_2) \end{bmatrix}$$

02:03

Similarly, if we toss two fair coins together, then expected value of $X_1 = 1$

AND

if we toss a fair die, then expected value of $X_2 = 3.5$

06:48

If $E(X_1)$ and $E(X_2)$ exist, then the expected value of X is given by

$$E(X) = \begin{bmatrix} E(X_1) \\ E(X_2) \end{bmatrix} = \begin{bmatrix} 1 \\ 3.5 \end{bmatrix}$$

07:38

If we consider the two experiments together, then the expected value of X is given by

$$E[X] = \begin{bmatrix} E(X_1) \\ E(X_2) \end{bmatrix} = \begin{bmatrix} 1 \\ 3.5 \end{bmatrix}$$

00:19

Topic No. 96

Transformation in the case
of a Bivariate Probability Mass Function
(explained through an example)

00:35

- Let (X_1, X_2) be a random vector.
- Suppose we know the joint distribution of (X_1, X_2) and we seek the distribution of a **transformation** of (X_1, X_2) , say, $Y = g(X_1, X_2)$ or $Y_1 = g_1(X_1, X_2)$, $Y_2 = g_2(X_1, X_2)$
- We may be able to obtain the cdf of Y . Another way is to use a transformation.

02:08

It is best to discuss the discrete and continuous cases **separately**.

02:17

DISCRETE CASE:

- Let $p_{X_1, X_2}(x_1, x_2)$ be the joint pmf of two discrete-type random variables X_1 and X_2 with S the (two-dimensional) set of points at which $p_{X_1, X_2}(x_1, x_2) > 0$; i.e., S is the **support** of (X_1, X_2) .
- Let $y_1 = u_1(x_1, x_2)$ and $y_2 = u_2(x_1, x_2)$ define a **one-to-one transformation** that maps S onto T

04:22

The joint pmf of the two new random variables

$$Y_1 = u_1(X_1, X_2) \text{ and } Y_2 = u_2(X_1, X_2)$$

is given by

$$p_{Y_1, Y_2}(y_1, y_2) = \begin{cases} p_{X_1, X_2}[w_1(y_1, y_2), w_2(y_1, y_2)] & (y_1, y_2) \in T \\ 0 & \text{elsewhere,} \end{cases}$$

where $x_1 = w_1(y_1, y_2)$, $x_2 = w_2(y_1, y_2)$

is the single-valued inverse of

$$y_1 = u_1(X_1, X_2) \text{ and } y_2 = u_2(X_1, X_2).$$

05:40

In using this **change of variable technique**, it should be emphasized that we need **two “new” variables** to replace the two “old” variables.

06:27

After we have found the joint pmf $p_{Y_1, Y_2}(y_1, y_2)$, we may obtain the **marginal** pmf of Y_1 by summing on y_2 or the **marginal** pmf Y_2 by summing on y_1 .

00:10

Topic No. 97

**Transformation in the case
of a Bivariate Probability Mass Function
(explained through an example)**

00:22 **Example:**

Let X_1 and X_2 have the joint pmf

$$p_{X_1, X_2}(x_1, x_2) = \begin{cases} \frac{\mu_1^{x_1} \mu_2^{x_2} e^{-\mu_1 - \mu_2}}{x_1! x_2!}, & x_1 = 0, 1, 2, 3, \dots, x_2 = 0, 1, 2, 3, \dots \\ \text{zero elsewhere} \end{cases}$$

where μ_1 and μ_2 are fixed positive real numbers.

Note: The space S is the set of points (x_1, x_2) , where each of x_1 and x_2 is a **non-negative integer**.

02:32

- Suppose that we wish to find the pmf of $Y_1 = X_1 + X_2$.
- If we use the change of variable technique, we need to define a **second random variable** Y_2 .

03:43

Because Y_2 is of **no interest** to us, let us choose it in such a way that we have a **simple** one-to-one transformation.

For example, take $Y_2 = X_2$.

Then $y_1 = x_1 + x_2$ and $y_2 = x_2$ represent a one-to-one transformation that maps S onto

$$T = \{(y_1, y_2) : y_2 = 0, 1, \dots, y_1 \text{ and } y_1 = 0, 1, 2, \dots\}$$

Note that if $(y_1, y_2) \in T$, then $0 \leq y_2 \leq y_1$.

06:16

The inverse functions are given by

$$x_1 = y_1 - y_2 \quad \text{and} \quad x_2 = y_2.$$

Thus the joint pmf of Y_1 and Y_2 is

$$p_{Y_1, Y_2}(y_1, y_2) = \begin{cases} \frac{\mu_1^{y_1 - y_2} \mu_2^{y_2} e^{-\mu_1 - \mu_2}}{(y_1 - y_2)! y_2!}, & (y_1, y_2) \in T, \\ \text{zero elsewhere} \end{cases}$$

00:54

Topic No. 98

**Transformation in the case
of a Bivariate Probability Density Function
using the Jacobian of Transformation
(explained through an example)**

01:12

Example:

Suppose (X_1, X_2) have the joint pdf

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} 1 & 0 < x_1 < 1, 0 < x_2 < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

The support of (X_1, X_2) is then the set

$$S = \{(x_1, x_2) : 0 < x_1 < 1, 0 < x_2 < 1\}$$

03:00

Suppose $Y_1 = X_1 + X_2$ and $Y_2 = X_1 - X_2$.

where,

$$y_1 = u_1(x_1, x_2) = x_1 + x_2$$

$$y_2 = u_2(x_1, x_2) = x_1 - x_2.$$

04:16

- This transformation is **one-to-one**.
- We first determine the set T in the $y_1 y_2$ -plane that is the mapping of S under this transformation.

04:30

- Inverse function

$$\text{Now } y_1 = x_1 + x_2, y_2 = x_1 - x_2$$

implies that

$$y_1 + y_2 = 2x_1 \Rightarrow x_1 = w_1(y_1, y_2) = \frac{y_1 + y_2}{2}$$

&

$$y_1 - y_2 = 2x_2 \Rightarrow x_2 = w_2(y_1, y_2) = \frac{y_1 - y_2}{2}.$$

06:37

Next, the Jacobian is given by

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix} = -\frac{1}{4} - \frac{1}{4} = -\frac{2}{4} = -\frac{1}{2}$$

$$\Rightarrow |J| = \frac{1}{2}.$$

10:47

Now, using the inequalities $0 < x_1 < 1$ and $0 < x_2 < 1$ we can write

$$0 < \frac{y_1 + y_2}{2} < 1 \quad \text{and} \quad 0 < \frac{y_1 - y_2}{2} < 1$$

and it is easy to see that these are equivalent to

$$-y_1 < y_2, \quad y_2 < 2 - y_1, \quad y_2 < y_1, \quad y_1 - 2 < y_2$$

and these define the set T.

14:35

Hence, the joint pdf of (Y_1, Y_2) is given by

$$f_{Y_1, Y_2}(y_1, y_2) = \begin{cases} f_{X_1, X_2}\left(\frac{y_1 + y_2}{2}, \frac{y_1 - y_2}{2}\right) |J|, & (y_1, y_2) \in T \\ 0 & \text{elsewhere} \end{cases}$$

or

$$f_{Y_1, Y_2}(y_1, y_2) = \begin{cases} \frac{1}{2}, & (y_1, y_2) \in T \\ 0 & \text{elsewhere} \end{cases}$$

17:25

In other words, the joint pdf of (Y_1, Y_2) is given by

$$f_{Y_1, Y_2}(y_1, y_2) = \begin{cases} \frac{1}{2}, & (y_1, y_2) \in T \\ 0 & \text{elsewhere} \end{cases}$$

00:12

Topic No. 99

**Transformation in the case of
a Bivariate Probability Mass Function
using the MGF technique
(explained through an example)**

00:27

Example:

Here X_1 and X_2 have the joint pmf

$$p_{X_1, X_2}(x_1, x_2) = \begin{cases} \frac{\mu_1^{x_1} \mu_2^{x_2} e^{-\mu_1} e^{-\mu_2}}{x_1! x_2!} & x_1 = 0, 1, 2, 3, \dots, x_2 = 0, 1, 2, 3, \dots, \\ 0 & \text{elsewhere,} \end{cases}$$

where μ_1 and μ_2 are fixed positive real numbers.

04:38

MGF of X_1 and X_2

$$E(e^{tX_1}) = \left[e^{\mu_1(e^t - 1)} \right]$$

$$E(e^{tX_2}) = \left[e^{\mu_2(e^t - 1)} \right]$$

05:34

Let $Y = X_1 + X_2$ and consider

$$\begin{aligned} E(e^{tY}) &= \sum_{x_1=0}^{\infty} \sum_{x_2=0}^{\infty} e^{t(x_1+x_2)} p_{X_1, X_2}(x_1, x_2) \\ &= \sum_{x_1=0}^{\infty} e^{tx_1} \frac{\mu_1^{x_1} e^{-\mu_1}}{x_1!} \sum_{x_2=0}^{\infty} e^{tx_2} \frac{\mu_2^{x_2} e^{-\mu_2}}{x_2!} \end{aligned}$$

08:49

$$\begin{aligned} E(e^{tY}) &= \left[e^{-\mu_1} \sum_{x_1=0}^{\infty} \frac{(e^t \mu_1)^{x_1}}{x_1!} \right] \left[e^{-\mu_2} \sum_{x_2=0}^{\infty} \frac{(e^t \mu_2)^{x_2}}{x_2!} \right] \\ &= \left[e^{\mu_1(e^t - 1)} \right] \left[e^{\mu_2(e^t - 1)} \right] \\ &= e^{(\mu_1 + \mu_2)(e^t - 1)}. \end{aligned}$$

Note that the factors in the brackets in the next-to-least equality are mgfs of X_1 and X_2 , respectively.

Hence, the mgf of Y is the same as that of X_1 except μ_1 has been replaced by $\mu_1 + \mu_2$.

12:03

Therefore,

By the uniqueness of mgfs, the pmf of Y must be

$$p_Y(y) = e^{-(\mu_1 + \mu_2)} \frac{(\mu_1 + \mu_2)^y}{y!}, \quad y = 0, 1, 2, \dots,$$

00:15

Topic No. 100

Concept of Conditional distribution with the help of an example

00:21

Introduction

The conditional probability of even A , given event B , as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

provided $P(B) \neq 0$.

01:16

Suppose now that A and B are the events $X=x$ and $Y=y$ so that we can write

$$P(X=x|Y=y) = \frac{P(X=x, Y=y)}{P(Y=y)} = \frac{f(x, y)}{h(y)}$$

provided $P(Y=y) = h(y) \neq 0$, where $f(x, y)$ is the value of the **joint probability distribution** of X and Y at (x, y) .

03:45

Definition

If $f(x, y)$ is the value of the joint probability distribution of the discrete random variables X and Y at (x, y) , and $h(y)$ is the value of the **marginal distribution** of Y of y , the function given by

$$f(x|y) = \frac{f(x, y)}{h(y)}, \quad h(y) \neq 0$$

for each x within the range of X , is called the conditional distribution of X given $Y=y$.

04:55

Example:

Let $f_{1|2}(x_1|x_2) = 2x_1/x_2^2, 0 < x_1 < x_2, 0 < x_2 < 1$ zero elsewhere, and $f_2(x_2) = 5x_2^4, 0 < x_2 < 1$, zero elsewhere, denote, respectively, the conditional pdf of X_1 , given $X_2 = x_2$, and the marginal pdf of X_2 .

Determine: (d) $P\left(\frac{1}{4} < X_1 < \frac{1}{2}\right)$.

08:24

Solution:

First we will find the marginal of x_1

$$\begin{aligned} f_{x_1}(x_1) &= \int_{x_1} 10x_1x_2^2 dx_2 = 10x_1 \int_{x_1} x_2^2 dx_2 \\ &= 10x_1 \left[\frac{x_2^3}{3} \right]_{x_1}^1 = 10x_1 \left(\frac{1}{3} - \frac{x_1^3}{3} \right) = \frac{10x_1}{3} - \frac{10x_1^4}{3} = \frac{10}{3} x_1 (1 - x_1^3) \end{aligned}$$

11:21

Now,

$$\begin{aligned} P\left(\frac{1}{4} < X_1 < \frac{1}{2}\right) &= \frac{10}{3} \int_{1/4}^{1/2} (x_1 - x_1^4) dx_1 = \frac{10}{3} \left(\left[\frac{x_1^2}{2} \right]_{1/4}^{1/2} - \left[\frac{x_1^5}{5} \right]_{1/4}^{1/2} \right) \\ &= \frac{10}{3} \left[\left(\frac{1}{4 \times 2} - \frac{1}{16 \times 2} \right) - \left(\frac{1}{32 \times 5} - \frac{1}{1024 \times 5} \right) \right] \end{aligned}$$

12:30

$$\begin{aligned} P\left(\frac{1}{4} < X_1 < \frac{1}{2}\right) &= \frac{10}{3} \left[\left(\frac{1}{8} - \frac{1}{32} \right) - \left(\frac{1}{160} - \frac{1}{5120} \right) \right] \\ &= \frac{10}{3} \left[\left(\frac{4-1}{32} \right) - \left(\frac{32-1}{5120} \right) \right] = \frac{10}{3} \left[\frac{3}{32} - \frac{31}{5120} \right] \\ &= \frac{10}{3} \left(\frac{480-31}{5120} \right) = \frac{4490}{3 \times 5120} = \frac{449}{1536}. \end{aligned}$$

00:45

Topic No. 101

The Concept of Conditional Expectation, Conditional Mean and Conditional Variance (explained through an example)

00:49

If $u(X_2)$ is a function of X_2 , the conditional expectation of $u(X_2)$, given that $X_1=x_1$, if it exists, is given by

$$E[u(X_2)|x_1] = \int_{-\infty}^{\infty} u(x_2) f_{2|1}(x_2|x_1) dx_2.$$

Note that $E[u(X_2)|x_1]$ is a function of x_1 .

02:48

If they do it exists, then $E(X_2|x_1)$ is the mean and

04:49

$$E\left\{\left[X_2 - E(X_2|x_1)\right]^2 | x_1\right\},$$

is the variance of the conditional distribution of X_2 given $X_1=x_1$, which can be written more simply as $\text{Var}(X_2|x_1)$.

05:37

It is convenient to refer to these as the “conditional mean” and the “conditional variance” of X_2 , given $X_1=x_1$.

Also, we have

$$\text{Var}(X_2|x_1) = E(X_2^2|x_1) - [E(X_2|x_1)]^2$$

from an earlier result.

06:44

The conditional expectation of $u(X_1)$, given $X_2=x_2$, if it exists, is given by

$$E[u(X_1)|x_2] = \int_{-\infty}^{\infty} u(x_1) f_{1|2}(x_1|x_2) dx_1.$$

In the case of when random variables of the discrete type, these conditional probabilities and conditional expectations are computed by using summation instead of integration

07:27

Example:

Let X_1 and X_2 have the joint pdf

$$f(x_1, x_2) = \begin{cases} 2 & 0 < x_1 < x_2 < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

07:58 Then the marginal probability density functions are, respectively,

$$f_1(x_1) = \begin{cases} \int_{x_1}^1 2 dx_2 = 2(1-x_1) & 0 < x_1 < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

and

$$f_2(x_2) = \begin{cases} \int_0^{x_2} 2 dx_1 = 2x_2 & 0 < x_2 < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

10:52

The conditional pdf of X_1 , given $X_2=x_2$, $0 < x_2 < 1$, is

$$f_{1|2}(x_1|x_2) = \begin{cases} \frac{2}{2x_2} = \frac{1}{x_2} & 0 < x_1 < x_2 < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

11:15

Here the conditional mean and the conditional variance of X_1 , given $X_2=x_2$, are respectively,

$$\begin{aligned} E[X_1|x_2] &= \int_{-\infty}^{\infty} x_1 f_{1|2}(x_1|x_2) dx_1 \\ &= \int_0^{x_2} x_1 \left(\frac{1}{x_2}\right) dx_1 = \frac{x_2}{2}, \quad 0 < x_2 < 1, \end{aligned}$$

14:28

and

$$\begin{aligned} \text{Var}(X_1|x_2) &= \int_0^{x_2} \left(x_1 - \frac{x_2}{2}\right)^2 \left(\frac{1}{x_2}\right) dx_1 \\ &= \frac{x_2^2}{12}, \quad 0 < x_2 < 1. \end{aligned}$$

00:15

Topic No. 102

**Rigorous proof of the theorem
containing the results**
 $E[E(X_2|X_1)] = E(X_2)$

01:07

Theorem:

Let (X_1, X_2) be a random vector such that $E(X_2)$ is finite.

Then,

$$E[E(X_2|X_1)] = E(X_2).$$

01:54

Proof:

$$E(X_2) = \int_{-\infty}^{\infty} x_2 h(x_2) dx_2$$

But

$$h(x_2) = \int_{-\infty}^{\infty} f(x_1, x_2) dx_1$$

03:29

Therefore, $E(X_2)$ can be written as

$$\begin{aligned} E(X_2) &= \int_{-\infty}^{\infty} x_2 \left(\int_{-\infty}^{\infty} f(x_1, x_2) dx_1 \right) dx_2 \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_2 f(x_1, x_2) dx_1 dx_2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_2 f(x_1, x_2) dx_2 dx_1 \end{aligned}$$

05:03

Multiplying and dividing by $f_1(x_1)$, $E(X_2)$ can be written as

$$E(X_2) = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x_2 \frac{f(x_1, x_2)}{f_1(x_1)} dx_2 \right] f_1(x_1) dx_1$$

$$\text{But } \int_{-\infty}^{\infty} x_2 \frac{f(x_1, x_2)}{f_1(x_1)} dx_2 = E(X_2 | x_1)$$

Therefore

$$E(X_2) = \int_{-\infty}^{\infty} E(X_2 | x_1) f_1(x_1) dx_1 = E[E(X_2 | x_1)].$$

08:02

Hence, we can write

$$E(X_2) = E[E(X_2 | X_1)].$$

This is sometimes called **the law of total expectation** and sometimes it is called **the law of iterated expectations**.

00:20

Topic No. 103

The Correlation Coefficient

(explained through an example)

00:43

- Let X and Y have the joint pdf $f(x, y)$.
- If $u(x, y)$ is a function of x and y , then $E[u(X, Y)]$ can be determined, if it exists.
- Here, let us assume that all mathematical expectations **exist**.

01:40

The means of X and Y , say μ_1 and μ_2 , are obtained by taking $u(x, y)$ to be x and y , respectively;

i.e.

$$\mu_1 = E[u(X, Y)] = E[X]$$

&

$$\mu_2 = E[u(X, Y)] = E[Y]$$

and

the variances of X and Y , say σ_1^2 and σ_2^2 , are obtained by setting the function $u(X, Y)$ equal to $(x - \mu_1)^2$ and $(y - \mu_2)^2$, respectively.

03:59

Covariance

- Consider the mathematical expectation

$$\begin{aligned} E[(X - \mu_1)(Y - \mu_2)] &= E(XY - \mu_2 X - \mu_1 Y + \mu_1 \mu_2) \\ &= E(XY) - \mu_2 E(X) - \mu_1 E(Y) + \mu_1 \mu_2 \\ &= E(XY) - \mu_1 \mu_2 \end{aligned}$$

- This number is called the **covariance** of X and Y and is often denoted by $\text{cov}(X, Y)$.

09:03 Correlation

If each of σ_1 and σ_2 is positive, the number

$$\rho = \frac{E[(X - \mu_1)(X - \mu_2)]}{\sigma_1 \sigma_2} = \frac{\text{cov}(X, Y)}{\sigma_1 \sigma_2}$$

is called the correlation coefficient of X and Y .

10:23 Example:

Let the random variables X and Y have the joint pdf

$$f(x, y) = \begin{cases} x + y & 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

We next compute the correlation coefficient ρ of X and Y .

12:14

Now

$$\mu_1 = E(X) = \int_0^1 \int_0^1 x(x + y) dx dy = \frac{7}{12}$$

&

$$\sigma_1^2 = E(X^2) - \mu_1^2 = \int_0^1 \int_0^1 x^2(x + y) dx dy - \left(\frac{7}{12}\right)^2 = \frac{11}{144}.$$

13:54

Similarly,

$$\mu_2 = E(Y) = \frac{7}{12} \quad \text{and} \quad \sigma_2^2 = E(Y^2) - \mu_2^2 = \frac{11}{144}.$$

14:19

Therefore, the covariance of X and Y is

$$E(XY) - \mu_1 \mu_2 = \int_0^1 \int_0^1 xy(x + y) dx dy - \left(\frac{7}{12}\right)^2 = -\frac{11}{144}.$$

15:33

Accordingly, the correlation coefficient of X and Y is

$$\rho = \frac{-\frac{11}{144}}{\sqrt{\left(\frac{11}{144}\right)\left(\frac{11}{144}\right)}} = -\frac{1}{11} = -0.09$$

Implying that there exists a very weak negative linear correlation between X and Y .

00:12

Topic No. 104

Properties of the Correlation Coefficient

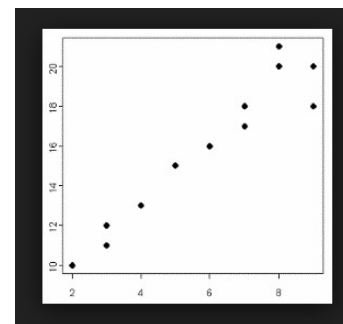
00:25

1. Coefficient of Correlation lies between -1 and +1:

The coefficient of correlation cannot take value less than -1 or more than one +1. Symbolically,

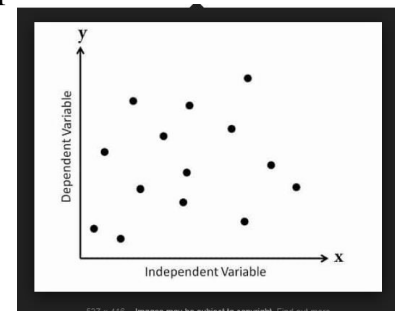
$$-1 \leq \rho \leq +1 \text{ or } |\rho| < 1.$$

01:22 ‘Upward-going’ scatter-diagram: $0 < \rho \leq 1$

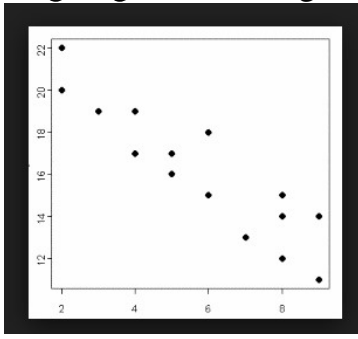


02:44

Neither upward nor downward scatter-diagram: $\rho = 0$



03:18 Downward-going' scatter-diagram: $-1 \leq \rho < 0$



00:12

Topic No. 105

Properties of the Correlation Coefficient

00:18

1. Coefficients of Correlation are independent of Change of Origin
2. Coefficient of Correlation is independent of Change of Scale

00:45

2. Coefficients of Correlation are independent of Change of Origin:

This property reveals that
if we subtract any constant from all the values of X (or add any constant to all the values of X)
and
subtract any constant from all the values of Y (or add any constant to all the values of Y),
it will not affect the coefficient of correlation.

01:49

In other words,

$$\rho_{D_X D_Y} = \rho_{XY}$$

where

$$D_X = X - A \quad \& \quad D_Y = Y - B$$

03:37

2. Coefficient of Correlation is independent of Change of Scale:

This property reveals that if we divide or multiply all the values of X by any constant and divide or multiply all the values of Y by any constant, it will not affect the coefficient of correlation.

$$\rho_{UV} = \rho_{XY}$$

where

$$U = \frac{X}{A} \quad \& \quad V = \frac{Y}{B}$$

00:25

Topic No. 106

Properties of the Correlation Coefficient

00:35

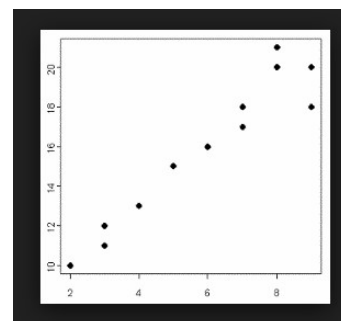
4. Coefficient of Correlation possess the property of symmetry

$$\rho_{XY} = \rho_{YX}$$

03:50

5. Co-efficient of correlation measures only **linear** correlation between X and Y .

04:25 'Upward-going' scatter-diagram: $0 < \rho \leq 1$



6. If two variables X and Y are **independent**, coefficient of correlation between them will be **zero**.

Topic No. 107

Another example of Computation of Correlation Coefficient In the case of joint PDF of two random variables



Computation of Correlation Coefficient

Example:
Let $f(x, y) = \begin{cases} 2, & 0 < x < y, 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$
be the joint pdf of X and Y .
Show that the correlation coefficient of X and Y is $\rho = -1/2$.



Computation of Correlation Coefficient

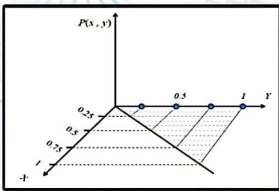
1. "Complicated" domain:

In this regard, the first thing to note is that $0 < x < y, 0 < y < 1$

can be re-written as $0 < x < y < 1$

and that $0 < x < y < 1$ can be re-written as $0 < x < 1, x < y < 1$

Computation of Correlation Coefficient



Computation of Correlation Coefficient

First and foremost, we need to find each of the two marginal distributions:

Computation of Correlation Coefficient

Because of the nature of the support of the bivariate density function, when finding marginal distributions:

the limits of the integral with respect to x are to be 0 to y (and this will give us the marginal distribution of y i.e. $f_Y(y)$)

and

the limits of the integral with respect to y are to be x to 1. (and this will give us the marginal distribution of x i.e. $f_X(x)$)

Computation of Correlation Coefficient

As such:

Marginal distribution of X :

$$f_X(x) = \int_x^1 2dy = 2[y]_x^1 = 2(1-x), 0 < x < 1$$

and

Marginal distribution of Y :

$$f_Y(y) = \int_0^y 2dx = 2[x]_0^y = 2(y-0) = 2y, 0 < y < 1$$

Computation of Correlation Coefficient

Next

$$E(X) = \int_0^1 x \cdot f_X(x) dx = \int_0^1 x \cdot 2(1-x) dx = 2 \int_0^1 (x - x^2) dx$$

$$= 2 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = 2 \left[\left(\frac{1}{2} - 0 \right) - \left(\frac{1}{3} - 0 \right) \right] = 2 \left[\left(\frac{1}{2} \right) - \left(\frac{1}{3} \right) \right] = \frac{1}{3}$$

Computation of Correlation Coefficient

and

$$E(X^2) = \int_0^1 x^2 \cdot f_X(x) dx = \int_0^1 x^2 \cdot 2(1-x) dx = 2 \int_0^1 (x^2 - x^3) dx$$

$$= 2 \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = 2 \left[\left(\frac{1}{3} - 0 \right) - \left(\frac{1}{4} - 0 \right) \right] = 2 \left[\left(\frac{1}{3} \right) - \left(\frac{1}{4} \right) \right] = \frac{1}{6}$$

Computation of Correlation Coefficient

Therefore,

$$\begin{aligned}\text{Var}(X) &= E(X^2) - [E(X)]^2 \\ &= \frac{1}{6} - \left(\frac{1}{3}\right)^2 = \frac{1}{6} - \frac{1}{9} = \frac{9-6}{54} = \frac{3}{54} = \frac{1}{18}\end{aligned}$$

Computation of Correlation Coefficient

Now, the product moment $E(XY)$:

$$\begin{aligned}E(XY) &= \int_0^1 \int_0^y xy \cdot f(x, y) dx dy = \int_0^1 \int_0^y xy(2) dx dy = 2 \int_0^1 \int_0^y xy dx dy \\ &= 2 \int_0^1 y \left(\frac{x^2}{2} \right)_0^y dx = 2 \int_0^1 y \left(\frac{y^2}{2} - 0 \right) dx = \int_0^1 y^3 dx = \left(\frac{y^4}{4} \right)_0^1 = \frac{1}{4} - 0 = \frac{1}{4}.\end{aligned}$$

Computation of Correlation Coefficient

Hence

$$\begin{aligned}\text{Cov}(X, Y) &= E(XY) - E(X)E(Y) \\ &= \frac{1}{4} - \frac{1}{3} \cdot \frac{2}{3} = \frac{1}{4} - \frac{2}{9} = \frac{9-8}{36} = \frac{1}{36}.\end{aligned}$$

Computation of Correlation Coefficient

Therefore,

ρ , the correlation coefficient of X and Y ,
is given by

$$\rho_{x,y} = \frac{\text{Cov}(X, Y)}{\sqrt{\sigma_x^2 \sigma_y^2}} = \frac{\frac{1}{36}}{\sqrt{\left(\frac{1}{18}\right)\left(\frac{1}{18}\right)}} = \frac{\frac{1}{36}}{\sqrt{\left(\frac{1}{18}\right)^2}} = \frac{\frac{1}{36}}{\frac{1}{18}} = \frac{1/2}{1} = \frac{1}{2}.$$

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Probability Distributions

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